

Cloudera on premises / CDP Private Cloud (PvC) Installation & Setup

:: Openshift Setup Guide ::

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In partnership with:

CLOUDERA

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INSTALL/CONFIGURE OPENSIFT

Install OpenShift using the Assisted Installer

Installation parameters:

Parameter	Setting/Value
Cluster name	ocp
Base domain	redhat.local
OpenShift version	4.19.12
CPU architecture	x86_64
Integrate with external partner platforms	No platform integration
Number of control plane nodes	3 (highly available cluster)
Hosts' network configuration	Static IP, bridges, and bonds
Configure via	YAML view
Networking stack type	IPv4
DNS	192.168.1.210
Machine network	192.168.2.0/24
Default gateway	192.168.2.1
api.ocp.redhat.local (API IP)	192.168.2.183
*.apps.ocp.redhat.local (Ingress IP)	192.168.2.184

STEP 1. Create a new OpenShift cluster using the OpenShift Assisted Installer from the Red Hat Hybrid Cloud Console:

<https://console.redhat.com/openshift/assisted-installer/clusters/~new>

Enter the cluster details using the installation parameters provided, then click **Next**.

Install OpenShift with the Assisted Installer

[Assisted Installer documentation](#) [What's new in Assisted Installer?](#)

1 Cluster details

- 2 Static network configurations >
- 3 Operators
- 4 Host discovery
- 5 Storage
- 6 Networking
- 7 Review and create

Cluster details

☐ I'm installing on a disconnected/air-gapped/secured environment Developer Preview

Cluster name *

ocp

Base domain *

redhat.local

Enter the name of your domain [domainname] or [domainname.com]. This cannot be changed after cluster installed. All DNS records must include the cluster name and be subdomains of the base you enter. The full cluster address will be:
ocp.redhat.local

OpenShift version *

OpenShift 4.19.12

[Learn more about OpenShift releases](#)

CPU architecture

x86_64

☐ Edit pull secret ?

Integrate with external partner platforms

No platform integration

Number of control plane nodes ?

3 (highly available cluster)

☐ Include custom manifests ?

Additional manifests will be applied at the install time for advanced configuration of the cluster.

Hosts' network configuration

☐ DHCP only ☒ Static IP, bridges, and bonds

Encryption of installation disks

☒ Control plane nodes

☐ Workers

☐ Arbiter

Next

Cancel

NETWORK CONFIGURATION NOTE:

The three host servers used here have a dual-port 10Gb network adapter. Two YAML NMState configuration examples are provided to utilize both network adapter ports using a network bond:

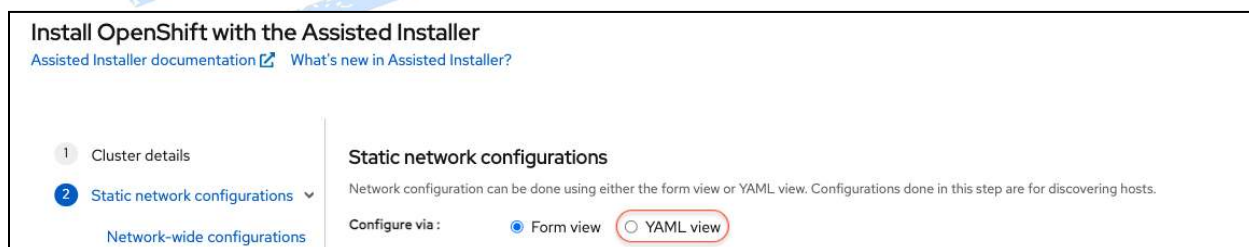
- **Balance-rr aggregate mode:** both network adapters/ports are used simultaneously for network traffic. This does require a static EtherChannel to be configured/enabled on the Ethernet switch (not LACP-negotiated).
- **Active-Backup fail-over mode:** only one network adapter/port will be used at a time for network traffic. This does not require any special network switch configuration.

In my environment, I will be using the **Balance-rr** network bonding mode, and I have configured static EtherChannels for the two ports for each server on my Ethernet switch. If you do not have an Ethernet switch that is capable, you may use the **Active-Backup** network bonding mode, as it does not require any special network switch configuration.

You may also optionally configure OpenShift to use only a single network adapter/port without using a network bond. In this case, it is also possible to assign the second adapter/port for selected traffic types, such as only for storage access, only for use by VMs, or only for VM live migration traffic.

Both configurations provided also set the network mtu to 9000, which requires that jumbo Ethernet frames be enabled on the Ethernet switch for the server network ports in use.

STEP 2. Select the button to configure via the **YAML view**.



The screenshot shows the 'Install OpenShift with the Assisted Installer' window. On the left, a sidebar lists three steps: '1 Cluster details', '2 Static network configurations' (which is highlighted with a blue circle and a dropdown arrow), and 'Network-wide configurations'. The main area is titled 'Static network configurations' and contains the text: 'Network configuration can be done using either the form view or YAML view. Configurations done in this step are for discovering hosts.' Below this text, there is a 'Configure via:' label followed by two radio buttons: 'Form view' (which is currently selected) and 'YAML view' (which is circled in red).

STEP 3. Enter the static network configuration for Host 1 using the following YAML code example along with the information provided in the table.

NOTE: You will need to provide the MAC addresses for your server's network adapters, as they will be different from those provided.

From the YAML view, click on "**Start from scratch**" and then copy/paste the YAML NMState configuration provided below, while editing the IP address to use for the host. Enter the MAC address and Interface name for the first network adapter in the host.

Click on **“Add another MAC to interface name mapping”** and then enter the MAC address and Interface name for the second network adapter in the host.

When you have finished entering the static network configuration for Host 1, select the check box for **“Copy the YAML content”** and click on **“Add another host configuration”**.

Static network configurations

Network configuration can be done using either the form view or YAML view. Configurations done in this step are for discovering hosts.

Upload, drag and drop, or copy and paste a YAML file that contains NMState into the editor for network configurations. Each host also needs the MAC to interface name mapping. [Learn more about NMState](#)

▼ Host 1

```

1 dns-resolver:
2   config:
3     server:
4       - 192.168.1.210
5   interfaces:
6     - ipv4:
7         address:
8           - ip: 192.168.2.170 # change this address per host
9             prefix-length: 24
10        dhcp: false
11        enabled: true
12      ipv6:
13        enabled: false
14      link-aggregation:
15        mode: balance-rr
16        options:
17          miimon: "140"
18      port:
19        - eno1
20        - eno2
21      name: bond0
22      state: up
  
```

MAC to interface name mapping ⓘ

MAC address *	Interface name *
70:DF:2F:F6:A1:2C	eno1
70:DF:2F:F6:A1:2D	eno2

⊕ Add another MAC to interface name mapping

Add another host configuration ☒ Copy the YAML content

Next Back Cancel View cluster events

Parameter	Setting/Value
Host 1 (c240m4-01.redhat.local)	
IP address (IPv4)	192.168.2.170
MAC Address: 70:DF:2F:F6:A1:2C	Interface name: eno1
MAC Address: 70:DF:2F:F6:A1:2D	Interface name: eno2
Host 2 (c240m4-02.redhat.local)	
IP address (IPv4)	192.168.2.171

MAC Address: 70:DF:2F:F6:A2:2C	Interface name: eno1
MAC Address: 70:DF:2F:F6:A2:2D	Interface name: eno2
Host 3 (C240m4-03.redhat.local)	
IP address (IPv4)	192.168.2.172
MAC Address: 70:DF:2F:F6:A3:2C	Interface name: eno1
MAC Address: 70:DF:2F:F6:A3:2D	Interface name: eno2

The following NMState configuration examples are provided for reference only.

These are also in the OpenShift Assisted Installer documentation: [10.4.1. NMState configuration](#) and [10.4.3. Additional NMState configuration examples](#).

YAML NMState configuration using Balance-rr network bonding mode 0.

```
dns-resolver:
  config:
    server:
      - 192.168.1.210
interfaces:
- ipv4:
    address:
      - ip: 192.168.2.170 # change this address per host
        prefix-length: 24
    dhcp: false
    enabled: true
  ipv6:
    enabled: false
  link-aggregation:
    mode: balance-rr
    options:
      miimon: "140"
    port:
      - eno1
      - eno2
  name: bond0
  state: up
  type: bond
  mtu: 9000
routes:
```



```
config:
- destination: 0.0.0.0/0
  next-hop-address: 192.168.2.1
  next-hop-interface: bond0
  table-id: 254
```

YAML NMState configuration using Active-Backup network bonding mode 1

```
dns-resolver:
  config:
    server:
      - 192.168.1.210
interfaces:
- ipv4:
    address:
      - ip: 192.168.2.170 # change this address per host
        prefix-length: 24
    dhcp: false
    enabled: true
  ipv6:
    enabled: false
  link-aggregation:
    mode: active-backup
    options:
      miimon: "140"
    port:
      - eno1
      - eno2
  name: bond0
  state: up
  type: bond
  mtu: 9000
routes:
  config:
    - destination: 0.0.0.0/0
      next-hop-address: 192.168.2.1
      next-hop-interface: bond0
      table-id: 254
```

STEP 4. Enter the static network configuration for Host 2 using the following.

Edit the IP address, and enter the MAC address and Interface name for the first network adapter in the host.

Click on **“Add another MAC to interface name mapping”** and then enter the MAC address and Interface name for the second network adapter in the host.

When you have finished entering the static network configuration for Host 2, select the check box for **“Copy the YAML content”** and click on **“Add another host configuration”**.

1 Cluster details

2 Static network configurations

3 Operators

4 Host discovery

5 Storage

6 Networking

7 Review and create

Static network configurations

Network configuration can be done using either the form view or YAML view. Configurations done in this step are for discovering hosts.

Upload, drag and drop, or copy and paste a YAML file that contains NMState into the editor for network configurations. Each host also needs the MAC to interface name mapping. [Learn more about NMState](#)

> Host 1

MAC to interface name mapping 70:DF:2F:F6:A1:2C -> eno1 1 left

Host 2

```
1 dns-resolver:
2   config:
3     server:
4       - 192.168.1.210
5   interfaces:
6     - ipv4:
7       address:
8         - ip: 192.168.2.171 # change this address per host
9         prefix-length: 24
10      dhcp: false
11      enabled: true
12    ipv6:
13      enabled: false
14    link-aggregation:
15      mode: balance-rr
16      options:
17        miimon: "140"
18      port:
19        - eno1
20        - eno2
21      name: bond0
22      state: up
```

MAC to interface name mapping (?)

MAC address *	Interface name *
70:DF:2F:F6:A2:2C	eno1
MAC address *	Interface name *
70:DF:2F:F6:A2:2D	eno2

+ Add another MAC to interface name mapping

Add another host configuration ☒ Copy the YAML content

STEP 5. Enter the static network configuration for Host 3.

Enter the MAC address and Interface name for the first network adapter in the host.

Click on **“Add another MAC to interface name mapping”** and then enter the MAC address and Interface name for the second network adapter in the host.

When you have finished entering the static network configuration for Host 3, click **“Next”**.

5

Storage

6

Networking

7

Review and create

Host 1

MAC to interface name mapping70:DF:2F:F6:A1:2C -> eno11 left

Host 2

MAC to interface name mapping70:DF:2F:F6:A2:2C -> eno11 left

Host 3

1

dns-resolver:

2

config:

3

server:

4

- 192.168.1.210

5

interfaces:

6

- ipv4:

7

address:

8

- ip: 192.168.2.172 # change this address per host

9

prefix-length: 24

10

dhcp: false

11

enabled: true

12

ipv6:

13

enabled: false

14

link-aggregation:

15

mode: balance-rr

16

options:

17

miimon: "140"

18

port:

19

- eno1

20

- eno2

21

name: bond0

22

state: up

YAML

MAC to interface name mapping

MAC address70:DF:2F:F6:A3:2CInterface nameeno1

MAC address70:DF:2F:F6:A3:2DInterface nameeno2

Add another MAC to interface name mapping

Add another host configurationCopy the YAML content

NextBackCancelView cluster events

STEP 6. For Operators to install, we will install these separately. So, without selecting any here, click “**Next**”.

1

Cluster details

2

Static network configurations

3

Operators

4

Host discovery

5

Storage

6

Networking

7

Review and create

Install OpenShift with the Assisted Installer

Assisted Installer documentationWhat's new in Assisted Installer?

Operators

Find bundles or operators

Bundles

Virtualization

Run virtual machines alongside containers on one platform.

OpenShift AI

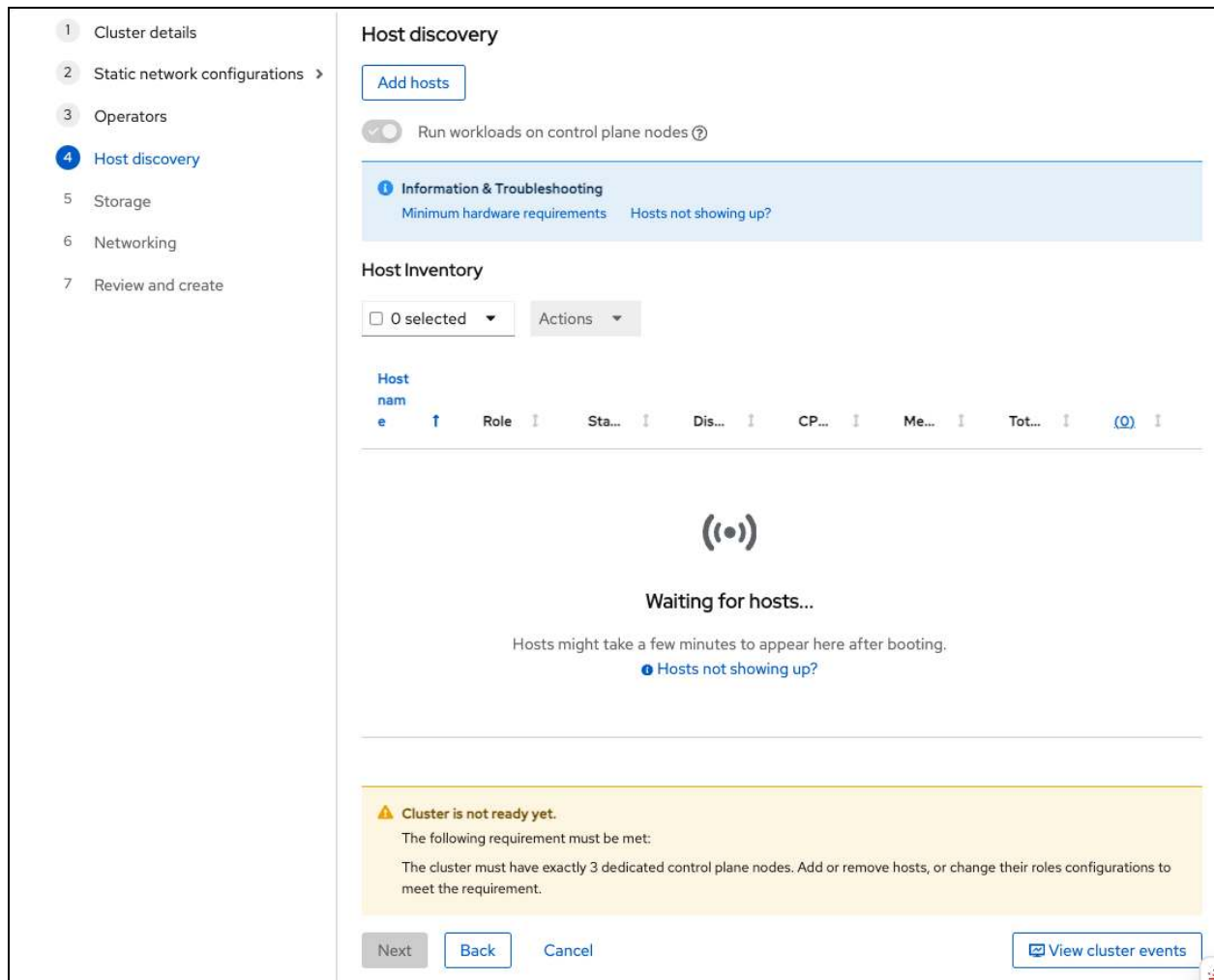
Train, serve, monitor and manage AI/ML models and applications using GPUs.

Developer Preview

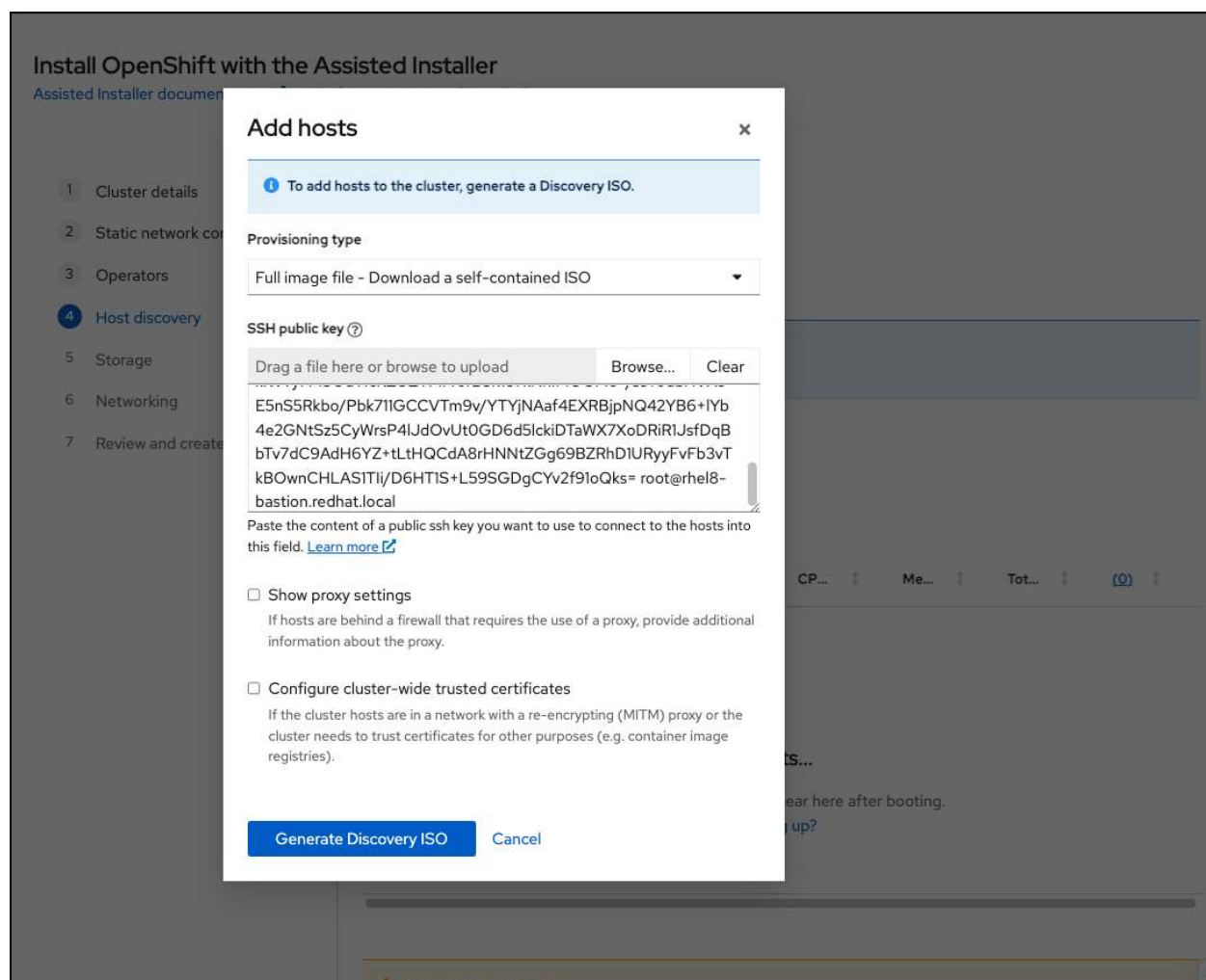
Single Operators (26 | 0 selected)

NextBackCancelView cluster events

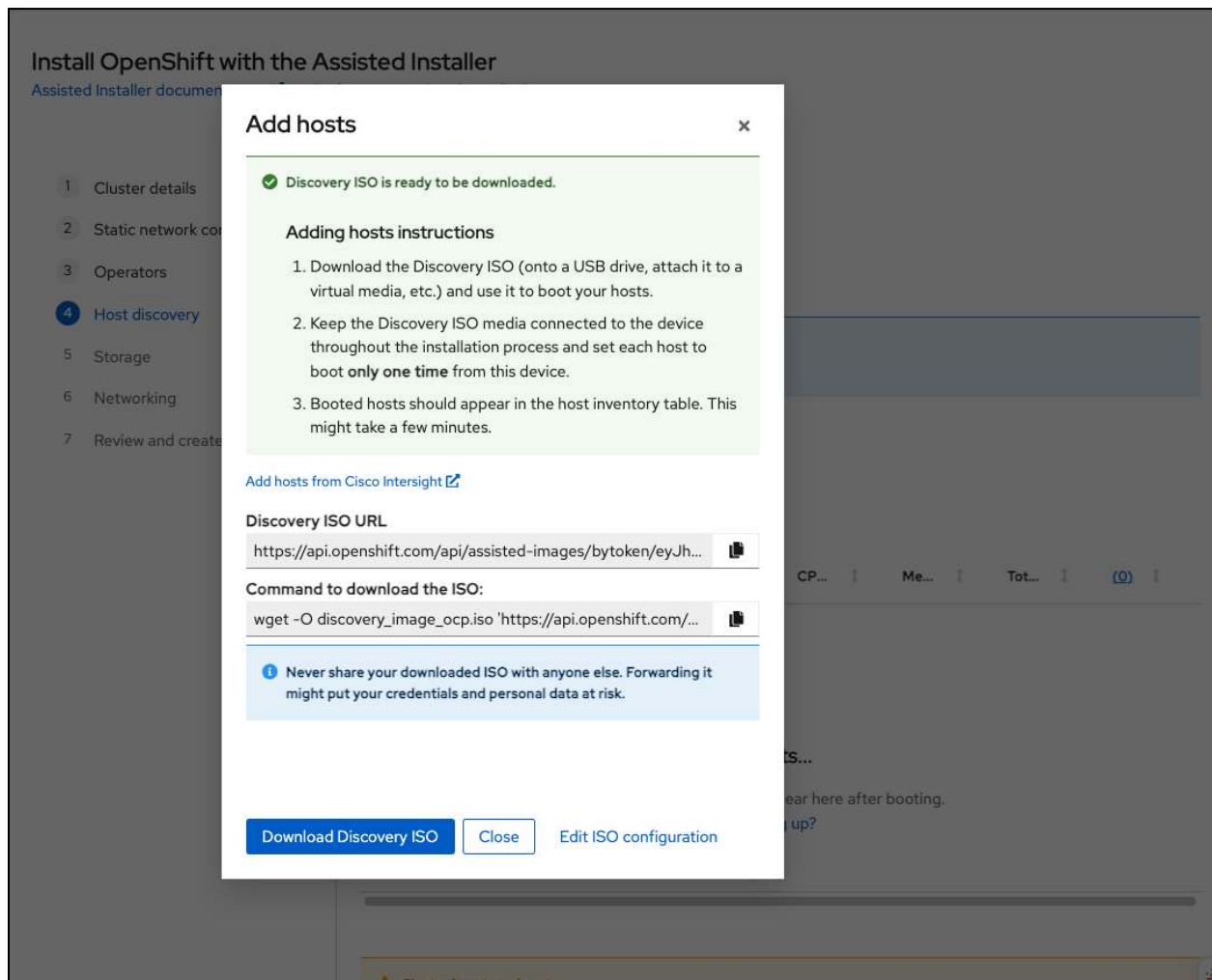
STEP 7. From the Host discovery screen, select “**Add hosts**” to generate a discovery ISO image.



STEP 8. From the “Add hosts” screen, select the **Full image file**, and paste your SSH public key (from ~/.ssh/id_rsa.pub file), then click on “**Generate Discovery ISO**”.



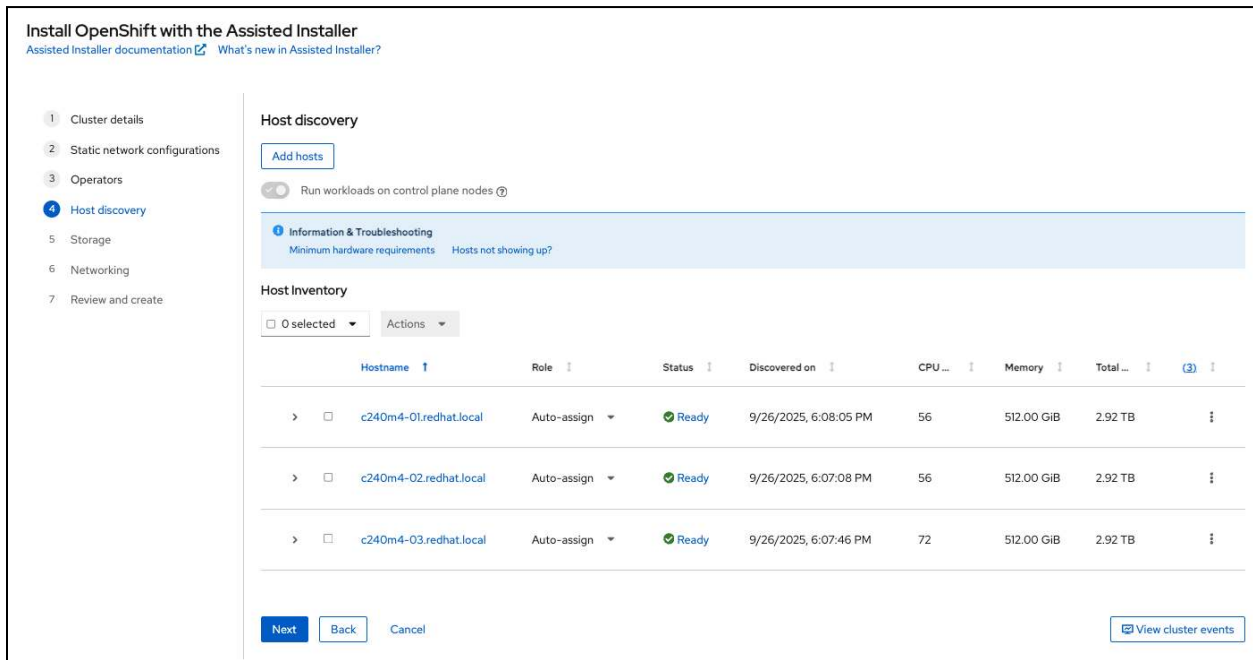
STEP 9. Next, click on the “**Download Discovery ISO**” to download the discovery ISO image. Each server host will boot this image to start the server discovery process. When you have finished downloading the ISO image, click on “**Close**”.



STEP 10. Use one of several methods to boot each server host from this ISO image. For my lab, I will be using this image with the Virtual Media function provided by the server's system management processor. You can also write this ISO image to a USB flash drive, or DVD, in order to boot from physical media types.

For my lab, I will be copying this ISO image file to a web server so that each server's system management processor can mount it as virtual media. Each server is then configured to boot from this virtual media as a one-time boot device.

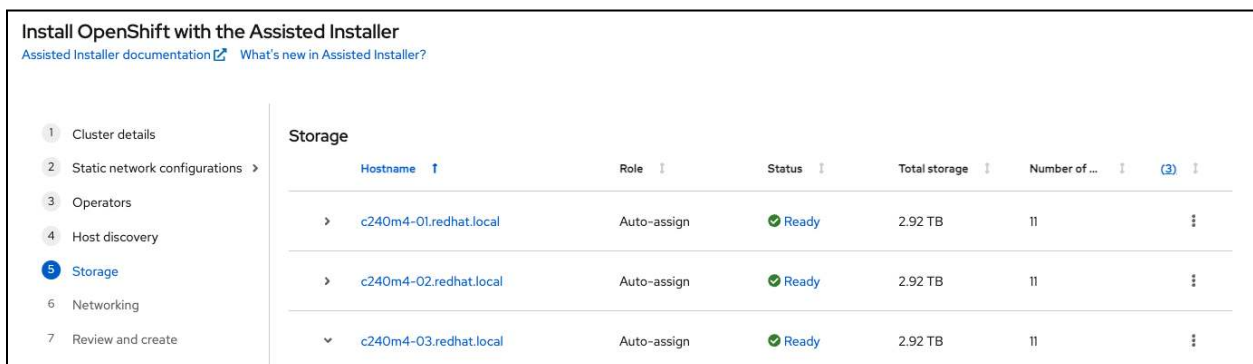
Once all three servers show up on the Host Inventory list, and the cluster is ready, click on **"Next"**.



```
sudo ip link set dev eno1 mtu 9000
sudo ip link set dev eno2 mtu 9000
sudo ip link set dev bond0 mtu 9000
sudo nmcli
```

```
sudo sgdisk -Z /dev/sdX
sudo wipefs -a /dev/sdX
```

STEP 11. From the Storage screen, expand each of the discovered hosts to ensure that the storage devices were discovered properly. Check that the disk that you want to install RHCOS onto is correctly set with a Role of “**installation disk**”. If not, use the drop-down selection to change the Role to the correct disk. In the screenshot, the installation disk is correctly set to a 120GB SSD. Click on “**Next**” to continue.



11 Disks

Name	Role	Limitat...	Format?	Drive t...	Size	Serial	Model	WWN ⓘ
sda	None	▼ 1	☐	SSD	0.00 B	20111102-00000002	vKVM-Mapped_vHDD	
sdb	None	▼ 1	☐	SSD	0.00 B	20111102-00000002	vKVM-Mapped_vFDD	
sdc	None	▼ 1	☐	SSD	0.00 B	20111102-00000002	CIMC-Mapped_vHDD	
sdd (bootable)	Installation disk		☒	SSD	120.03 GB	PHWA638304H4120CGN	INTEL_SSDSC2BB12	0x55cd2e414d4af529
sde	None	▼	☐	SSD	400.09 GB	S18PNWAGB03349	MZ6ER400HAGL_003	0x5002538475bfda0
sdf	None	▼	☐	SSD	400.09 GB	S18PNEAFC01271	MZ6ER400HAGL_003	0x5002538454c716e0
sdg	None	▼	☐	SSD	1.60 TB	PHDV6325005YIP6EGN	INTEL_SSDSC2BB01	0x55cd2e414d453bd4
sdh	None	▼	☐	SSD	200.05 GB	S18NNEAF800120	MZ6ER200HAGM_003	0x500253845481bfd0
sdi	None	▼	☐	SSD	200.05 GB	S18NNEAFA00354	MZ6ER200HAGM_003	0x5002538454a497d0
sr0	None	▼ 2	☐	ODD	1.07 GB	20111102-00000002	vKVM-Mapped_vDVD	
srl (bootable)	None	▼ 2	☐	ODD	1.32 GB	20111102-00000002	CIMC-Mapped_vDVD	

⚠ All bootable disks, except for read-only disks, will be formatted during installation. Make sure to back up any critical data before proceeding.

Next
Back
Cancel
View cluster events

STEP 12. From the Networking screen, enter the following parameters, and click on **Next** to continue.

Parameter	Setting/Value
Network Management	Cluster-Managed Networking
Machine Network	192.168.2.0/24 (192.168.2.0 - 192.168.2.255)
API IP	192.168.2.183
Ingress IP	192.168.2.184
Use advanced networking	Not selected
Use the same host discovery SSH key	Selected

Install OpenShift with the Assisted Installer

[Assisted Installer documentation](#) [What's new in Assisted Installer?](#)

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- 7 Review and create

Networking

Network Management

☒ Cluster-Managed Networking ☐ User-Managed Networking [?](#)

Networking stack type

☒ IPv4 [?](#) ☐ Dual-stack [?](#)

Machine network *

192.168.2.0/24 (192.168.2.0 - 192.168.2.255)

API IP [?](#) *

192.168.2.183

Ingress IP [?](#) *

192.168.2.184

☐ Use advanced networking

Configure advanced networking properties (e.g. CIDR ranges).

Host SSH Public Key for troubleshooting after installation

☒ Use the same host discovery SSH key

Host inventory

Hostname	Role	Stat...	Acti...	IPv4 addre...	IPv...	MAC address	(3)
> c240m4-01.redhat.local	Auto-assign	Ready	bond0	192.168.2.170/24	-	70:df:2f:f6:a1:2c	⋮
> c240m4-02.redhat.local	Auto-assign	Ready	bond0	192.168.2.171/24	-	70:df:2f:f6:a2:2c	⋮
> c240m4-03.redhat.local	Auto-assign	Ready	bond0	192.168.2.172/24	-	70:df:2f:f6:a3:2c	⋮

Next

Back

Cancel

[View cluster events](#)

STEP 13. From the Review and create screen, verify the installation parameters and click on ***"Install cluster"*** to start the installation.

Install OpenShift with the Assisted Installer

[Assisted Installer documentation](#) [What's new in Assisted Installer?](#)

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- 7 **Review and create**

Review and create

> Preflight checks ☒ Cluster preflight checks ☒ Host preflight checks ☒ Cluster support level: Full

Cluster summary

Cluster details

Cluster address ocp.redhat.local
OpenShift version 4.19.12
CPU architecture x86_64
Hosts' network configuration Static IP

Host inventory

Hosts 3
Total cores 184
Total memory 1.50 TiB
Total storage 8.76 TB

▼ Networking

Networking management type

Cluster-managed networking

Stack type

IPv4

Machine networks CIDR

192.168.2.0/24

API IP

192.168.2.183

Ingress IP

192.168.2.184

Advanced networking settings

Cluster network CIDR

10.128.0.0/14

Cluster network host prefix

23

Service network CIDR

172.30.0.0/16

Networking type

Open Virtual Network (OVN)

Install cluster

Back

Cancel

View cluster events

STEP 14. Monitor the progress of the installation. Select “View cluster events” and the “Status” field to the right of each server host.

OpenShift

[Cluster List](#) > [Assisted Clusters](#) > ocp

ocp

Installation progress

Started on

9/25/2025, 5:50:00 PM

Preparing for installation

0%

✔

Control Planes

0 control plane nodes installed

Ⓢ

Initialization

Pending

Abort installation

Download kubeconfig

View cluster events

Download Installation Logs

> Host inventory (3) ✔

> Cluster summary

[Back to all clusters](#)

[Cluster List](#) > [Assisted Clusters](#) > ocp

ocp

Installation progress

Started on
9/25/2025, 5:50:00 PM

Preparing for installation0%

Control Planes
Installing 3 control plane nodes

Initialization
Pending

Abort installation

Download kubeconfig

View cluster events

Download Installation Logs

▼ Host inventory (3)

Hostname	Role	Status	Discovered on	CPU Cor...	Memory	Total s...
c240m4-01.redhat.local	Control plane node, Worker	Preparing for installation	9/25/2025, 4:46:35 PM	56	512.00 GiB	2.92 TB
c240m4-02.redhat.local	Control plane node, Worker (bootstrap)	Preparing for installation	9/25/2025, 4:44:40 PM	56	512.00 GiB	2.92 TB
c240m4-03.redhat.local	Control plane node, Worker	Preparing for installation	9/25/2025, 4:49:23 PM	72	512.00 GiB	2.92 TB

Cluster summary

ocp

Installation progress

Started on
9/25/2025, 5:50:00 PM

Installing

Control Planes
Installing 3 control plane nodes

Initialization
Pending

Abort installation

Download kubeconfig

View cluster events

Download Installation Logs

Download and save your Assisted Installer's service account

▼ Host inventory (3)

Hostname	Role	Status	Discovered on	CPU Cor...	Memory	Total s...
c240m4-01.redhat.local	Control plane node, Worker	Preparing for installation	9/25/2025, 4:46:35 PM	56	512.00 GiB	2.92 TB
c240m4-02.redhat.local	Control plane node, Worker (bootstrap)	Preparing for installation	9/25/2025, 4:44:40 PM	56	512.00 GiB	2.92 TB
c240m4-03.redhat.local	Control plane node, Worker	Preparing for installation	9/25/2025, 4:49:23 PM	72	512.00 GiB	2.92 TB

Cluster summary

Cluster Events

▼ Hosts

▼ Severity

Filter by text

Time	Message
9/25/2025, 5:51:32 PM	Host: c240m4-03.redhat.local, reached installation stage Starting installation: master
9/25/2025, 5:51:32 PM	Host c240m4-03.redhat.local: updated status from installing to installing-in-progress (Starting installation)
9/25/2025, 5:51:32 PM	c240m4-03.redhat.local: Performing quick format of disk sdd(/dev/disk/by-id/wwn-0x55cd2e41d4d4af529)
9/25/2025, 5:51:21 PM	Host c240m4-03.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:21 PM	Host c240m4-02.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:21 PM	Host c240m4-01.redhat.local: updated status from preparing-successful to installing (Installation is in progress)
9/25/2025, 5:51:17 PM	Updated status of the cluster to installing
9/25/2025, 5:51:13 PM	Host c240m4-02.redhat.local: updated status from preparing-for-installation to preparing-successful (Host finished successfully to prepare for installation)

1 - 10 of 611 of 7

Close

STEP 15. When the installation is complete, download the kubeconfig file, and record the password for the kubeadmin user.

Started on
9/25/2025, 5:50:00 PM

Installed on 9/25/2025, 7:00:11 PM

Control Planes
3 control plane nodes
installed

Initialization
Completed

Installation completed successfully

Launch OpenShift Console

Download kubeconfig

View cluster events

Download Installation Logs

Web Console URL
<https://console-openshift-console.apps.ocp.redhat.local>
Not able to access the Web Console?

Username
kubeadmin

Password
.....

Download and save your kubeconfig file in a safe place. This file will be automatically deleted from Assisted Installer's service in 20 days.

Add new hosts by generating a new Discovery ISO under your cluster's "Add hosts" tab on console.redhat.com/openshift

Host inventory (3)

Hostname	Role	Status	Discovered on	CPU Cores	Memory	Total st...
c240m4-01.redhat.local	Control plane node, Worker	Installed	9/25/2025, 4:46:35 PM	56	512.00 GiB	2.92 TB
c240m4-02.redhat.local	Control plane node, Worker (bootstrap)	Installed	9/25/2025, 4:44:40 PM	56	512.00 GiB	2.92 TB
c240m4-03.redhat.local	Control plane node, Worker	Installed	9/25/2025, 4:49:23 PM	72	512.00 GiB	2.92 TB

Cluster summary

Cluster support level: Full
Your installed cluster will be fully supported

Cluster details

Cluster address

ocp.redhat.local

OpenShift version

4.19.12

CPU architecture

x86_64

Hosts' network configuration

Static IP

Host inventory

Hosts

3

Total cores

184

Total memory

1.50 TiB

Total storage

8.76 TB

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▼ Networking

Networking management type	Cluster-managed networking
Stack type	IPv4
Machine networks CIDR	192.168.2.0/24
API IP	192.168.2.183
Ingress IP	192.168.2.184

Advanced networking settings

Cluster network CIDR	10.128.0.0/14
Cluster network host prefix	23
Service network CIDR	172.30.0.0/16
Networking type	Open Virtual Network (OVN)

STEP 16. Login to the OpenShift console, click on the Web Console URL:

<https://console-openshift-console.apps.ocp.redhat.local/>

When prompted for the Username and Password, enter the kubeadmin user and the password recorded previously from the installation progress screen.

Log in to your account

Username *

kubeadmin

Password *

.....

Log in



Welcome to Red Hat OpenShift

OpenShift Credentials

Web Console URL: <https://console-openshift-console.apps.ocp.redhat.local/>

<https://console.redhat.com/openshift/assisted-installer/clusters/2e9aece0-66f8-42d5-92e0-6f802344fa1d>

Username: [your_openshift_username] (select login with kube:admin)

Password: [your_openshift_password]

Username: admin (select login with ldap)

Password: redhat123

Verify MTU setting on all nodes:

Run the following command to verify that the cluster installation set the MTU as expected:

```
[root@rhel8-bastion ~]# for i in $(oc get node -o jsonpath='{.items[*].metadata.name}'); do oc debug node/${i} -- chroot /host ip -d link |grep -1 maxmtu|grep -1 bond0; done
```

Sample output

```
To use host binaries, run `chroot /host`
Link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a1:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
openvswitch numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65536 tso_max_segs 65535 gro_max_size 65536
gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
10: bond0: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 9000 qdisc noqueue master ovs-system state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a1:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 65535

Removing debug pod ...
Starting pod/c240m4-02redhatlocal-debug-4ds4j ...
To use host binaries, run `chroot /host`
Link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a2:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
openvswitch numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 65536 tso_max_segs 65535 gro_max_size 65536
gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
10: bond0: <BROADCAST,MULTICAST,MASTER,UP,LOWER_UP> mtu 9000 qdisc noqueue master ovs-system state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a2:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 65535

Removing debug pod ...
Starting pod/c240m4-03redhatlocal-debug-hq9tl ...
To use host binaries, run `chroot /host`
Link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 promiscuity 0 allmulti 0 minmtu 0 maxmtu 0 numtxqueues 1 numrxqueues 1 gso_max_size 65536 gso_max_segs 65535 tso_max_size 524280 tso_max_segs 65535 gro_max_size 65536 gso_ipv4_max_size 65536 gro_ipv4_max_size 65536
2: eno1: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
```



```
Link/ether 70:df:2f:f6:a3:2c brd ff:ff:ff:ff:ff:ff promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
--
--
3: eno2: <BROADCAST,MULTICAST,SLAVE,UP,LOWER_UP> mtu 9000 qdisc mq master bond0 state UP mode DEFAULT group default qlen 1000
Link/ether 70:df:2f:f6:a3:2c brd ff:ff:ff:ff:ff:ff permaddr 70:df:2f:f6:a3:2d promiscuity 1 allmulti 0 minmtu 68 maxmtu 9000
```

DRAFT

MACHINECONFIG:

[6.10.3. Automatically allocating resources for nodes](#)

KubeletConfig

OpenShift Container Platform can automatically determine the optimal **system-reserved** CPU and memory resources for nodes associated with a specific machine config pool and update the nodes with those values when the nodes start. By default, the **system-reserved** CPU is **500m** and **system-reserved** memory is **1Gi**.

To automatically determine and allocate the **system-reserved** resources on nodes, create a **KubeletConfig** custom resource (CR) to set the **autoSizingReserved: true** parameter. A script on each node calculates the optimal values for the respective reserved resources based on the installed CPU and memory capacity on each node. The script takes into account that increased capacity requires a corresponding increase in the reserved resources.

Automatically determining the optimal **system-reserved** settings ensures that your cluster is running efficiently and prevents node failure due to resource starvation of system components, such as CRI-O and kubelet, without needing to manually calculate and update the values.

This feature is disabled by default.

Procedure

STEP 1. Create a custom resource (CR) for your configuration change:

Sample configuration for a resource allocation CR

```
[root@bastion]# cat << EOF | oc apply -f -
```

```
apiVersion: machineconfiguration.openshift.io/v1
kind: KubeletConfig
metadata:
  name: dynamic-node-master
spec:
  machineConfigPoolSelector:
    matchLabels:
      pools.operator.machineconfiguration.openshift.io/master: ""
  autoSizingReserved: true
```

EOF

```
[root@bastion ]# cat << EOF | oc apply -f -
```

```
apiVersion: machineconfiguration.openshift.io/v1
kind: KubeletConfig
metadata:
  name: dynamic-node-worker
spec:
  machineConfigPoolSelector:
    matchLabels:
      pools.operator.machineconfiguration.openshift.io/worker: ""
  autoSizingReserved: true
```

EOF

The previous example enables automatic resource allocation on all worker nodes. OpenShift Container Platform drains the nodes, applies the kubelet config, and restarts the nodes.

WARNING: This will trigger a serial restart of nodes using this MachineConfigPool. Refer also to the documentation for [Modifying nodes](#) to see how to modify the reservations with those values

Verification

STEP 1. View the /etc/node-sizing.env file on a node(s) you configured by entering the following command:

```
[root@bastion ]# for i in $(oc get node -o jsonpath='{
.items[*].metadata.name }'); do oc debug node/${i} -- chroot /host cat
/etc/node-sizing.env; done
```

Temporary namespace openshift-debug-spjgv is created for debugging node...

Starting pod/c240m4-01redhatlocal-debug-x56qq ...

To use host binaries, run `chroot /host`

SYSTEM_RESERVED_MEMORY=17Gi

SYSTEM_RESERVED_CPU=0.91

SYSTEM_RESERVED_ES=1Gi

Removing debug pod ...

Temporary namespace openshift-debug-spjgv was removed.

Temporary namespace openshift-debug-8wxrm is created for debugging node...

Starting pod/c240m4-02redhatlocal-debug-6k858 ...

To use host binaries, run `chroot /host`

SYSTEM_RESERVED_MEMORY=17Gi

SYSTEM_RESERVED_CPU=0.91

SYSTEM_RESERVED_ES=1Gi

Removing debug pod ...

Temporary namespace openshift-debug-8wxrm was removed.

Temporary namespace openshift-debug-m7tpk is created for debugging node...

Starting pod/c240m4-03redhatlocal-debug-s65tx ...

To use host binaries, run `chroot /host`

SYSTEM_RESERVED_MEMORY=17Gi

SYSTEM_RESERVED_CPU=0.91

SYSTEM_RESERVED_ES=1Gi

Removing debug pod ...
Temporary namespace openshift-debug-m7tpk was removed.

Or run it on a single node:

```
[root@bastion]# oc debug node/c240m4-03.redhat.local -- chroot /host cat /etc/node-sizing.env
```

INSTALL/CONFIGURE OPERATORS

Install Local Storage Operator

[2.1. Install Local Storage Operator](#)

NOTE: If re-installing, ensure that the SSDs are empty, and wiped before configuring ODF.

Access each node using: `ssh core@c240m4-0x`, or `oc debug node/c240m4-0x`, then run the following commands:

```
[root@bastion]# ssh core@c240m4-01  
[core@c240m4-01]# sudo sgdisk -Z /dev/sdX  
[core@c240m4-01]# sudo wipefs -a /dev/sdX
```

Use the `sudo lsblk` command to identify the disks' device names, and use them in place of `/dev/sdX` above.

STEP 1. In OperatorHub, search on local storage, select the **Local Storage** Operator.

Project: All Projects ▾

OperatorHub



Discover Operators from the Kubernetes community and Red Hat partners, curated by Red Hat. You can purchase commercial software through [Red Hat Marketplace](#). You can install Operators on your clusters to provide optional add-ons and shared services to your developers. After installation, the Operator capabilities will appear in the [Software Catalog](#), providing a self-service experience.

All Items

AI/Machine Learning
Application Runtime
Big Data
Cloud Provider
Database
Developer Tools
Development Tools
Drivers and plugins
Integration & Delivery
Logging & Tracing
Modernization & Migration
Monitoring
Networking
Observability
OpenShift Optional
OpenShift Optional
Security
Storage
Streaming & Messaging
Other

All Items

Q local storage x

2 items



Red Hat

Local Storage
provided by Red Hat

Configure and use local storage
volumes.



Red Hat

LVM Storage
provided by Red Hat

Logical volume manager storage
provides dynamically provisioned
local storage for container...

STEP 2. Click on *Install*.

Project: All Projects

OperatorHub

Discover Operators from [Red Hat Marketplace](#).
Operator capabilities will

Local Storage
4.19.0-202509091807 provided by Red Hat

Install

Channel
stable

Version
4.19.0-202509...

Capability level
☒ Basic Install
☒ Seamless Upgrades
☒ Full Lifecycle
☐ Deep Insights
☐ Auto Pilot

Source
Red Hat

Provider
Red Hat

Infrastructure features
Disconnected
Designed for FIPS
Proxy-aware

Valid Subscriptions
OpenShift Kubernetes Engine
OpenShift Virtualization Engine

Operator that configures local storage volumes for use in Kubernetes and OpenShift. OpenShift 4.2 and above are the only supported OpenShift versions.

STEP 3. Click **Install**.

[OperatorHub](#) > Operator Installation

Install Operator ★

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel * ⓘ
stable

Version *
4.19.0-202509091807

Installation mode *
☐ All namespaces on the cluster (default)
This mode is not supported by this Operator
☒ A specific namespace on the cluster
Operator will be available in a single Namespace only.
Installed Namespace *
☒ Operator recommended Namespace: **PR** openshift-local-storage
☐ Select a Namespace

Namespace creation
Namespace **openshift-local-storage** does not exist and will be created.

☒ Enable Operator recommended cluster monitoring on this Namespace

Update approval * ⓘ
☒ Automatic
☐ Manual

Install

Cancel

 **Local Storage**
provided by Red Hat

Provided APIs

LV Local Volume
Manage local storage volumes for OpenShift

LVS Local Volume Set
A Local Volume set allows you to filter a set of storage volumes, group them and create a dedicated storage class to consume storage from the set of volumes.

LVD Local Volume Discovery
Discover list of potentially usable disks on the chosen set of nodes

STEP 4. Ensure that the operator installs successfully. Then continue with the next step by installing ODF. The installation process for ODF will configure the Local Storage operator.

 **Local Storage**
local-storage-operator.v4.19.0-202509091807 provided by Red Hat ✓

Operator installed successfully
The Operator has installed successfully. Ready for use.

View Operator

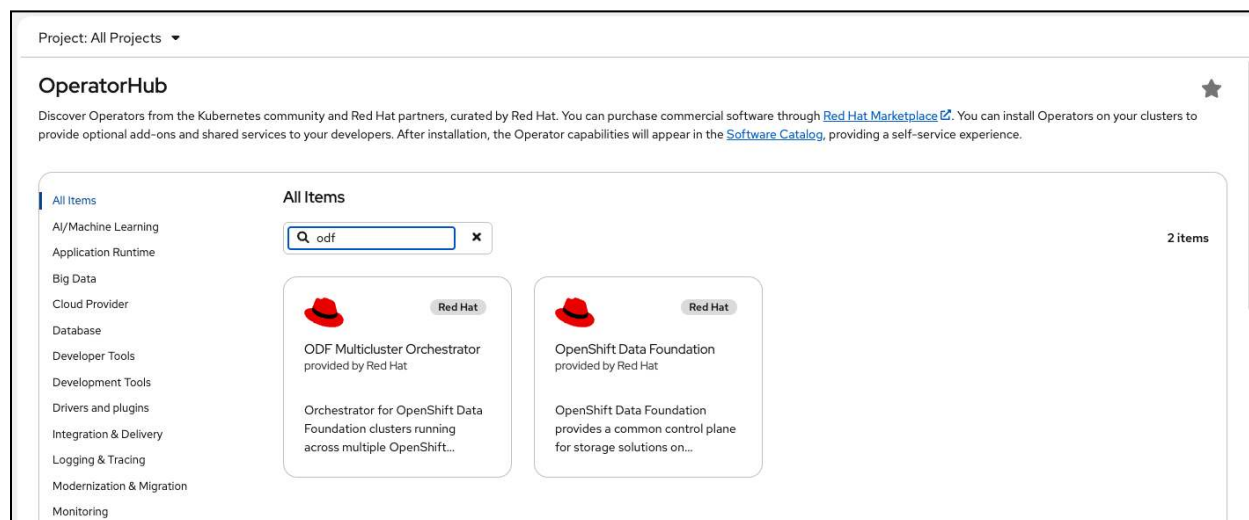
[View installed Operators in Namespace openshift-local-storage](#)

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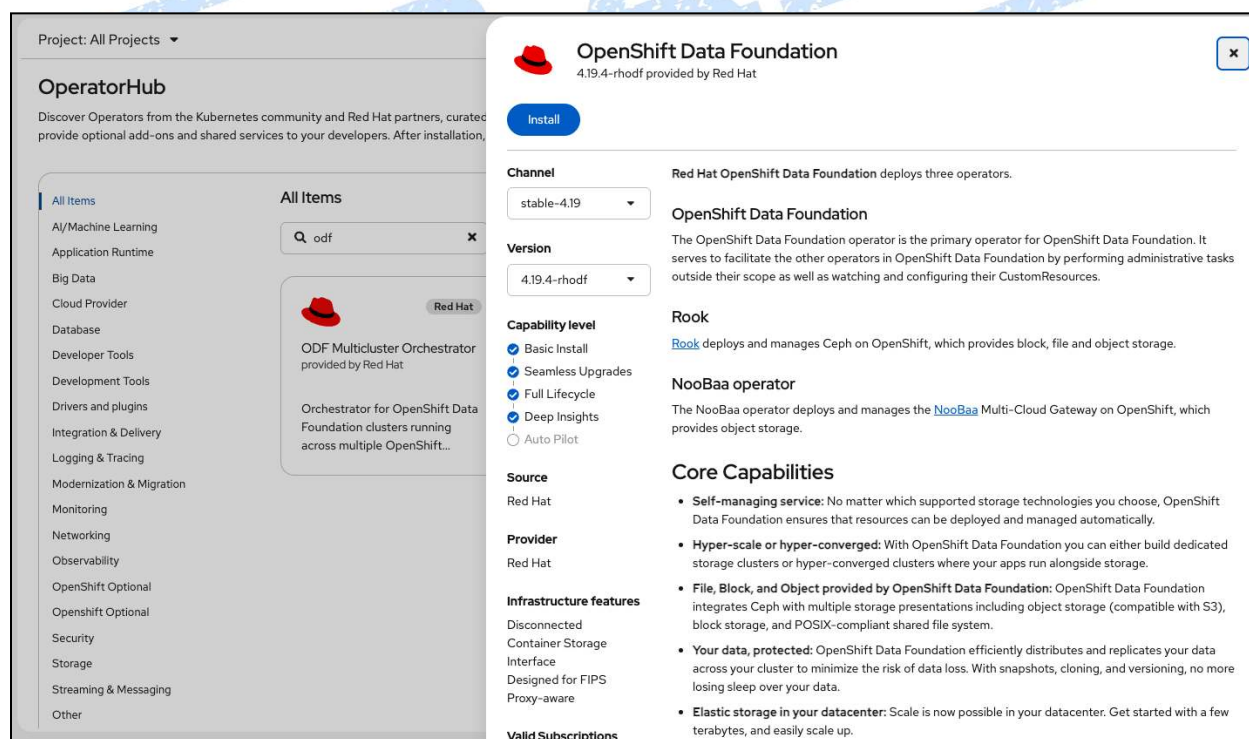
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Install OpenShift Data Foundation (Internal mode)

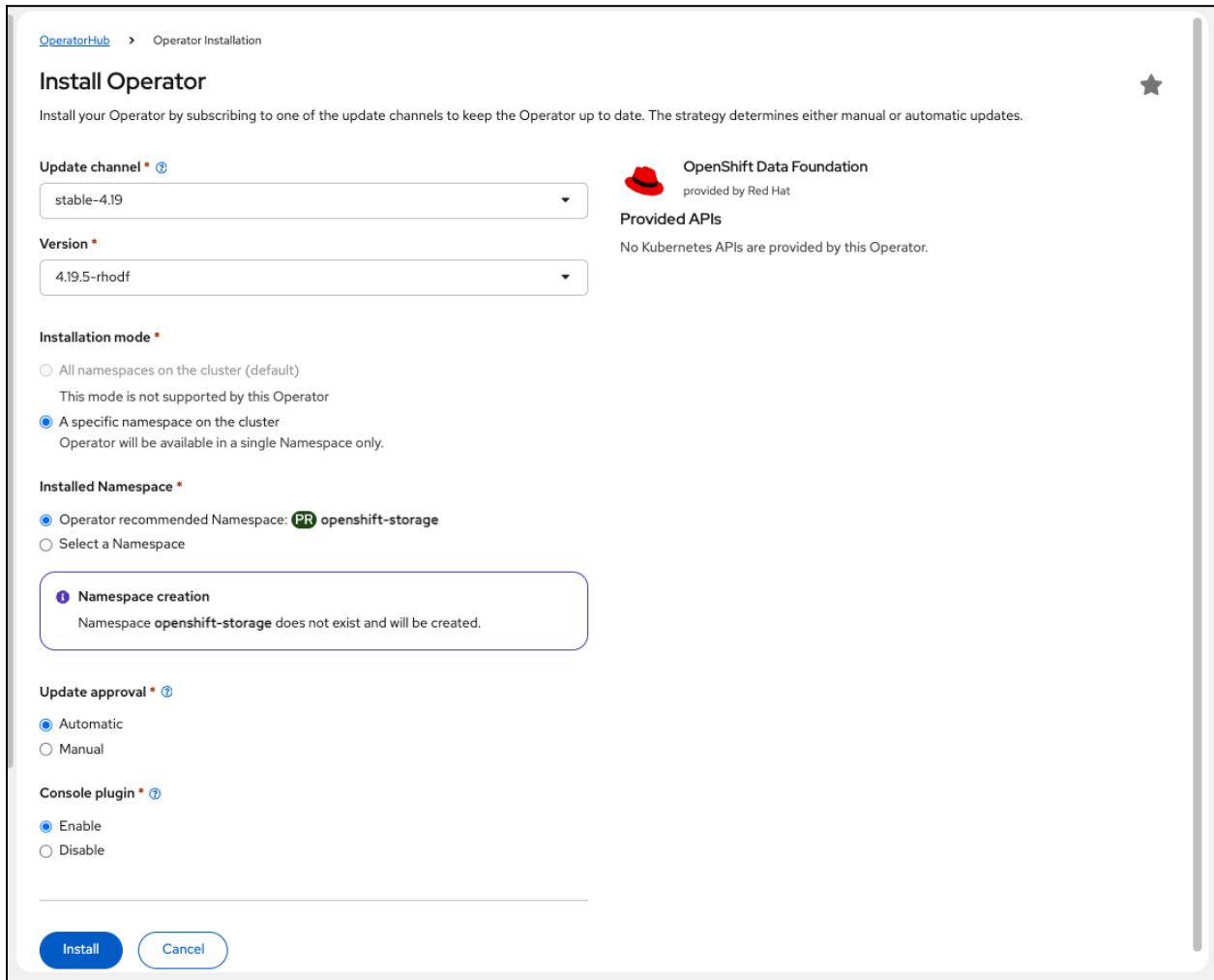
STEP 1. In OperatorHub, search on odf, select “*OpenShift Data Foundation*”.



STEP 2. Click on “*Install*”.



STEP 3. Click on “**Install**”.



OperatorHub > Operator Installation

Install Operator

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel * ⓘ

stable-4.19

Version *

4.19.5-rhodef

Installation mode *

☐ All namespaces on the cluster (default)
This mode is not supported by this Operator

☒ A specific namespace on the cluster
Operator will be available in a single Namespace only.

Installed Namespace *

☒ Operator recommended Namespace: **PR** openshift-storage

☐ Select a Namespace

Namespace creation
Namespace openshift-storage does not exist and will be created.

Update approval * ⓘ

☒ Automatic

☐ Manual

Console plugin * ⓘ

☒ Enable

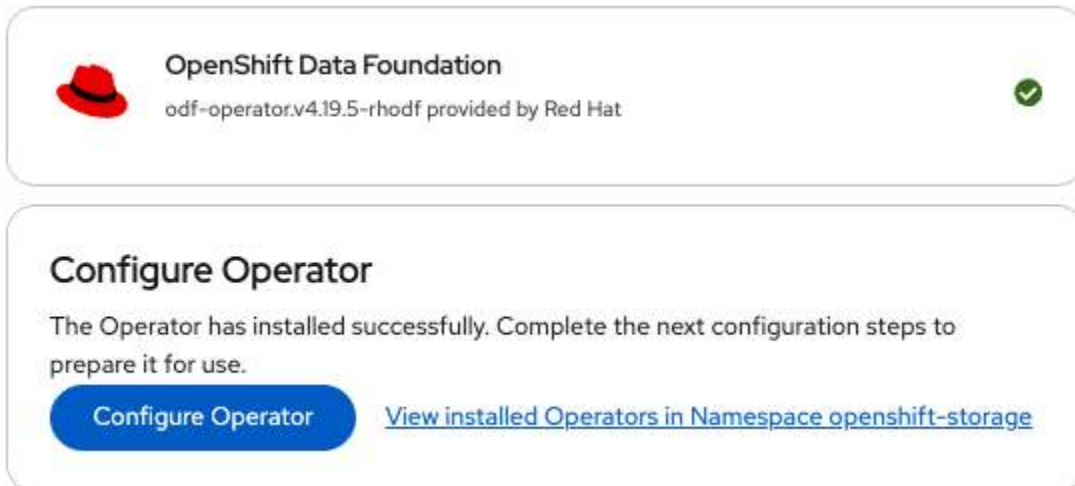
☐ Disable

Install Cancel

OpenShift Data Foundation
provided by Red Hat

Provided APIs
No Kubernetes APIs are provided by this Operator.

STEP 4. Ensure that the operator installs successfully. Then, click on “**Configure Operator**”.



OpenShift Data Foundation
odf-operator.v4.19.5-rhodef provided by Red Hat

Configure Operator

The Operator has installed successfully. Complete the next configuration steps to prepare it for use.

Configure Operator [View installed Operators in Namespace openshift-storage](#)

STEP 4. Select “*Create a new StorageClass using local storage devices*”, “*Use Ceph RBD as the default StorageClass*”, and “*Set default StorageClass for virtualization*” then select “*Next*”.

Create StorageSystem

Create a StorageSystem to represent your Data Foundation system and all its required storage and computing resources.

Namespace: openshift-storage

1 Backing storage

2 Create local volume set

3 Capacity and nodes

4 Security and network

5 Review and create

Deployment type

Full deployment

Backing storage type

☐ Use an existing StorageClass

Data Foundation will use an existing StorageClass available on your hosting platform.

☒ Create a new StorageClass using local storage devices

Data Foundation will use a StorageClass provided by the Local Storage Operator (LSO) on top of your attached drives. This option is available on any platform with devices attached to nodes.

☐ Connect an external storage platform

Data Foundation will create a dedicated StorageClass.

☐ Enable network file system (NFS)

Allow NFS to use low resources by default.

☒ Use Ceph RBD as the default StorageClass

Configure default RBD StorageClass to avoid adding manual annotations within a StorageClass and selecting a specific StorageClass when making storage requests or provisions in your PVCs.

☒ Set default StorageClass for virtualization

If enabled, RBD virtualization StorageClass will be marked as the default for KubeVirt VM disks (persistent volumes) upon installation.

☐ Use external PostgreSQL

Allow Noobaa to connect to an external postgres server

Next **Back** **Cancel**

STEP 5. Enter a name for the LocalVolumeSet and StorageClass, and select the appropriate settings for the storage across your cluster.

In this example, a 3-node OpenShift cluster is being used, with each node having 3 SSDs installed, for a total of 9 Disks.

Then click on “**Next**”.

Create StorageSystem

Create a StorageSystem to represent your Data Foundation system and all its required storage and computing resources.

Namespace: openshift-storage

- 1 Backing storage
- 2 Create local volume set
- 3 Capacity and nodes
- 4 Security and network
- 5 Review and create

LocalVolumeSet name *

odf-localvolumeset

A LocalVolumeSet will be created to allow you to filter a set of disks, group them and create a dedicated StorageClass to consume storage from them.

StorageClass name

odf-localvolumeset

Filter disks by

☒ Disks on all nodes (3 node)

Uses the available disks that match the selected filters on all nodes.

☐ Disks on selected nodes

Uses the available disks that match the selected filters only on selected nodes.

Disk type

SSD / NVMe

i Disk type is set to SSD/NVMe

Data Foundation supports only SSD/NVMe disk type for internal mode deployment.

> Advanced

Next

Back

Cancel

Selected capacity

3 Node | 9 Disk

6.55 TiB

Out of 6.55 TiB

STEP 6. Select “Yes” to continue.

Create LocalVolumeSet

x

After the LocalVolumeSet is created you won't be able to edit it.

Note: If you wish to use the Arbiter stretch cluster, a minimum of 4 nodes (2 different zones, 2 nodes per zone) and 1 additional zone with 1 node is required. All nodes must be pre-labeled with zones in order to be validated on cluster creation.

Are you sure you want to continue?

Yes

Cancel

STEP 7. On the “Capacity and nodes” screen you will see confirmation of the selected capacity, and nodes. Select the amount of resources to assign to ODF; either **Lean**, **Balanced** or **Performance** mode. For most cases, you would select the default “**Balanced**” mode setting. Select “**Next**”.

Create StorageSystem

Create a StorageSystem to represent your Data Foundation system and all its required storage and computing resources.

Namespace: openshift-storage

1 Backing storage

2 Create local volume set

3 Capacity and nodes

4 Security and network

5 Review and create

Selected capacity

Available raw capacity

6.54 TiB

The available capacity is based on all attached disks associated with the selected StorageClass odf-localvolumeset

Selected nodes

Selected nodes are based on the StorageClass odf-localvolumeset and with a recommended requirement of 14 CPU and 34 GiB RAM per node.

If not labeled, the selected nodes are labeled cluster.ocs.openshift.io/openshift-storage="" to make them target hosts for Data Foundation's components.

Name	Role	CPU	Memory	Zone
c240m4-01.redhat.local	control-plane, master, worker	56	503.8 GiB	-
c240m4-02.redhat.local	control-plane, master, worker	56	503.6 GiB	-
c240m4-03.redhat.local	control-plane, master, worker	72	503.8 GiB	-

3 node selected (184 CPUs and 1.48 TiB on 0 zone)

Configure performance

Select a profile to customise the performance of the Data Foundation cluster to meet your requirements.

Balanced mode

Lean mode

Balanced mode

Performance mode

CPUs required: 29, Memory required: 81 GiB

CPUs required: 36, Memory required: 87 GiB

CPUs required: 57, Memory required: 120 GiB

When the nodes in the selected StorageClass are spread across fewer than 3 availability zones, the StorageCluster will be deployed with the host based failure domain.

Next

Back

Cancel

STEP 8. Optionally enable encryption or to specify advanced networking settings, then select **“Next”**.

Create StorageSystem

Create a StorageSystem to represent your Data Foundation system and all its required storage and computing resources.

Namespace: openshift-storage

1 Backing storage

2 Create local volume set

3 Capacity and nodes

4 Security and network

5 Review and create

Encryption

☐ Enable data encryption for block and file storage

Data encryption for block and file storage. MultiCloud Object Gateway supports encryption for objects by default.

☐ In-transit encryption

Encrypts all Ceph traffic including data, using Ceph msgrv2.

Network

☒ Default (Pod) ⓘ

☐ Isolate network using Multus

Multus allows a network separation between the data operations and the control plane operations.

Next

Back

Cancel

STEP 9. Click on “**Create StorageSystem**”.

Create StorageSystem

Create a StorageSystem to represent your Data Foundation system and all its required storage and computing resources.

Namespace: openshift-storage

1 Backing storage

2 Create local volume set

3 Capacity and nodes

4 Security and network

5 Review and create

Backing storage

Deployment type: Full deployment

Network file system: Disabled

Ceph RBD as the default StorageClass: Enabled

Default StorageClass for virtualization : Enabled

Backing storage type: odf-localvolumeset

Capacity and nodes

Cluster capacity: 6.54 TiB

Selected nodes: 3 node

CPU and memory: 184 CPU and 1.48 TiB memory

Performance profile: Balanced

Automatic capacity scaling: Disabled

Zone: 0 zone

Taint nodes: Disabled

Security and network

Encryption: Disabled

In-transit encryption: Disabled

Network: Default (OVN)

Create StorageSystem

Back

Cancel

STEP 10. When complete, you will see a **Ready** status.

Data Foundation

Overview

Topology

Storage Systems

StorageSystems

★ Create StorageSystem

Name

Search by name...

Name	Status	Raw Capacity	Used capacity	IOPS	Throughput	Latency
 ocs-storagecluster	Phase:  Ready	6.55 TiB	33.01 GiB	10.73 IOPS	28.4 KBps	1.09 ms

Create a Replica 2 Block Pool and StorageClass for ODF

Perform the following from the OpenShift console to create the Replica 2 Block Pool:

STEP 1. Click **Storage > Data Foundation**.

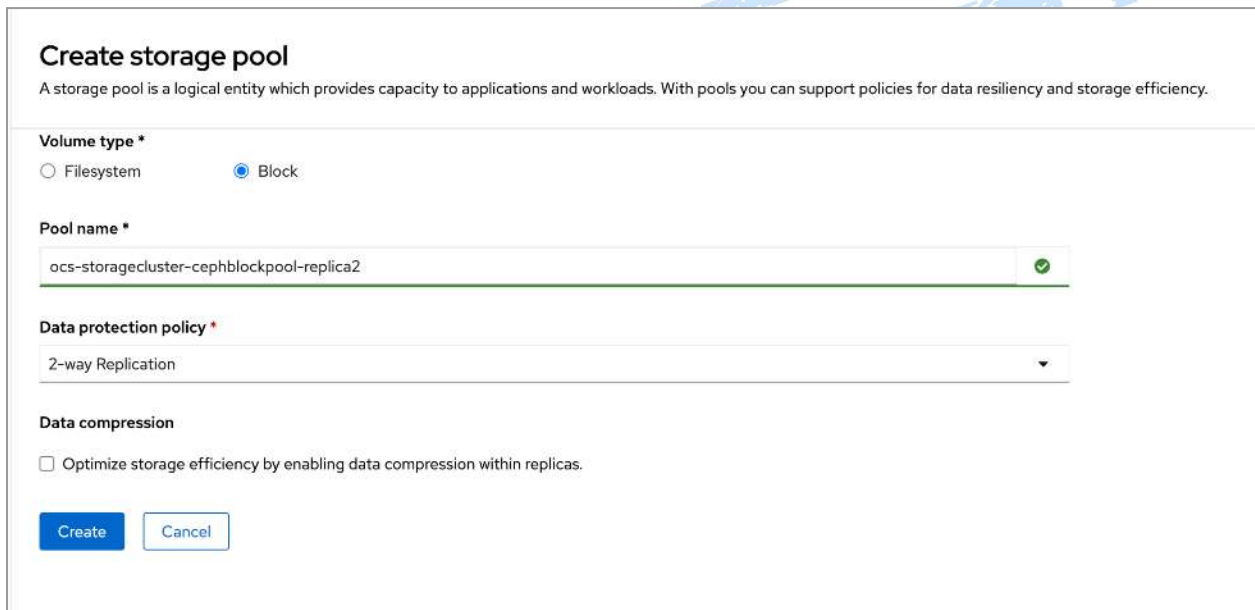
STEP 2. In the **Storage systems** tab, select the storage system and then click the **Storage pools** tab.

STEP 3. Click **Create storage pool**.

STEP 4. Select **Volume type** as **Block**.

STEP 5. Enter **Pool name**, such as "**ocs-storagecluster-cephblockpool-replica2**".

STEP 6. Select Data protection policy as **2-way Replication**.



Create storage pool

A storage pool is a logical entity which provides capacity to applications and workloads. With pools you can support policies for data resiliency and storage efficiency.

Volume type *

☐ Filesystem ☒ Block

Pool name *

ocs-storagecluster-cephblockpool-replica2

Data protection policy *

2-way Replication

Data compression

☐ Optimize storage efficiency by enabling data compression within replicas.

Create **Cancel**

STEP 7. Click **Create**.

After the Pool is created, perform the following from the OpenShift console to create the Replica 2 StorageClass:

STEP 1. Navigate to **Storage > StorageClasses**.

STEP 2. Click **Create Storage Class**.

STEP 3. Enter a **name**, such as "**ocs-storagecluster-ceph-replica2-rbd**"; then select the **openshift-storage.rbd,csi.ceph.com** provisioner, and select the Storage Pool entered in the previous step, such as "**ocs-storagecluster-cephblockpool-replica2**".

StorageClass

[Edit YAML](#)**Name *****Description****Reclaim policy ***

Determines what happens to persistent volumes when the associated persistent volume claim is deleted. Defaults to "Delete"

Volume binding mode *

Determines when persistent volume claims will be provisioned and bound. Defaults to "WaitForFirstConsumer"

Provisioner *

Determines what volume plugin is used for provisioning PersistentVolumes.

Storage system *

Select a StorageSystem for your workloads.

Storage Pool *

Storage pool into which volume data shall be stored

☐ Enable Encryption

An encryption key will be generated for each PersistentVolume created using this StorageClass.

Additional parameters

Specific fields for the selected provisioner.

Parameter

Value

[+ Add Parameter](#)☒ Allow PersistentVolumeClaims to be expanded

STEP 4. Click **Create**.

Configuring the Image Registry Operator to use CephFS storage with Red Hat OpenShift Data Foundation

Procedure

STEP 1. Create a PVC to use the cephfs storage class. For example:

```
[root@bastion ]# cat <<EOF | oc apply -f -
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: registry-storage-pvc
  namespace: openshift-image-registry
spec:
  accessModes:
  - ReadWriteMany
  resources:
    requests:
      storage: 100Gi
  storageClassName: ocs-storagecluster-cephfs
EOF
```

STEP 2. Configure the image registry to use the CephFS file system storage by entering the following command:

```
[root@bastion ]# oc patch config.image/cluster -p
'{"spec":{"managementState":"Managed","replicas":2,"storage":{"managementState":"Unmanaged","pvc":{"claim":"registry-storage-pvc"}}}}' --type=merge
```

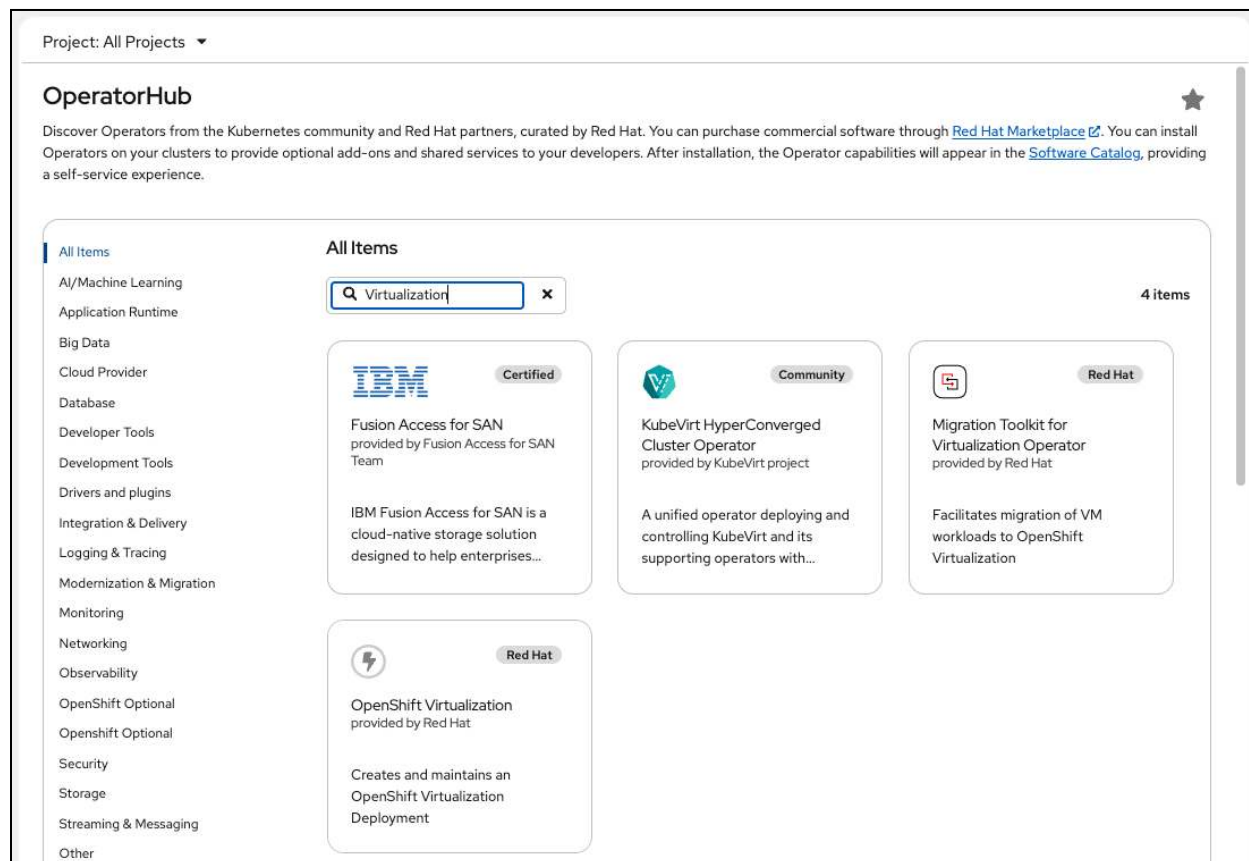
Example

```
apiVersion: imageregistry.operator.openshift.io/v1
kind: Config
metadata:
  name: cluster
spec:
  logLevel: Normal
  managementState: Managed
  replicas: 2
  storage:
    managementState: Unmanaged
  pvc:
    claim: registry-storage-pvc
```

Also for reference: [Configuring Image Registry to use OpenShift Data Foundation](#)

Install OpenShift Virtualization Operator

STEP 1. Search on *Virtualization*, select *OpenShift Virtualization*. Click on “*Install*”.



STEP 2. Click on “*Install*”.

The screenshot shows the OpenShift Virtualization page in the OperatorHub. The page is titled "OpenShift Virtualization" and "4.19.6 provided by Red Hat". There is a blue "Install" button at the top left. Below it, there are two dropdown menus: "Channel" set to "stable" and "Version" set to "4.19.6". To the right of these dropdowns is a "Requirements" section stating: "Your cluster must be installed on bare metal infrastructure with Red Hat Enterprise Linux CoreOS workers." Below the requirements is a "Details" section with text about OpenShift Virtualization extending Red Hat OpenShift Container Platform and managing virtualized workloads. On the left side of the page, there is a sidebar with "OperatorHub" and a list of categories like "All Items", "AI/Machine Learning", "Application Runtime", etc. Below the categories is a "Source" section with checkboxes for "Red Hat (2)", "Certified (1)", and "Community (1)".

Project: All Projects ▾

OperatorHub

Discover Operators from the Kubernetes community and install Operators on your clusters to provide optional add-on functionality and a self-service experience.

All Items

- AI/Machine Learning
- Application Runtime
- Big Data
- Cloud Provider
- Database
- Developer Tools
- Development Tools
- Drivers and plugins
- Integration & Delivery
- Logging & Tracing
- Modernization & Migration
- Monitoring
- Networking
- Observability
- OpenShift Optional
- OpenShift Optional
- Security
- Storage
- Streaming & Messaging
- Other

Source

- ☐ Red Hat (2)
- ☐ Certified (1)
- ☐ Community (1)

OpenShift Virtualization

4.19.6 provided by Red Hat

Install

Channel

stable ▾

Version

4.19.6 ▾

Capability level

- ☒ Basic Install
- ☒ Seamless Upgrades
- ☒ Full Lifecycle
- ☒ Deep Insights
- ☐ Auto Pilot

Source

Red Hat

Provider

Red Hat

Infrastructure features

- Disconnected
- Proxy-aware
- Designed for FIPS
- Single Node Clusters
- Container Storage
- Interface
- Container Network
- Interface
- Configurable TLS ciphers

Valid Subscriptions

OpenShift Kubernetes

Requirements

Your cluster must be installed on bare metal infrastructure with Red Hat Enterprise Linux CoreOS workers.

Details

OpenShift Virtualization extends Red Hat OpenShift Container Platform, allowing you to host and manage virtualized workloads on the same platform as container-based workloads. From the OpenShift Container Platform web console, you can import a VMware virtual machine from vSphere, create new or clone existing VMs, perform live migrations between nodes, and more. You can use OpenShift Virtualization to manage both Linux and Windows VMs.

The technology behind OpenShift Virtualization is developed in the [KubeVirt](#) open source community. The KubeVirt project extends [Kubernetes](#) by adding additional virtualization resource types through [Custom Resource Definitions](#) (CRDs). Administrators can use Custom Resource Definitions to manage [VirtualMachine](#) resources alongside all other resources that Kubernetes provides.

STEP 2. Select “*Install*”.

Install Operator

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel *

stable

Version *

4.19.6

Installation mode *

- ☐ All namespaces on the cluster (default)
This mode is not supported by this Operator
- ☒ A specific namespace on the cluster
Operator will be available in a single Namespace only.

Installed Namespace *

- ☒ Operator recommended Namespace: **PR** openshift-cnv
- ☐ Select a Namespace

1 Namespace creation

Namespace **openshift-cnv** does not exist and will be created.

Update approval *

- ☒ Automatic
- ☐ Manual

Install

Cancel



OpenShift Virtualization
provided by Red Hat

Provided APIs



OpenShift
Virtualization
Deployment

Required

Represents the deployment of
OpenShift Virtualization

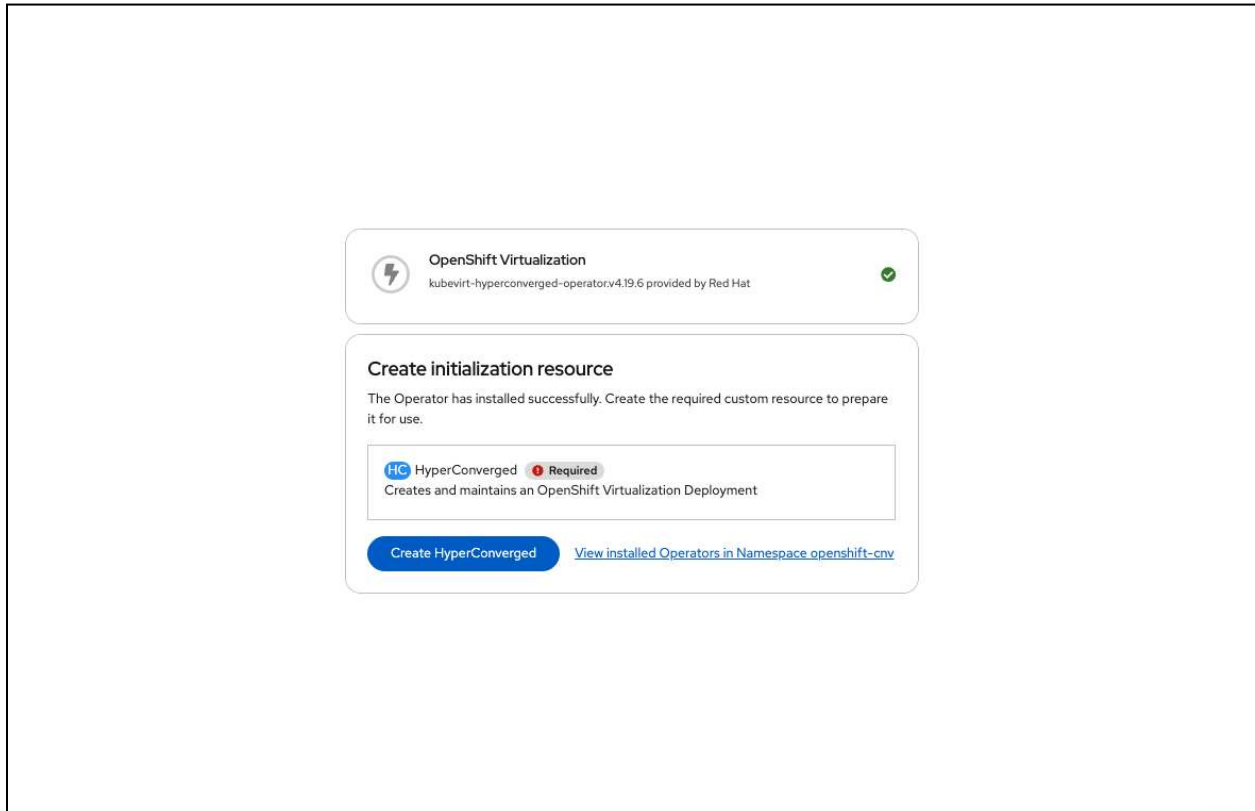


HostPathProvisioner
Deployment

Represents the deployment of
HostPathProvisioner

DRAFT

STEP 3. Once the operator installation is completed, select “**Create HyperConverged**” to create a deployment of OpenShift Virtualization.



STEP 4. Create a deployment using the default settings; scroll down to the bottom of the page, and click on “**Create**”.

Project: openshift-cnv ▾

Create HyperConverged

Create by completing the form. Default values may be provided by the Operator authors.

Configure via: ☒ Form view ☐ YAML view

Note: Some fields may not be represented in this form view. Please select "YAML view" for full control.



OpenShift Virtualization Deployment

provided by Red Hat

Represents the deployment of OpenShift Virtualization

Name *

kubevirt-hyperconverged

Labels

app=frontend

infra

infra HyperConvergedConfig influences the pod configuration (currently only placement) for all the infra components needed on the virtualization enabled cluster but not necessarily directly on each node running VMs/VMIs.

workloads

workloads HyperConvergedConfig influences the pod configuration (currently only placement) of components which need to be running on a node where virtualization workloads should be able to run. Changes to Workloads HyperConvergedConfig can be applied only without existing workload.

storageImport

StorageImport contains configuration for importing containerized data

enableCommonBootImageImport

☒ enableCommonBootImageImport

Opt-in to automatic delivery/updates of the common data import cron templates.

There are two sources for the data import cron templates: hard coded list of common templates, and custom (user defined) templates that can be added to the dataImportCronTemplates field. This field only controls the common templates. It is possible to use custom templates by adding them to the dataImportCronTemplates field.

defaultRuntimeClass

DefaultRuntimeClass defines a cluster default for the RuntimeClass to be used for VMIs pods if not set there. Default RuntimeClass can be changed when kubevirt is running, existing VMIs are not impacted till the next restart/live-migration when they are eventually going to consume the new default RuntimeClass.

virtualMachineOptions

VirtualMachineOptions holds the cluster level information regarding the virtual machine.

higherWorkloadDensity

HigherWorkloadDensity holds configurataion aimed to increase virtual machine density

defaultCPUModel

DefaultCPUModel defines a cluster default for CPU model: default CPU model is set when VMI doesn't have any CPU model.

When VMI has CPU model set, then VMI's CPU model is preferred.

When default CPU model is not set and VMI's CPU model is not set too, host-model will be set.

Default CPU model can be changed when kubevirt is running.

tektonPipelinesNamespace

TektonPipelinesNamespace defines namespace in which example pipelines will be deployed.

If unset, then the default value is the operator namespace.

Deprecated: This field is ignored.

obsoleteCPUs

ObsoleteCPUs allows avoiding scheduling of VMs for obsolete CPU models

liveMigrationConfig

Live migration limits and timeouts are applied so that migration processes do not overwhelm the cluster.

tektonTasksNamespace

TektonTasksNamespace defines namespace in which tekton tasks will be deployed. If unset, then the default value is the operator namespace.
 Deprecated: This field is ignored.

certConfig

certConfig holds the rotation policy for internal, self-signed certificates

tuningPolicy

Select tuningPolicy

TuningPolicy allows to configure the mode in which the RateLimits of kubevirt are set. If TuningPolicy is not present the default kubevirt values are used.
 It can be set to 'annotation' for fine-tuning the kubevirt queryPerSeconds (qps) and burst values. Qps and burst values are taken from the annotation hco.kubevirt.io/tuningPolicy

enableApplicationAwareQuota

☐ enableApplicationAwareQuota

EnableApplicationAwareQuota if true, enables the Application Aware Quota feature

mediatedDevicesConfiguration

MediatedDevicesConfiguration holds information about MDEV types to be defined on nodes, if available

scratchSpaceStorageClass

Override the storage class used for scratch space during transfer operations. The scratch space storage class is determined in the following order:
 value of scratchSpaceStorageClass, if that doesn't exist, use the default storage class, if there is no default storage class, use the storage class of the DataVolume, if no storage class specified, use no storage class for scratch space

logVerbosityConfig

LogVerbosityConfig configures the verbosity level of Kubevirt's different components. The higher the value - the higher the log verbosity.

commonTemplatesNamespace

CommonTemplatesNamespace defines namespace in which common templates will be deployed. It overrides the default openshift namespace.

tlsSecurityProfile

TLSecurityProfile specifies the settings for TLS connections to be propagated to all kubevirt-hyperconverged components.
 If unset, the hyperconverged cluster operator will consume the value set on the APIServer CR on OCP/OKD or Intermediate if on vanilla k8s.
 Note that only Old, Intermediate and Custom profiles are currently supported, and the maximum available MinTLSVersions is VersionTLS12.

applicationAwareConfig

ApplicationAwareConfig set the AAQ configurations

ksmConfiguration

KSMConfiguration holds the information regarding the enabling the KSM in the nodes (if available).

instancetypeConfig

InstancetypeConfig holds the configuration of instance type related functionality within KubeVirt.

evictionStrategy

Select evictionStrategy ▾

EvictionStrategy defines at the cluster level if the VirtualMachineInstance should be migrated instead of shut-off in case of a node drain. If the VirtualMachineInstance specific field is set it overrides the cluster level one.

Allowed values:

- "None" no eviction strategy at cluster level.
- "LiveMigrate" migrate the VM on eviction; a not live migratable VM with no specific strategy will block the drain of the node until manually evicted.
- "LiveMigrateIfPossible" migrate the VM on eviction if live migration is possible, otherwise directly evict.
- "External" block the drain, track eviction and notify an external controller.

Defaults to LiveMigrate with multiple worker nodes, None on single worker clusters.

featureGates

featureGates is a map of feature gate flags. Setting a flag to "true" will enable the feature. Setting "false" or removing the feature gate, disables the feature.

localStorageClassName

Deprecated: LocalStorageClassName the name of the local storage class.

workloadUpdateStrategy

WorkloadUpdateStrategy defines at the cluster level how to handle automated workload updates

deployVmConsoleProxy

☐ deployVmConsoleProxy

deploy VM console proxy resources in SSP operator

uninstallStrategy

BlockUninstallIfWorkloadsExist ▾

UninstallStrategy defines how to proceed on uninstall when workloads (VirtualMachines, DataVolumes) still exist.

BlockUninstallIfWorkloadsExist will prevent the CR from being removed when workloads still exist.

BlockUninstallIfWorkloadsExist is the safest choice to protect your workloads from accidental data loss, so it's strongly advised.

RemoveWorkloads will cause all the workloads to be cascading deleted on uninstallation.

WARNING: please notice that RemoveWorkloads will cause your workloads to be deleted as soon as this CR will be, even accidentally, deleted.

Please correctly consider the implications of this option before setting it.

BlockUninstallIfWorkloadsExist is the default behaviour.

vddkInitImage

VDDK Init Image eventually used to import VMs from external providers

Deprecated: please use the Migration Toolkit for Virtualization

vmStateStorageClass

VMStateStorageClass is the name of the storage class to use for the PVCs created to preserve VM state, like TPM.

kubeSecondaryDNSNameServerIP

KubeSecondaryDNSNameServerIP defines name server IP used by KubeSecondaryDNS

dataImportCronTemplates

DataImportCronTemplates holds list of data import cron templates (golden images)

permittedHostDevices

PermittedHostDevices holds information about devices allowed for passthrough

commonBootImageNamespace

CommonBootImageNamespace override the default namespace of the common boot images, in order to hide them.

If not set, HCO won't set any namespace, letting SSP to use the default. If set, use the namespace to create the DataImportCronTemplates and the common image streams, with this namespace. This field is not set by default.

CommonInstancetypesDeployment

CommonInstancetypesDeployment holds the configuration of common-instancetypes deployment within KubeVirt.

resourceRequirements

ResourceRequirements describes the resource requirements for the operand workloads.

filesystemOverhead

FilesystemOverhead describes the space reserved for overhead when using Filesystem volumes.
A value is between 0 and 1, if not defined it is 0.055 (5.5 percent overhead)

Create

Cancel

Install NMState Operator

STEP 1. In OperatorHub, search on **NMState**, select the **Kubernetes NMState Operator**.

OperatorHub

Discover Operators from the Kubernetes community and Red Hat partners, curated by Red Hat. You can purchase commercial software through [Red Hat Marketplace](#). You can install Operators on your clusters to provide optional add-ons and shared services to your developers. After installation, the Operator capabilities will appear in the [Software Catalog](#), providing a self-service experience.

All Items

AI/Machine Learning

Application Runtime

Big Data

Cloud Provider

Database

Developer Tools

Development Tools

Drivers and plugins

Integration & Delivery

Logging & Tracing

Modernization & Migration

Monitoring

Networking

Observability

OpenShift Optional

OpenShift Optional

Security

Storage

Streaming & Messaging

Other

All Items

nmstate

nmstate

Red Hat

Kubernetes NMState Operator
provided by Red Hat, Inc.

Kubernetes NMState is a declarative means of configuring NetworkManager.

1 items

STEP 2. Click on ***“Install”***.

The screenshot shows the OperatorHub interface. On the left, there's a sidebar with a search bar and a list of categories. The main area displays the details for the 'Kubernetes NMState Operator' (version 4.19.0-202509240557 provided by Red Hat, Inc.). A blue 'Install' button is prominently displayed at the top. Below it, the configuration options are shown:

- Channel:** A dropdown menu set to 'stable'.
- Version:** A dropdown menu set to '4.19.0-202509240557'.
- Capability level:** Radio buttons for 'Basic Install' (selected), 'Seamless Upgrades', 'Full Lifecycle', 'Deep Insights', and 'Auto Pilot'.
- Source:** 'Red Hat'.
- Provider:** 'Red Hat, Inc.'.
- Infrastructure features:** 'Disconnected' and 'Designed for FIPS'.
- Valid Subscriptions:** A list including 'OpenShift Kubernetes Engine', 'OpenShift Virtualization Engine', 'OpenShift Container Platform', and 'OpenShift Platform Plus'.

STEP 3. Select ***“Install”***.

You are logged in as a temporary administrative user. Update the [cluster OAuth configuration](#) to allow others to log in.

OperatorHub > Operator Installation

Install Operator

★

Install your Operator by subscribing to one of the update channels to keep the Operator up to date. The strategy determines either manual or automatic updates.

Update channel *

stable

Version *

4.19.0-202509240557

Installation mode *

☐ All namespaces on the cluster (default)
This mode is not supported by this Operator

☒ A specific namespace on the cluster
Operator will be available in a single Namespace only.

Installed Namespace *

☒ Operator recommended Namespace: **PR** openshift-nmstate

☐ Select a Namespace

Namespace creation

Namespace **openshift-nmstate** does not exist and will be created.

Update approval *

☒ Automatic

☐ Manual

Install

Cancel

 Kubernetes NMState Operator
provided by Red Hat, Inc.

Provided APIs

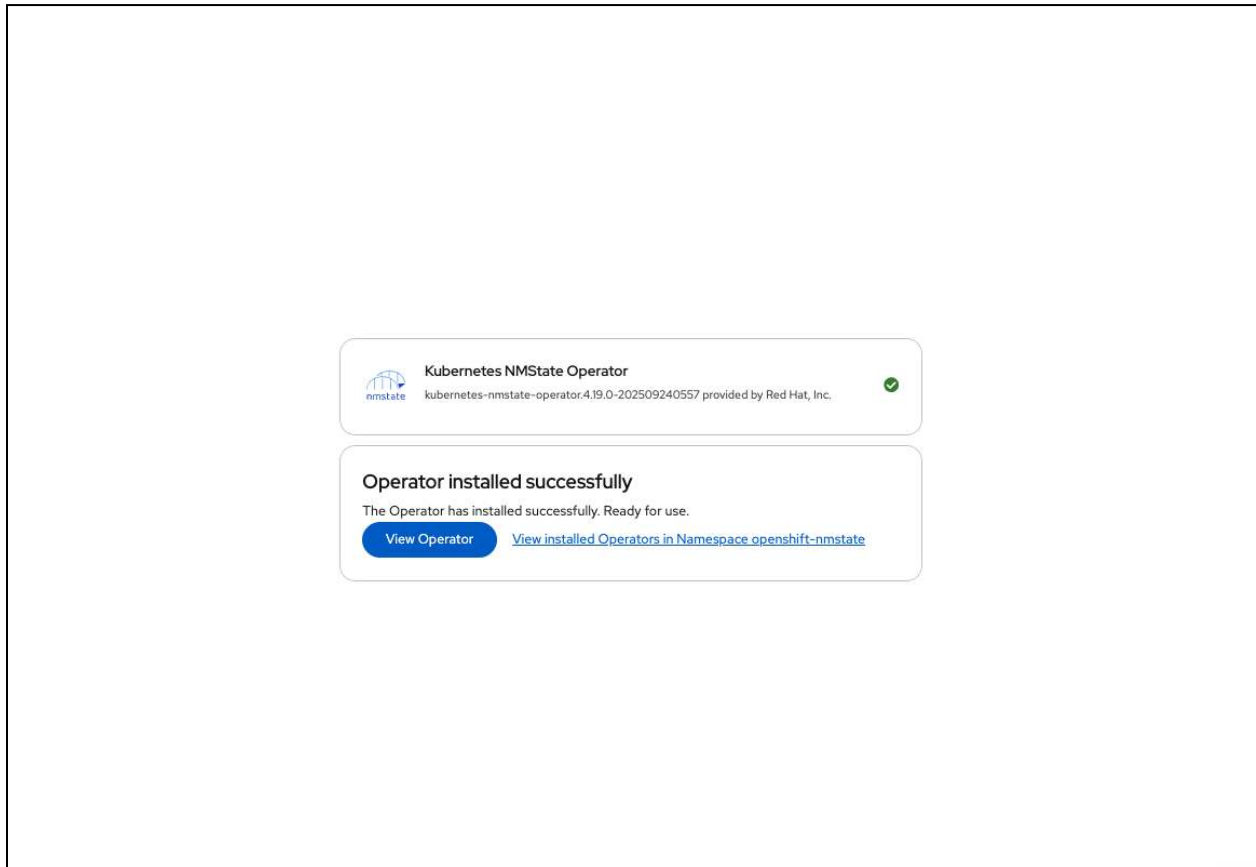
NMS NMState

Represents an NMState deployment.

STEP 4. Once the operator installation is completed, select **“View Operator”**.

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STEP 5. Select ***Create instance***.

Project: openshift-nmstate

Installed Operators

Operator details



Kubernetes NMState Operator

4.19.0-202509240557 provided by Red Hat, Inc.

★

Actions

Details

YAML

Subscription

Events

NMState

Provided APIs

NMS NMState

Represents an NMState deployment.

Create instance

Provider

Red Hat, Inc.

Created at

1 minute ago

Links

Kubernetes Nmstate Operator

<https://github.com/nmstate/kubernetes-nmstate>

Maintainers

Red Hat support@redhat.com

Description

A Kubernetes Operator to install Kubernetes NMState

ClusterServiceVersion details

Name

kubernetes-nmstate-operator.4.19.0-202509240557

Namespace

NS openshift-nmstate

Labels

olm.managed=true

operatorframework.io/arch.amd64=supported

operatorframework.io/arch.arm64=supported

operatorframework.io/arch.ppc64le=supported

Status

Succeeded

Status reason

install strategy completed with no errors

Operator Deployments

[nmstate-operator](#)

Operator ServiceAccounts

STEP 6. Select “Create”.

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Page 50 of 190

Project: openshift-nmstate ▾

Create NMState

Create by completing the form. Default values may be provided by the Operator authors.

Configure via: ☒ Form view ☐ YAML view

Note: Some fields may not be represented in this form view. Please select "YAML view" for full control.


NMState
 provided by Red Hat, Inc.
 Represents an NMState deployment.

Name *

nmstate

Labels

app=frontend

affinity >
 Affinity is an optional affinity selector that will be added to handler DaemonSet manifest.

infraAffinity >
 InfraAffinity is an optional affinity selector that will be added to webhook, metrics & console-plugin Deployment manifests.

infraTolerations >
 InfraTolerations is an optional list of tolerations to be added to webhook, metrics & console-plugin Deployment manifests. If InfraTolerations is specified, the webhook, metrics and the console plugin will be able to be scheduled on nodes with corresponding taints.

probeConfiguration >
 ProbeConfiguration is an optional configuration of NMState probes testing various functionalities. If ProbeConfiguration is specified, the handler will use the config defined here instead of its default values.

selfSignConfiguration >
 SelfSignConfiguration defines self signed certificate configuration

tolerations >
 Tolerations is an optional list of tolerations to be added to handler DaemonSet manifest. If Tolerations is specified, the handler daemonset will be also scheduled on nodes with corresponding taints.

Create Cancel

Create Secondary Network

OpenShift Container Platform 4.18: Virtualization – Networking – [10.9 Connecting a virtual machine to an OVN-Kubernetes secondary network](#)

Create the following **NodeNetworkConfigurationPolicy** and **NetworkAttachmentDefinition**.

STEP 1. From the OpenShift console, navigate to **Networking > NodeNetworkConfigurationPolicy**, then click on **“Create NodeNetworkConfigurationPolicy”**. Click on **“Edit YAML”**, and then copy/paste the following YAML into the editor window. When complete, click **“Create”**.

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: localnet-network-mapping
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ''
  desiredState:
    ovn:
```

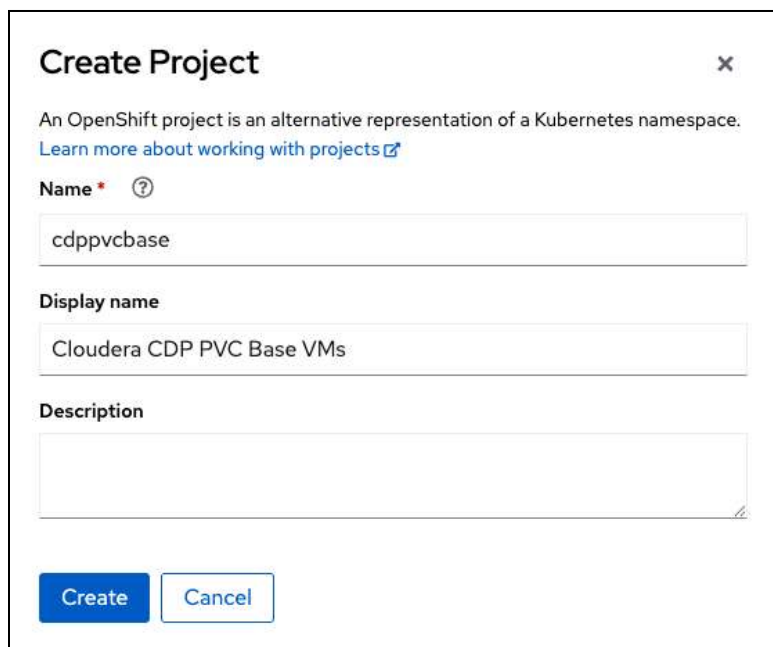
```
bridge-mappings:
- localnet: localnet-network
  bridge: br-ex
  state: present
```

STEP 2. From the OpenShift console, navigate to **Networking > NetworkAttachmentDefinition**, then click on **“Create NetworkAttachmentDefinition”**. Click on **“YAML view”**, and then copy/paste the following YAML into the editor window. When complete, click **“Create”**.

```
apiVersion: k8s.cni.cncf.io/v1
kind: NetworkAttachmentDefinition
metadata:
  name: localnet-network
  namespace: default
spec:
  config: |2
    {
      "cniVersion": "0.3.1",
      "name": "localnet-network",
      "type": "ovn-k8s-cni-overlay",
      "topology": "localnet",
      "netAttachDefName": "default/localnet-network"
    }
  }
```


Create Project for CDP PVC Base VMs

STEP 1. Navigate to **Home > Projects** from the OpenShift Console. Select the **Create Project** button to create the *cdppvcbase* project:



The 'Create Project' dialog box in the OpenShift console. It has a title bar with a close button. Below the title, there is a brief explanation of OpenShift projects and a link to learn more. The form contains three fields: 'Name' (with a required asterisk and help icon) containing 'cdppvcbase', 'Display name' containing 'Cloudera CDP PVC Base VMs', and 'Description' which is empty. At the bottom are 'Create' and 'Cancel' buttons.

Create Project ✕

An OpenShift project is an alternative representation of a Kubernetes namespace.
[Learn more about working with projects](#)

Name * ?

cdppvcbase

Display name

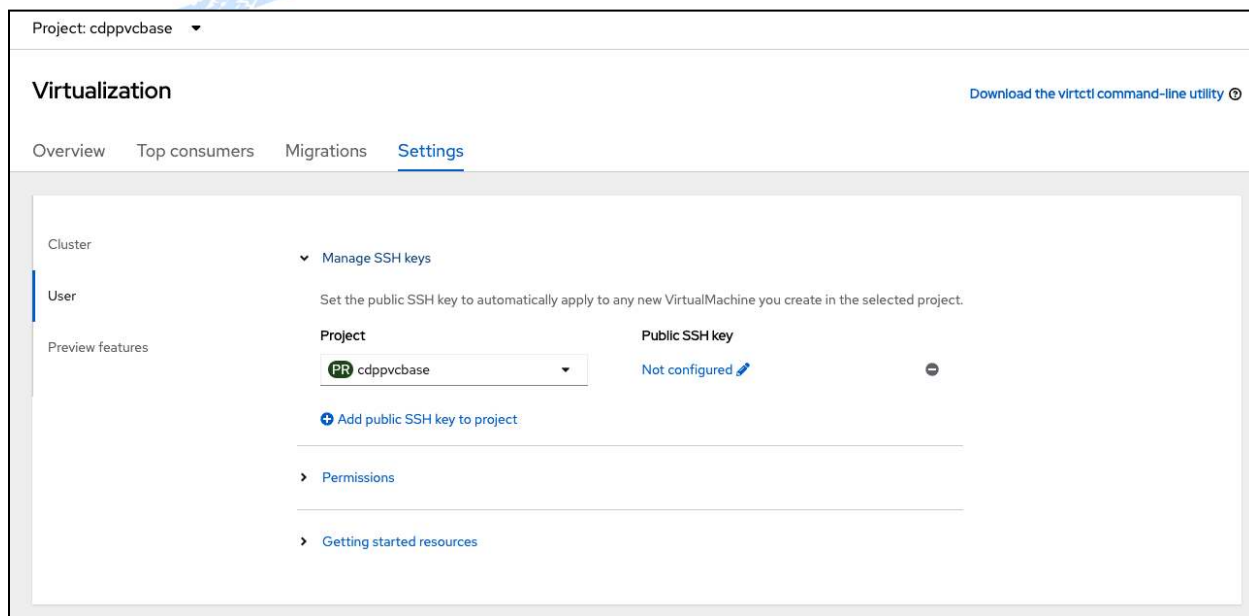
Cloudera CDP PVC Base VMs

Description

Create Cancel

Create SSH Keys in Project

STEP 1. Navigate to **Virtualization > Overview > Settings > User > Manage SSH Keys**. Select the project *cdppvcbase*, then click on *Not configured* to add the public SSH key to the project.



The 'Virtualization' settings page for the 'cdppvcbase' project. It shows tabs for Overview, Top consumers, Migrations, and Settings. The 'Manage SSH keys' section is expanded, showing instructions to set a public SSH key. A dropdown menu shows the selected project 'cdppvcbase'. The 'Public SSH key' field is currently 'Not configured' with an edit icon. Below this are links for 'Add public SSH key to project', 'Permissions', and 'Getting started resources'.

Project: cdppvcbase ▼

Virtualization Download the virtctl command-line utility

Overview Top consumers Migrations Settings

Cluster

User

Preview features

▼ Manage SSH keys

Set the public SSH key to automatically apply to any new VirtualMachine you create in the selected project.

Project

cdppvcbase

Public SSH key

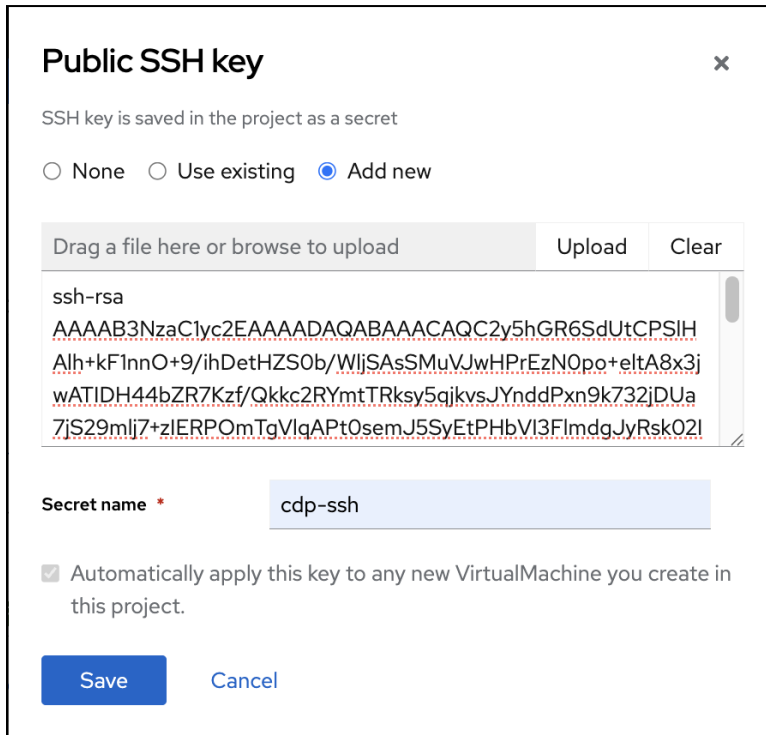
Not configured ✎

[Add public SSH key to project](#)

[Permissions](#)

[Getting started resources](#)

STEP 2. Select *Add new*, and paste the contents of the public key into the page, provide a *secret name*, and click on *Save*.



Public SSH key ×

SSH key is saved in the project as a secret

☐ None ☐ Use existing ☒ Add new

Drag a file here or browse to upload Upload Clear

```
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQCAQC2y5hGR6SdUtCPSIH
Alh+kF1nnO+9/ihDetHZSOb/WljSAsSMuVJwHPREzN0po+eltA8x3j
wATIDH44bZR7Kzf/Qkkc2RYmtTRksy5qjkvsJYnddPxn9k732jDUa
7jS29mlj7+zIERPomTgVlqAPtOsemJ5SyEtPHbVI3FlmdgJyRsk02l
```

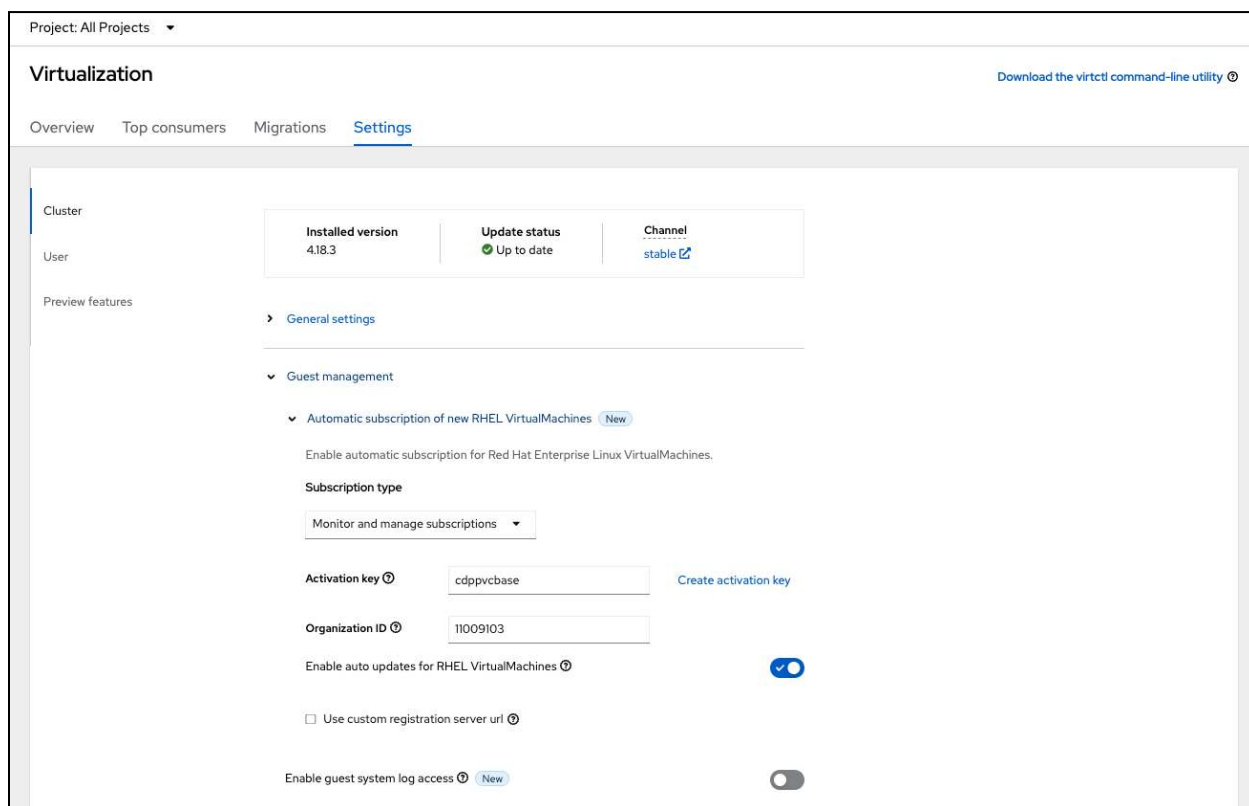
Secret name *

☒ Automatically apply this key to any new VirtualMachine you create in this project.

Save Cancel

Auto Register RHEL VMs

STEP 1. From the OpenShift console, navigate to *Virtualization > Overview > Settings > Cluster > Guest Management > Automatic subscription of new RHEL VirtualMachines*. Select *Monitor and manage subscriptions*. Enter the *Activation key*, *Organization ID* and select *Enable auto updates for RHEL VirtualMachines*.



For more information, refer to [Blog: Subscribing RHEL VMs in OpenShift Virtualization](#)

To create an activation key for your Red Hat Account and Org, you can follow the link provided at: <https://console.redhat.com/settings/connector/activation-keys>

This registration only happens during VM creation via the CloudInit script.

NOTE: In case the VM does not register with Red Hat Insights, you can use the following command to manually register the VM: `rhc connect`

Example:

```
# rhc connect --activation-key [your_activation_key] --organization [your_org_#]
```

This can also be performed by using the ansible command, in case it isn't applied during VM creation.

```
ansible all -m shell -a "subscription-manager register --org=[your_org_#] --activationkey=[your_activation_key]"
```

Create VMs on OpenShift Virtualization

The following table is provided as a reference for the VMs that will be created for CDP Base:

Hostname	CPU	RAM	Disk	IP Address
cldr-mngr.<domain_name>	32	64 GiB	600 GiB	10.1.49.100/23
cldr-utility.<domain_name>	32	64 GiB	600 GiB	10.1.49.101/23
pvcbase-master[01-03].<domain_name>	32	64 GiB	500 GiB	10.1.49.102-104/23
pvcbase-worker[01-96].<domain_name>	32	64 GiB	500 GiB	10.1.49.105-199/23

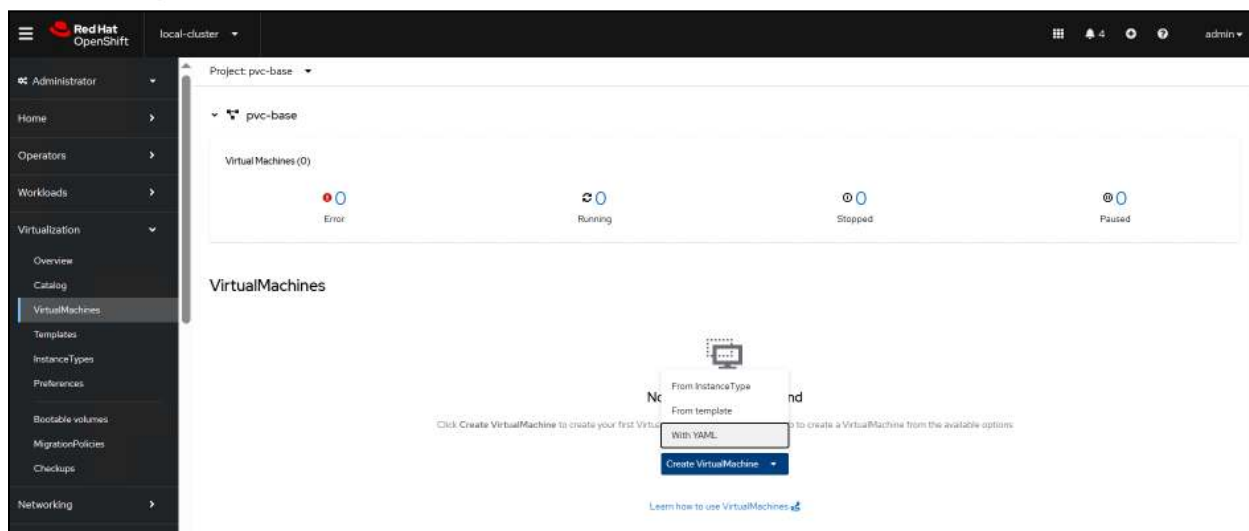
<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

The IPA server was created in a server (or a VM) running outside from OpenShift Virtualization:

ipaserver.<domain_name>	16	32 GiB	250 GiB	10.1.49.1/23
-------------------------	----	--------	---------	--------------

STEP 1. We will quickly create the VirtualMachines using the YAML definition provided below.

From the OpenShift console, navigate to **Virtualization > VirtualMachines**. Make sure that the “**cdppvcbase**” Project is shown in the upper left part of the screen. Click on the “**Create VirtualMachine**” button and select “**With YAML**”.

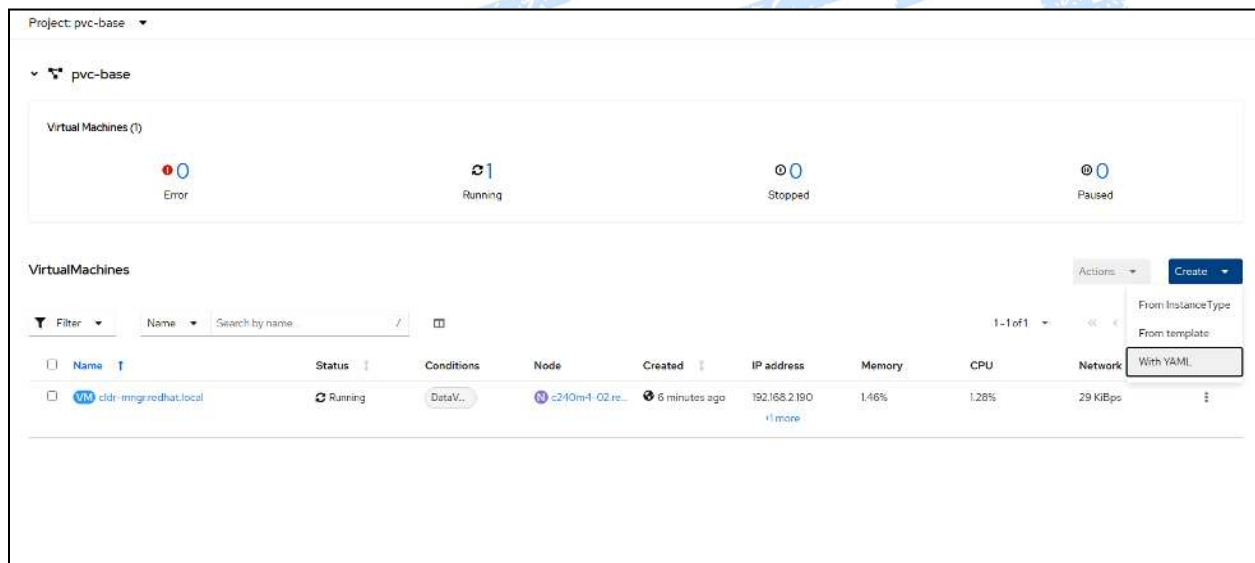


STEP 2. From the Create VirtualMachine screen, copy/paste the YAML code provided below into the editor window replacing the code that is initially shown.

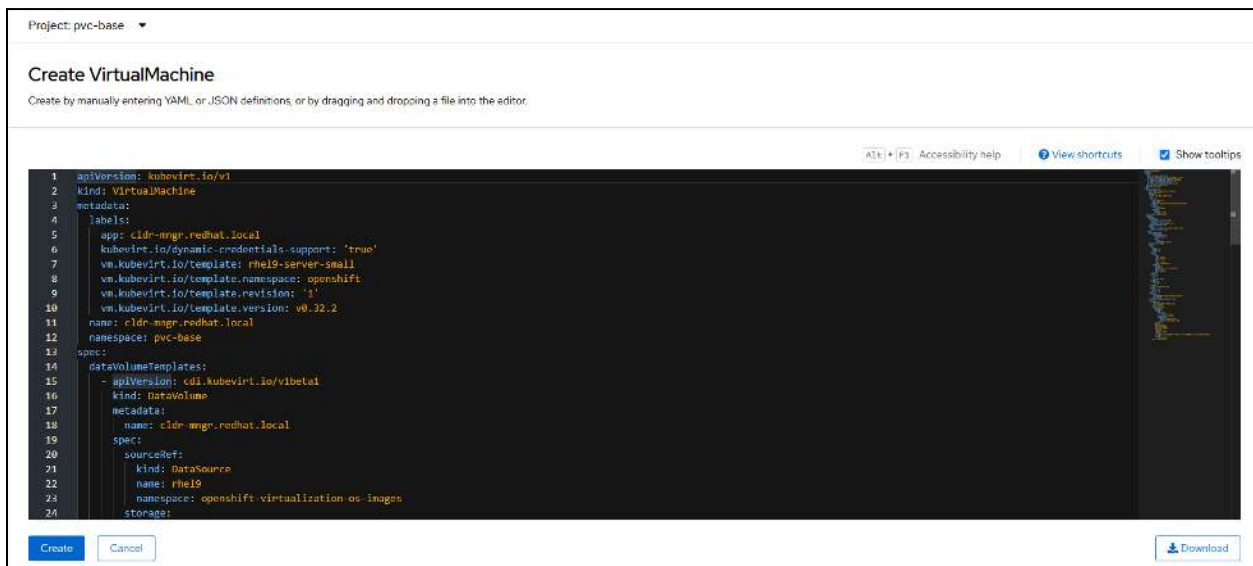
STEP 3. To create the VM for `cldr-mngr`, no changes are needed to the provided YAML code. Click on “Create”.

STEP 4. Next, we will create the other VMs; using the VirtualMachine YAML code for the remaining VMs that is provided at the bottom of this section; after step 8.

STEP 5. Navigate back to *Virtualization* > *VirtualMachines*, and click on the “Create” button on the right side of the screen and then select “With YAML”.

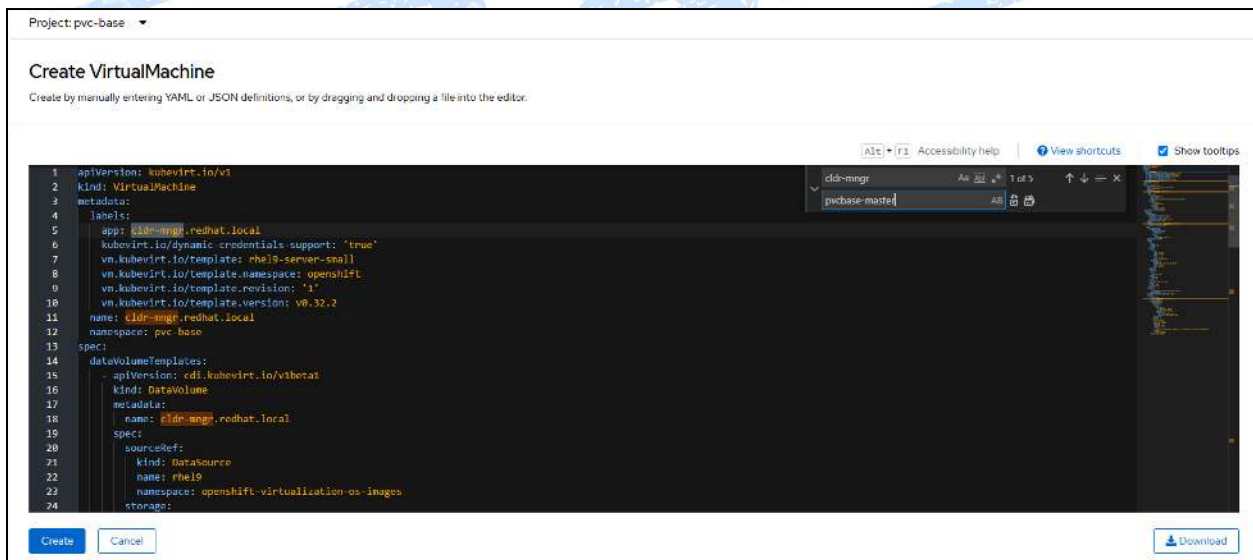


STEP 6. Again, copy/paste the provided YAML code into the editor window, replacing the code that is initially shown.



NOTE: Use the VirtualMachine YAML code for each of the remaining VMs that is provided at the bottom of this section; after step 8.

STEP 7. When any additional/optional changes have been made to the configuration parameters, click on “Create”.



STEP 8. Repeat the last three steps until all of the VMs have been created.

The following is the cloud-init section that is included in each of the VM YAML definitions provided.

cloud-init section of the VirtualMachine YAML

```
- cloudInitNoCloud:
  networkData: |
    ethernets:
      eth0:
        addresses:
          - 10.1.49.100/23      # Set IP v4 address
        gateway4: 10.1.49.254  # Set default gateway
        nameservers:
          search: [cdp.rdu2.scalelab.redhat.com] # Set DNS search domain
          addresses: [10.1.49.1] # Set DNS server address
        version: 2
  userData: |-
    #cloud-config
    user: cloud-user          # Create user account
    password: r3dh4t!         # Set password for user account
    chpasswd: { expire: False } # Disable password expiration
    ssh_pwauth: True          # Enable password login for root over ssh
    hostname: cldr-mngr.cdp.rdu2.scalelab.redhat.com # set hostname for VM
    packages:
      - dnf-automatic
    runcmd:
      - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
      - subscription-manager release --set=9.5          # Set RHEL minor version
      - systemctl enable --now dnf-automatic-install.timer
      - |
        #!/bin/bash
        # Add entries for sysctl
        echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
        echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
        echo "vm.swappiness=1" >> /etc/sysctl.conf
        # Enable sysctl
        sysctl -p
```

The VirtualMachine YAML code for all 7 of the VMs are provided below. Refer to the section [OPTIONAL Method to create VMs](#) for the process to create the VM configuration and its associated YAML code.

cldr-mngr VirtualMachine YAML

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  annotations:
    kubemacpool.io/transaction-timestamp: '2025-10-27T20:30:50.413104701Z'
    kubevirt.io/latest-observed-api-version: v1
    kubevirt.io/storage-observed-api-version: v1
    vm.kubevirt.io/validations: |
      [
```



```

    {
      "name": "minimal-required-memory",
      "path": "jsonpath::.spec.domain.memory.guest",
      "rule": "integer",
      "message": "This VM requires more memory.",
      "min": 1610612736
    }
  ]
  resourceVersion: '8837376'
  name: cldr-mngr
  uid: 38f278e3-7b56-4f65-8586-c4e16adbc40e
  creationTimestamp: '2025-10-27T20:30:50Z'
  generation: 1
  managedFields:
    - apiVersion: kubevirt.io/v1
      fieldsType: FieldsV1
      fieldsV1:
        'f:metadata':
          'f:annotations':
            .: {}
            'f:kubemacpool.io/transaction-timestamp': {}
            'f:vm.kubevirt.io/validations': {}
          'f:labels':
            .: {}
            'f:app': {}
            'f:kubevirt.io/dynamic-credentials-support': {}
            'f:vm.kubevirt.io/template': {}
            'f:vm.kubevirt.io/template.namespace': {}
            'f:vm.kubevirt.io/template.revision': {}
            'f:vm.kubevirt.io/template.version': {}
        'f:spec':
          .: {}
          'f:dataVolumeTemplates': {}
          'f:runStrategy': {}
          'f:template':
            .: {}
            'f:metadata':
              .: {}
              'f:annotations':
                .: {}
                'f:vm.kubevirt.io/flavor': {}
                'f:vm.kubevirt.io/os': {}
                'f:vm.kubevirt.io/workload': {}
              'f:creationTimestamp': {}
              'f:labels':
                .: {}
                'f:affinity_group_1': {}
                'f:kubevirt.io/domain': {}
                'f:kubevirt.io/size': {}
            'f:spec':
              'f:accessCredentials': {}

```



```

    'f:volumes': {}
    'f:hostname': {}
    'f:architecture': {}
    .: {}
    'f:terminationGracePeriodSeconds': {}
    'f:domain':
      .: {}
      'f:cpu':
        .: {}
        'f:cores': {}
        'f:sockets': {}
        'f:threads': {}
      'f:devices':
        .: {}
        'f:disks': {}
        'f:interfaces': {}
        'f:networkInterfaceMultiqueue': {}
        'f:rng': {}
      'f:features':
        .: {}
        'f:acpi': {}
        'f:mmio':
          .: {}
          'f:enabled': {}
      'f:firmware':
        .: {}
        'f:bootloader':
          .: {}
          'f:efi': {}
      'f:machine':
        .: {}
        'f:type': {}
      'f:memory':
        .: {}
        'f:guest': {}
      'f:resources': {}
    'f:networks': {}
    'f:affinity':
      .: {}
      'f:podAntiAffinity':
        .: {}
        'f:requiredDuringSchedulingIgnoredDuringExecution': {}
  manager: OpenAPI-Generator
  operation: Update
  time: '2025-10-27T20:30:50Z'
- apiVersion: kubevirt.io/v1
  fieldsType: FieldsV1
  fieldsV1:
    'f:metadata':
      'f:annotations':
        'f:kubevirt.io/latest-observed-api-version': {}

```

```

      'f:kubevirt.io/storage-observed-api-version': {}
      'f:finalizers':
        .: {}
        'v:"kubevirt.io/virtualMachineControllerFinalize"': {}
    manager: virt-controller
    operation: Update
    time: '2025-10-27T20:30:50Z'
- apiVersion: kubevirt.io/v1
  fieldsType: FieldsV1
  fieldsV1:
    'f:status':
      'f:printableStatus': {}
      'f:runStrategy': {}
      'f:conditions': {}
      .: {}
      'f:ready': {}
      'f:volumeSnapshotStatuses': {}
      'f:observedGeneration': {}
      'f:created': {}
      'f:desiredGeneration': {}
    manager: virt-controller
    operation: Update
    subresource: status
    time: '2025-10-27T20:31:39Z'
namespace: cdppvcbase
finalizers:
- kubevirt.io/virtualMachineControllerFinalize
labels:
  app: cldr-mngr
  kubevirt.io/dynamic-credentials-support: 'true'
  vm.kubevirt.io/template: rhel9-server-large
  vm.kubevirt.io/template.namespace: openshift
  vm.kubevirt.io/template.revision: '1'
  vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
  - apiVersion: cdi.kubevirt.io/v1beta1
    kind: DataVolume
    metadata:
      creationTimestamp: null
      name: cldr-mngr-dv
    spec:
      source:
        pvc:
          name: rhel9.5
          namespace: cdppvcbase
      storage:
        resources:
          requests:
            storage: 600Gi
        storageClassName: ocs-storagecluster-ceph-replica2-rbd

```

```

runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: large
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      affinity_group_1: ''
      kubevirt.io/domain: cldr-mngr
      kubevirt.io/size: large
  spec:
    accessCredentials:
      - sshPublicKey:
          propagationMethod:
            noCloud: {}
          source:
            secret:
              secretName: cdp-ssh
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: affinity_group_1
                  operator: Exists
              topologyKey: kubernetes.io/hostname
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - disk:
              bus: virtio
              name: rootdisk
          - disk:
              bus: virtio
              name: cloudinitdisk
        interfaces:
          - bridge: {}
            #      macAddress: '02:00:e3:00:02:3f'
            model: virtio
            name: default
            state: up
            networkInterfaceMultiqueue: true
            rng: {}
    features:

```

```

    acpi: {}
    smm:
      enabled: true
  firmware:
    bootloader:
      efi: {}
  machine:
    type: pc-q35-rhel9.6.0
  memory:
    guest: 32Gi
  resources: {}
  hostname: cldr-mngr
  networks:
    - multus:
        networkName: default/localnet-network
        name: default
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: cldr-mngr-dv
        name: rootdisk
    - cloudInitNoCloud:
        networkData: |
          ethernet:
            eth0:
              addresses:
                - 10.1.49.100/23
              gateway4: 10.1.49.254
              nameservers:
                search: [cdp.rdu2.scalelab.redhat.com]
                addresses: [10.1.49.1]
        version: 2
  userData: |-
    #cloud-config
    user: cloud-user
    password: r3dh4t!
    chpasswd: { expire: False }
    ssh_pwauth: True
    hostname: cldr-mngr.cdp.rdu2.scalelab.redhat.com
    packages:
      - dnf-automatic
    runcmd:
      - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
      - subscription-manager release --set=9.5
      - systemctl enable --now dnf-automatic-install.timer
      - |
        #!/bin/bash
        # Add entries for sysctl
        echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf

```

```

        echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
        echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
        echo "vm.swappiness=1" >> /etc/sysctl.conf
        # Enable sysctl
        sysctl -p
name: cloudinitdisk

```

cldr-utility VirtualMachine YAML

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  annotations:
    kubemacpool.io/transaction-timestamp: '2025-10-27T20:21:32.838262833Z'
    kubevirt.io/latest-observed-api-version: v1
    kubevirt.io/storage-observed-api-version: v1
    vm.kubevirt.io/validations: |
      [
        {
          "name": "minimal-required-memory",
          "path": "jsonpath::.spec.domain.memory.guest",
          "rule": "integer",
          "message": "This VM requires more memory.",
          "min": 1610612736
        }
      ]
  resourceVersion: '8817926'
  name: cldr-utility
  uid: 610bbc39-d416-4c0c-8d53-a06cb55c459f
  creationTimestamp: '2025-10-27T20:21:32Z'
  generation: 1
  managedFields:
    - apiVersion: kubevirt.io/v1
      fieldsType: FieldsV1
      fieldsV1:
        'f:metadata':
          'f:annotations':
            .: {}
            'f:kubemacpool.io/transaction-timestamp': {}
            'f:vm.kubevirt.io/validations': {}
          'f:labels':
            .: {}
            'f:app': {}
            'f:kubevirt.io/dynamic-credentials-support': {}
            'f:vm.kubevirt.io/template': {}
            'f:vm.kubevirt.io/template.namespace': {}
            'f:vm.kubevirt.io/template.revision': {}
            'f:vm.kubevirt.io/template.version': {}

```

```

'f:spec':
  .: {}
  'f:dataVolumeTemplates': {}
  'f:runStrategy': {}
  'f:template':
    .: {}
    'f:metadata':
      .: {}
      'f:annotations':
        .: {}
        'f:vm.kubevirt.io/flavor': {}
        'f:vm.kubevirt.io/os': {}
        'f:vm.kubevirt.io/workload': {}
      'f:creationTimestamp': {}
      'f:labels':
        .: {}
        'f:affinity_group_1': {}
        'f:kubevirt.io/domain': {}
        'f:kubevirt.io/size': {}
    'f:spec':
      'f:accessCredentials': {}
      'f:volumes': {}
      'f:hostname': {}
      'f:architecture': {}
      .: {}
      'f:terminationGracePeriodSeconds': {}
      'f:domain':
        .: {}
        'f:cpu':
          .: {}
          'f:cores': {}
          'f:sockets': {}
          'f:threads': {}
        'f:devices':
          .: {}
          'f:disks': {}
          'f:interfaces': {}
          'f:networkInterfaceMultiqueue': {}
          'f:rng': {}
        'f:features':
          .: {}
          'f:acpi': {}
          'f:mmio':
            .: {}
            'f:enabled': {}
          'f:firmware':
            .: {}
            'f:bootloader':
              .: {}
              'f:efi': {}
            'f:machine':

```

```

      .: {}
      'f:type': {}
      'f:memory':
      .: {}
      'f:guest': {}
      'f:resources': {}
      'f:networks': {}
      'f:affinity':
      .: {}
      'f:podAntiAffinity':
      .: {}
      'f:requiredDuringSchedulingIgnoredDuringExecution': {}
manager: OpenAPI-Generator
operation: Update
time: '2025-10-27T20:21:32Z'
- apiVersion: kubevirt.io/v1
  fieldsType: FieldsV1
  fieldsV1:
    'f:metadata':
      'f:annotations':
        'f:kubevirt.io/latest-observed-api-version': {}
        'f:kubevirt.io/storage-observed-api-version': {}
      'f:finalizers':
        .: {}
        'v:"kubevirt.io/virtualMachineControllerFinalize"': {}
manager: virt-controller
operation: Update
time: '2025-10-27T20:21:32Z'
- apiVersion: kubevirt.io/v1
  fieldsType: FieldsV1
  fieldsV1:
    'f:status':
      'f:printableStatus': {}
      'f:runStrategy': {}
      'f:conditions': {}
      .: {}
      'f:ready': {}
      'f:volumeSnapshotStatuses': {}
      'f:observedGeneration': {}
      'f:created': {}
      'f:desiredGeneration': {}
manager: virt-controller
operation: Update
subresource: status
time: '2025-10-27T20:22:14Z'
namespace: cdppvcbase
finalizers:
- kubevirt.io/virtualMachineControllerFinalize
labels:
  app: cldr-utility
  kubevirt.io/dynamic-credentials-support: 'true'

```

```

vm.kubevirt.io/template: rhel9-server-large
vm.kubevirt.io/template.namespace: openshift
vm.kubevirt.io/template.revision: '1'
vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        creationTimestamp: null
        name: cldr-utility-dv
      spec:
        source:
          pvc:
            name: rhel9.5
            namespace: cdppvcbase
        storage:
          resources:
            requests:
              storage: 600Gi
          storageClassName: ocs-storagecluster-ceph-replica2-rbd
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: large
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        affinity_group_1: ''
        kubevirt.io/domain: cldr-utility
        kubevirt.io/size: large
    spec:
      accessCredentials:
        - sshPublicKey:
            propagationMethod:
              noCloud: {}
            source:
              secret:
                secretName: cdp-ssh
      affinity:
        podAntiAffinity:
          requiredDuringSchedulingIgnoredDuringExecution:
            - labelSelector:
                matchExpressions:
                  - key: affinity_group_1
                    operator: Exists
              topologyKey: kubernetes.io/hostname
      architecture: amd64
      domain:

```



```
cpu:
  cores: 1
  sockets: 8
  threads: 1
devices:
  disks:
    - disk:
        bus: virtio
        name: rootdisk
    - disk:
        bus: virtio
        name: cloudinitdisk
  interfaces:
    - bridge: {}
      #      macAddress: '02:00:e3:00:02:29'
      model: virtio
      name: default
      state: up
    networkInterfaceMultiqueue: true
    rng: {}
  features:
    acpi: {}
    smm:
      enabled: true
  firmware:
    bootloader:
      efi: {}
  machine:
    type: pc-q35-rhel9.6.0
  memory:
    guest: 32Gi
  resources: {}
hostname: cldr-utility
networks:
  - multus:
      networkName: default/localnet-network
      name: default
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: cldr-utility-dv
      name: rootdisk
  - cloudInitNoCloud:
      networkData: |
        ethernet:
          eth0:
            addresses:
              - 10.1.49.101/23
            gateway4: 10.1.49.254
            nameservers:
              search: [cdp.rdu2.scalelab.redhat.com]
```

```

        addresses: [10.1.49.1]
        version: 2
      userData: |-
        #cloud-config
        user: cloud-user
        password: r3dh4t!
        chpasswd: { expire: False }
        ssh_pwauth: True
        hostname: cldr-utility.cdp.rdu2.scalelab.redhat.com
        packages:
          - dnf-automatic
        runcmd:
          - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
          - subscription-manager release --set=9.5
          - systemctl enable --now dnf-automatic-install.timer
          - |
            #!/bin/bash
            # Add entries for sysctl
            echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
            echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
            echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
            echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
            echo "vm.swappiness=1" >> /etc/sysctl.conf
            # Enable sysctl
            sysctl -p
      name: cloudinitdisk

```

pvcbase-masterXX VirtualMachine YAML

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: pvcbase-master01
  generation: 1
  namespace: cdppvcbase
  finalizers:
    - kubevirt.io/virtualMachineControllerFinalize
  labels:
    app: pvcbase-master01
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:

```

```
creationTimestamp: null
name: pvcbase-master01-dv
spec:
  source:
    pvc:
      name: rhel9.5
      namespace: cdppvcbase
  storage:
    resources:
      requests:
        storage: 500Gi
      storageClassName: ocs-storagecluster-ceph-replica2-rbd
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: large
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      affinity_group_1: ''
      kubevirt.io/domain: pvcbase-master01
      kubevirt.io/size: large
  spec:
    accessCredentials:
      - sshPublicKey:
          propagationMethod:
            noCloud: {}
          source:
            secret:
              secretName: cdp-ssh
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: affinity_group_1
                  operator: Exists
              topologyKey: kubernetes.io/hostname
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - disk:
              bus: virtio
              name: rootdisk
```

```
- disk:
  bus: virtio
  name: cloudinitdisk
interfaces:
- bridge: {}
  macAddress: '02:00:e3:00:02:a1'
  model: virtio
  name: default
  state: up
  networkInterfaceMultiqueue: true
  rng: {}
features:
  acpi: {}
  smm:
    enabled: true
firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.6.0
memory:
  guest: 32Gi
resources: {}
hostname: pvcbase-master01
networks:
- multus:
  networkName: default/localnet-network
  name: default
terminationGracePeriodSeconds: 180
volumes:
- dataVolume:
  name: pvcbase-master01-dv
  name: rootdisk
- cloudInitNoCloud:
  networkData: |
    ethernets:
      eth0:
        addresses:
          - 10.1.49.102/23
        gateway4: 10.1.49.254
        nameservers:
          search: [cdp.rdu2.scalelab.redhat.com]
          addresses: [10.1.49.1]
  version: 2
userData: |-
  #cloud-config
  user: cloud-user
  password: r3dh4t!
  chpasswd: { expire: False }
  ssh_pwauth: True
  hostname: pvcbase-master01.cdp.rdu2.scalelab.redhat.com
```

```

    packages:
      - dnf-automatic
    runcmd:
      - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
      - subscription-manager release --set=9.5
      - systemctl enable --now dnf-automatic-install.timer
      - |
        #!/bin/bash
        # Add entries for sysctl
        echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
        echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
        echo "vm.swappiness=1" >> /etc/sysctl.conf
        # Enable sysctl
        sysctl -p
    name: cloudinitdisk

```

pvcbase-workerXX VirtualMachine YAML

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  name: pvcbase-worker01
  labels:
    app: pvcbase-worker01
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-large
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.34.0
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        creationTimestamp: null
        name: pvcbase-worker01-dv
      spec:
        source:
          pvc:
            name: rhel9.5
            namespace: cdppvcbase
        storage:
          resources:
            requests:
              storage: 500Gi
          storageClassName: ocs-storagecluster-ceph-replica2-rbd

```

```
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: large
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      affinity_group_1: ''
      kubevirt.io/domain: pvcbase-worker01
      kubevirt.io/size: large
  spec:
    accessCredentials:
      - sshPublicKey:
          propagationMethod:
            noCloud: {}
          source:
            secret:
              secretName: cdp-ssh
    affinity:
      podAntiAffinity:
        requiredDuringSchedulingIgnoredDuringExecution:
          - labelSelector:
              matchExpressions:
                - key: affinity_group_1
                  operator: Exists
              topologyKey: kubernetes.io/hostname
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - disk:
              bus: virtio
              name: rootdisk
          - disk:
              bus: virtio
              name: cloudinitdisk
        interfaces:
          - bridge: {}
            macAddress: '02:00:e3:00:02:42'
            model: virtio
            name: default
            state: up
            networkInterfaceMultiqueue: true
            rng: {}
    features:
```

```

    acpi: {}
    smm:
      enabled: true
  firmware:
    bootloader:
      efi: {}
  machine:
    type: pc-q35-rhel9.6.0
  memory:
    guest: 32Gi
  resources: {}
  hostname: pvcbase-worker01
  networks:
    - multus:
        networkName: default/localnet-network
        name: default
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcbase-worker01-dv
        name: rootdisk
    - cloudInitNoCloud:
        networkData: |
          ethernet:
            eth0:
              addresses:
                - 10.1.49.105/23
              gateway4: 10.1.49.254
              nameservers:
                search: [cdp.rdu2.scalelab.redhat.com]
                addresses: [10.1.49.1]
        version: 2
  userData: |-
    #cloud-config
    user: cloud-user
    password: r3dh4t!
    chpasswd: { expire: False }
    ssh_pwauth: True
    hostname: pvcbase-worker01.cdp.rdu2.scalelab.redhat.com
    packages:
      - dnf-automatic
    runcmd:
      - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
      - subscription-manager release --set=9.5
      - systemctl enable --now dnf-automatic-install.timer
      - |
        #!/bin/bash
        # Add entries for sysctl
        echo "net.ipv6.conf.all.disable_ipv6 = 1" >> /etc/sysctl.conf
        echo "net.ipv6.conf.default.disable_ipv6 = 1" >> /etc/sysctl.conf

```

```
echo "net.ipv6.conf.lo.disable_ipv6 = 0" >> /etc/sysctl.conf
echo "vm.overcommit_memory=0" >> /etc/sysctl.conf
echo "vm.swappiness=1" >> /etc/sysctl.conf
# Enable sysctl
sysctl -p
name: cloudinitdisk
```

OPTIONAL Method to create VMs

If needed, please refer to the optional method to create VMs from the VM Template Catalog:

[OPTIONAL Method to create VMs](#)

This method can be used to create the VirtualMachine YAML definitions.

DRAFT

CONFIGURE THE LDAP IDENTITY PROVIDER WITH OPENSIFT:

If needed, you can access the IPA Admin console:

<https://idm.redhat.local/ipa/ui/>

From the WebUI for IPAServer Administration, enter the same admin credentials used for CLI authentication: (i.e. *admin/redhat123*)


```
[root@idm ~]# vi /etc/openldap/ldap.conf
```

```
# File modified by ipa-client-install
. . .
# When no CA certificates are specified the Shared System Certificates
# are in use. In order to have these available along with the ones specified
# by TLS_CACERTDIR one has to include them explicitly:
#TLS_CACERT /etc/pki/tls/cert.pem

# System-wide Crypto Policies provide up to date cipher suite which should
# be used unless one needs a finer grinded selection of ciphers. Hence, the
# PROFILE=SYSTEM value represents the default behavior which is in place
# when no explicit setting is used. (see openssl-ciphers(1) for more info)
#TLS_CIPHER_SUITE PROFILE=SYSTEM

# Turning this off breaks GSSAPI used with krb5 when rdns = false
SASL_NOCANON      on

URI ldaps://ipaserver.cdp.rdu2.scalelab.redhat.com
BASE dc=cdp,dc=rdu2,dc=scalelab,dc=redhat,dc=com
SASL_MECH GSSAPI
```

If any changes are made to the IPA/IdM configuration, use the following command to restart the IPA server:

```
[root@idm ~]# ipactl restart
```

STEP 1. Create a secret with the Identity Management (IdM) admin user password. This secret will be used for the LDAP identity provider:

```
[root@bastion ]# oc create secret generic ldap-secret -n openshift-config \
--from-literal=bindPassword=redhat123
```

STEP 2. TLS communication needs certificate authority validation, in this case, the CA is obtained from IdM.

The certificate is stored on the ipaserver in the file: `/etc/ipa/ca.crt`

Even when IdM is configured not to create/deploy a Certificate Authority, but instead use certificates provided by an external CA, the installer still stores the CA certificate in the file: `/etc/ipa/ca.crt`

Use the following command to download the Certificate Authority's root certificate to a file named `ca.crt`.

```
[root@bastion ]# scp
root@ipaserver.cdp.rdu2.scalelab.redhat.com:/etc/ipa/ca.crt /root/ca.crt
```

or

```
[root@bastion ]# wget -c -nv  
http://ipaserver.cdp.rdu2.scalelab.redhat.com/ipa/config/ca.crt
```

STEP 3. Create a configuration map containing the IdM Certificate Authority's root certificate to make OpenShift trust the IdM certificates.
The ConfigMap key must be named ca.crt.

```
[root@bastion ]# oc create configmap ldap-ca-configmap -n openshift-config \  
--from-file=ca.crt=/root/ca.crt
```

STEP 4. Create a file to modify the OpenShift OAuth configuration by adding the LDAP identity provider and the other elements required for proper configuration.

```
[root@bastion ]# cat > ldap-cr.yaml << EOF
```

```
apiVersion: config.openshift.io/v1  
kind: OAuth  
metadata:  
  name: cluster  
spec:  
  identityProviders:  
    - name: ldap  
      mappingMethod: claim  
      type: LDAP  
      ldap:  
        attributes:  
          email: []  
          id:  
            - dn  
          name:  
            - cn  
          preferredUsername:  
            - uid  
        bindDN:  
'uid=admin,cn=users,cn=accounts,dc=cdp,dc=rdu2,dc=scalelab,dc=redhat,dc=com'  
        bindPassword:  
          name: ldap-secret  
        ca:  
          name: ldap-ca-configmap  
        insecure: false  
        url:  
'ldaps://ipaserver.cdp.rdu2.scalelab.redhat.com/cn=users,cn=accounts,dc=cdp,dc=rdu2,dc=scalelab,dc=redhat,dc=com?uid'
```

EOF

STEP 5. Apply the custom resource and wait until the pods in the openshift-authentication namespace are restarted so that the identity provider is active:

```
[root@bastion ]# oc apply -f ldap-cr.yaml
```

STEP 6. View the status, as the pods in the openshift-authentication namespace redeploy. This can take ten minutes or more before the pods are ready.

```
[root@bastion ]# oc get clusteroperator/authentication
```

```
[root@bastion ]# oc get pods -l app=oauth-openshift -n  
openshift-authentication
```

If there are any issues with the IdM integration, check the pod status and logs of the deployment.apps/oauth-openshift deployment in the openshift-authentication namespace.

```
[root@bastion ]# oc logs deployment.apps/oauth-openshift -n  
openshift-authentication
```

STEP 7. Add the cluster-admin role to the IdM admin user that needs cluster administration privileges.

```
[root@bastion ]# oc adm policy add-cluster-role-to-user cluster-admin admin  
Warning: User 'admin' not found  
clusterrole.rbac.authorization.k8s.io/cluster-admin added: "admin"
```

STEP 8. Verify that you are able to log in with the IdM admin user.

```
[root@bastion ]# oc login -u admin -p redhat123  
https://api.vlan601.rdu2.scalelab.redhat.com/:6443
```

```
[root@bastion ]# oc whoami
```

If necessary, the kubeadmin account is still available to use to login.

```
[root@bastion ]# oc login -u [your_openshift_username] -p  
[your_openshift_password] https://api.vlan601.rdu2.scalelab.redhat.com:6443
```

Create OpenShift cluster using the Assisted Installer

The Assisted Installer is used to create an OpenShift cluster for CDP Data Services on bare-metal server nodes.

The following table is a reference of the bare-metal servers that will be used for the OpenShift cluster for CDP Data Services:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
f36-h17-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:78:dd:91	10.1.49.216/23
f36-h18-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:78:d7:91	10.1.49.217/23
f36-h19-000-r640 (ocp master)	40	384 GiB	446 GiB	bc:97:e1:69:d2:41	10.1.49.218/23
d29-h11-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1c:07:61	10.1.49.219/23
d29-h13-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:d3:31	10.1.49.220/23
d30-h05-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:9a:a1	10.1.49.221/23
d30-h07-000-r750 (ocp worker)	56	512 GiB	446 GiB	00:62:0b:1d:d1:a1	10.1.49.222/23

<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

Installation parameters:

Parameter	Setting/Value
Cluster name	pvcocp
Base domain	cdp.rdu2.scalelab.redhat.com
OpenShift version	4.17.12
CPU architecture	x86_64
Integrate with external partner platforms	No platform integration
Number of control plane nodes	3 (highly available cluster)
Hosts' network configuration	Static IP, bridges, and bonds
Configure via	YAML view

Networking stack type	IPv4
DNS	10.1.49.1
Machine network	10.1.48.0/23
Default gateway	10.1.49.254
api.pvcocp.cdp.rdu2.scalelab.redhat.com (API IP)	10.1.49.5
*.apps.pvcocp.cdp.rdu2.scalelab.redhat.com (Ingress IP)	10.1.49.6

STEP 1. Create a new OpenShift cluster using the OpenShift Assisted Installer from the Red Hat Hybrid Cloud Console:

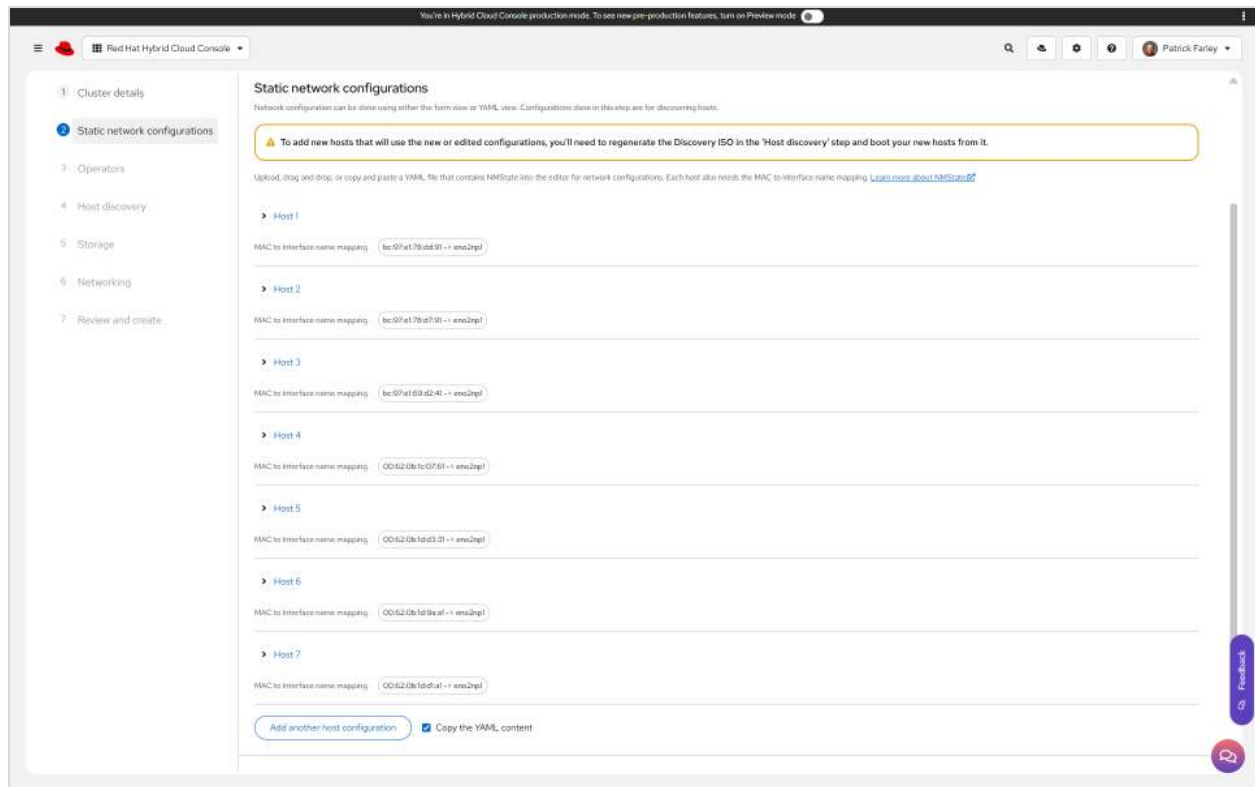
<https://console.redhat.com/openshift/assisted-installer/clusters/~new>

Enter the cluster details using the installation parameters provided, then click **Next**.

The screenshot displays the 'Cluster details' page in the Red Hat Hybrid Cloud Console. The page is titled 'Cluster details' and includes a sidebar with navigation links: 1. Cluster details, 2. Static network configuration, 3. Operators, 4. Host discovery, 5. Storage, 6. Networking, and 7. Review and create. The main content area contains the following fields and options:

- Cluster name:** pvcocp
- Base domain:** cdp.rdu2.scalelab.redhat.com
- OpenShift version:** OpenShift 4.17.42
- CPU architecture:** x86_64
- Integrate with external partner platforms:** No platform integration
- Number of control plane nodes:** 3
- Include custom manifests:** ☐ (Additional manifests will be applied at the install time for advanced configuration of the cluster)
- Hosts' network configuration:** ☒ DHCP only, ☐ Static IP, bridges, and bonds
- Encryption of installation disks:** ☒ Control plane nodes, ☒ Workers, ☒ Arbiter

At the bottom of the page, there is a 'Next' button and a 'Cancel' button. A 'View cluster events' link is also present in the bottom right corner.



R640

interfaces:

- ipv4:
 - address:
 - ip: 10.1.49.216 # Edit this IP address for each server
 - prefix-length: 23
 - dhcp: false
 - enabled: true
 - name: eno2np1
 - state: up
 - type: ethernet
 - mtu: 9000
- ipv4:
 - dhcp: false
 - enabled: false
 - name: eno1np0
 - state: down
 - type: ethernet

routes:

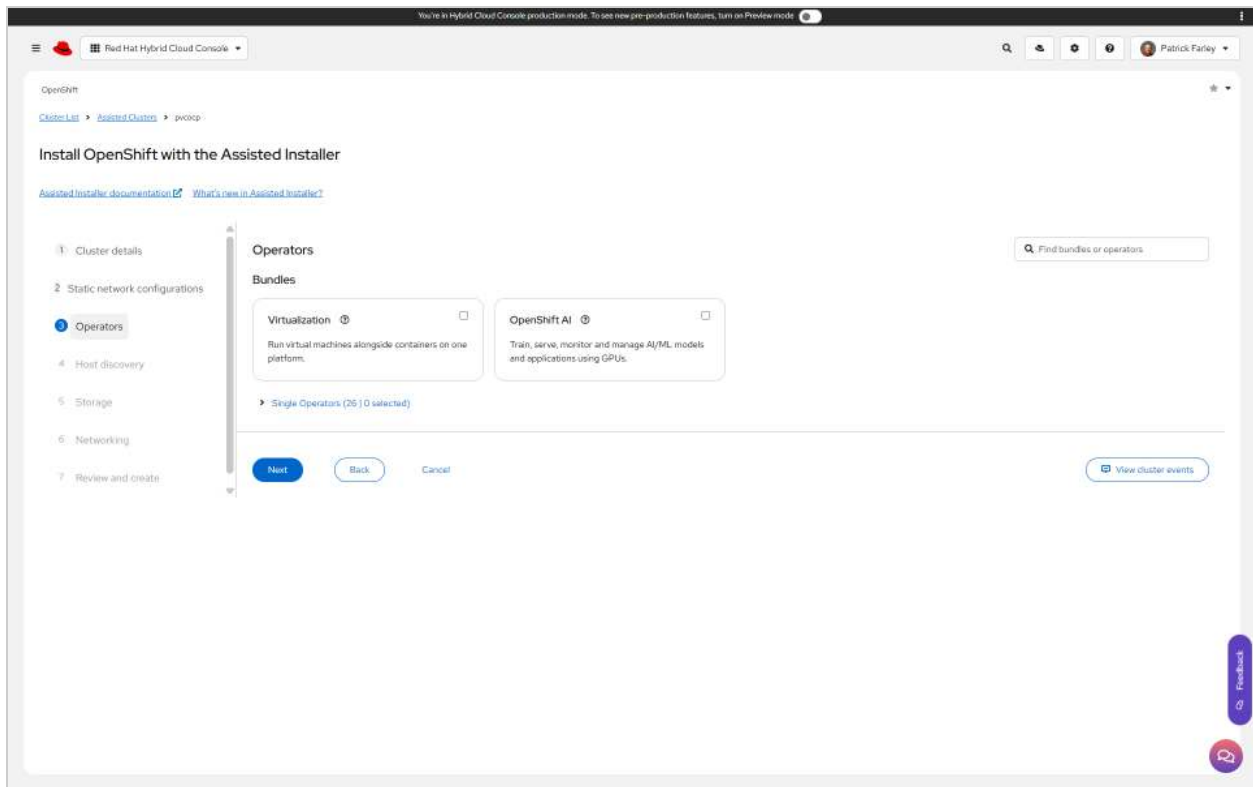
- config:
 - destination: 0.0.0.0/0
 - next-hop-address: 10.1.49.254
 - next-hop-interface: eno2np1
 - table-id: 254

dns-resolver:

```
config:
  server:
    - 10.1.49.1
  search:
    - cdp.rdu2.scalelab.redhat.com
```

R750

```
interfaces:
- ipv4:
  address:
    - ip: 10.1.49.216 # Edit this IP address for each server
    prefix-length: 23
  dhcp: false
  enabled: true
  name: eno12409np1
  state: up
  type: ethernet
  mtu: 9000
- ipv4:
  dhcp: false
  enabled: false
  name: eno12399np0
  state: down
  type: ethernet
routes:
  config:
    - destination: 0.0.0.0/0
      next-hop-address: 10.1.49.254
      next-hop-interface: eno2np1
      table-id: 254
dns-resolver:
  config:
    server:
      - 10.1.49.1
    search:
      - cdp.rdu2.scalelab.redhat.com
```



DISCOVERY ISO IMAGE

```
dnf install -y httpd
```

```
wget -O discovery_image_pvcocp.iso
```

```
'https://api.openshift.com/api/assisted-images/bytoken/eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3NjIzNjM2MDAsInN1YiI6IjY5YTEyMzk2LTAyMTQtNDkxZi1hMWM4LWJjODUzZjlmYzY0NiJ9.3s-YXD5-BpewK6MfWt3x9mR1wVaw7pPmJ0kWWisv6Cc/4.17/x86_64/full.iso'
```

```
cp discovery_image_pvcocp.iso /var/www/html
```

Verify that the discovery image is available at:

```
http://10.1.49.1/discovery_image_pvcocp.iso
```

BADFISH

Configure the servers to boot from the Discovery image:

```
export USER=quads
```

```
export PASS=rdu2@4618
```



```
dnf install -y podman
```

```
vi masterlist
```

```
10.6.61.148
```

```
10.6.61.152
```

```
10.6.61.153
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-off;done
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --mount-virtual-media  
http://10.1.49.1/discovery_image_pvcocp.iso;done
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-on;done
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-state;done
```

```
Get Current Boot mode setting
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --get-bios-attribute --attribute BootMode;done
```

```
Set UEFI mode
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value  
Uefi;done
```

```
Set BIOS mode
```

```
for HOST in `cat ~/masterlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value  
Bios;done
```

```
vi workerlist
```

```
10.1.37.21
```

```
10.1.64.196
10.1.65.226
10.1.65.96
10.1.65.139
10.1.64.17
10.1.65.3
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-off;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --mount-virtual-media
http://10.1.49.1/discovery_image_pvcocp.iso;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-on;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --power-state;done
```

Get Current Boot mode setting

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --get-bios-attribute --attribute BootMode;done
```

Set UEFI mode

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Uefi;done
```

Set BIOS mode

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish
-H $HOST -u $USER -p $PASS --set-bios-attribute --attribute BootMode --value
Bios;done
```

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

Red Hat Hybrid Cloud Console

Patrick Farley

1 Cluster details
2 Static network configurations
3 Operators
4 Host discovery
5 Storage
6 Networking
7 Review and create

Host discovery

[Add hosts](#)

Run workloads on control plane nodes

Information & Troubleshooting
[Minimum hardware requirements](#) [Hosts not showing up?](#)

Host Inventory

0 selected Actions

Hostname	Role	Status	Discovered on	CPUs	Memory	Total	...
q29-h18-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	04/11/2025, 22:51:43	112	512.00 GB	45.77 TB	
q29-h13-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	04/11/2025, 22:52:16	112	512.00 GB	45.77 TB	
q30-h05-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	04/11/2025, 22:52:15	112	512.00 GB	45.77 TB	
q30-h07-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	04/11/2025, 22:51:19	112	512.00 GB	45.77 TB	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	04/11/2025, 22:32:26	80	384.00 GB	958.58 GB	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	04/11/2025, 22:32:49	80	384.00 GB	958.58 GB	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	04/11/2025, 22:33:03	80	384.00 GB	958.58 GB	

Next Back Cancel

[View cluster events](#)

[Feedback](#)

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

Red Hat Hybrid Cloud Console

Patrick Farley

OpenShift

[Cluster List](#) [Assisted Cluster](#) [pvcsp](#)

Install OpenShift with the Assisted Installer

[Assisted installer documentation](#) [What's new in Assisted Installer?](#)

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Storage

Hostname	Role	Status	Total storage	Number of disks	...
q29-h18-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	45.77 TB	28	
q29-h13-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	45.77 TB	28	
q30-h05-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	45.77 TB	28	
q30-h07-000-r750.cdu.rdu2.scalelab.redhat.com	Worker	Ready	45.77 TB	28	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	958.58 GB	4	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	958.58 GB	4	
f36-h18-000-r640.cdu.rdu2.scalelab.redhat.com	Control plane node	Ready	958.58 GB	4	

All bootable disks, except for read-only disks, will be formatted during installation. Make sure to back up any critical data before proceeding.

Next Back Cancel

[View cluster events](#)

[Feedback](#)

mtu

3/0

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Red Hat Hybrid Cloud Console

OpenShift

Cluster ListAssisted Clustersjwcoep

Install OpenShift with the Assisted Installer

[Assisted installer documentation](#) [What's new in Assisted installer?](#)

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Networking

Network Management

- Cluster-Managed Networking
- User-Managed Networking

Networking stack type

- IPv4
- Dual-stack

Machine network

- 10.148.0/23 (10.148.0 - 10.149.255)

API IP

- 10.149.5

Ingress IP

- 10.149.6

☐ Use advanced networking

- Configure advanced networking properties (e.g. CIDR ranges)

Host SSH Public Key for troubleshooting after installation

- ☒ Use the same host discovery SSH key

Host inventory

Hostname	Role	Status	Active	IPv4 add.	IPv6	MAC address	
q29-h8-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Ready Some validations failed	eno12409np1	10.149.219/23	-	00:62:0b:1c:07:61	
q29-h3-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Ready Some validations failed	eno12409np1	10.149.220/23	-	00:62:0b:1d:d3:31	
q30-h05-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Ready Some validations failed	eno12409np1	10.149.221/23	-	00:62:0b:1d:9a:a1	
q30-h07-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Ready Some validations failed	eno12409np1	10.149.222/23	-	00:62:0b:1d:d1:a1	
f36-h18-000-r540.cdp.rdu2.scalelab.redhat.com	Control plane node	Ready Some validations failed	eno2npl	10.149.216/23	-	bc:97:e1:78:de:9f	
f36-h18-000-r540.cdp.rdu2.scalelab.redhat.com	Control plane node	Ready Some validations failed	eno2npl	10.149.217/23	-	bc:97:e1:78:d7:9f	
f36-h18-000-r540.cdp.rdu2.scalelab.redhat.com	Control plane node	Ready Some validations failed	eno2npl	10.149.218/23	-	bc:97:e1:69:d2:4f	

Next

Back

Cancel

View cluster events

Feedback

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

mtu 0/0

Red Hat Hybrid Cloud Console

4 Host discovery

5 Storage

6 Networking

7 Review and create

Cluster details

Cluster address: pvcocp.cdp.rdu2.sclablab.redhat.com

OpenShift version: 4.17.42

CPU architecture: x86_64

Hosts' network configuration: Static IP

Host inventory

Hosts: 7

Total cores: 688

Total memory: 313 TiB

Total storage: 185.95 TiB

Networking

Networking management type: Cluster-managed networking

Stack type: IPv4

Machine network CIDR: 10.148.0/23

API IP: 10.149.5

Ingress IP: 10.149.6

Advanced networking settings

Cluster network CIDR: 10.128.0.0/14

Cluster network host prefix: 23

Service network CIDR: 172.30.0.0/16

Networking type: Open Virtual Network (OVN)

Install cluster Back Cancel View cluster events

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

mtu 0/0

Red Hat Hybrid Cloud Console

OpenShift

Cluster List Advanced Cluster pvcocp

Installation progress

Started on: 04/11/2025, 22:02:51

Preparing for installation: 0%

Control Planes: 0 control plane nodes installed

Workers: 0 workers installed

Installation: Pending

Abort installation Download kubecfg View cluster events Download installation logs

Host inventory (7)

Hostname	Role	Status	Discovered on	CPU Cores	Memory	Total disk
d29-h11-000-r750.cdp.rdu2.sclablab.redhat.com	Worker	Ready Some validations failed	04/11/2025, 22:51:43	112	512.00 GiB	45.77 TiB
d29-h13-000-r750.cdp.rdu2.sclablab.redhat.com	Worker	Ready Some validations failed	04/11/2025, 22:52:36	112	512.00 GiB	45.77 TiB
d30-h05-000-r750.cdp.rdu2.sclablab.redhat.com	Worker	Ready Some validations failed	04/11/2025, 22:52:36	112	512.00 GiB	45.77 TiB
d30-h07-000-r750.cdp.rdu2.sclablab.redhat.com	Worker	Ready Some validations failed	04/11/2025, 22:51:19	112	512.00 GiB	45.77 TiB
f36-h17-000-r640.cdp.rdu2.sclablab.redhat.com	Control plane node (bootstrap)	Ready Some validations failed	04/11/2025, 22:32:26	80	384.00 GiB	958.58 GiB
f36-h18-000-r640.cdp.rdu2.sclablab.redhat.com	Control plane node	Ready Some validations failed	04/11/2025, 22:32:49	80	384.00 GiB	958.58 GiB
f36-h19-000-r640.cdp.rdu2.sclablab.redhat.com	Control plane node	Ready Some validations failed	04/11/2025, 22:33:03	80	384.00 GiB	958.58 GiB

Cluster summary

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

mtu 3/0

OpenShift

Cluster List > Assisted Clusters > pvccop

pvccop

Installation progress

Started on 04/11/2025, 23:02:51

Installing 40%

Control Planes Installing 3 control plane nodes

Workers Installing 4 workers

Initialization Pending

[Abort installation](#)
[Download kubeconfig](#)
[View cluster events](#)
[Download installation logs](#)

Download and save your kubeconfig file in a safe place. This file will be automatically deleted from Assisted Installer's service in 30 days.

Host Inventory (7)

Hostname	Role	Status	Discovered on	CPU Core	Memory	Total disk
d29-n13-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installing 4/2	04/11/2025, 22:51:43	102	512.00 GiB	45.77 TB
d29-n13-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installing 4/2	04/11/2025, 22:52:16	102	512.00 GiB	45.77 TB
d30-n05-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installing 4/2	04/11/2025, 22:52:15	102	512.00 GiB	45.77 TB
d30-n07-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installing 4/2	04/11/2025, 22:51:19	102	512.00 GiB	45.77 TB
f36-n17-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node (bootstrap)	Installing 2/2	04/11/2025, 22:32:26	80	384.00 GiB	958.58 GB
f36-n18-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node	Installing 2/2	04/11/2025, 22:32:49	80	384.00 GiB	958.58 GB
f36-n19-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node	Installing 2/2	04/11/2025, 22:33:03	80	384.00 GiB	958.58 GB

Cluster summary

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode.

Patrick Farley

pvccop

Installation progress

Started on 05/11/2025, 10:31:54

Installed on 05/11/2025, 11:09:05

Control Planes 3 control plane nodes installed

Workers 4 workers installed

Initialization Completed

Installation completed successfully

[Launch OpenShift Console](#)
[Download kubeconfig](#)
[View cluster events](#)
[Download installation logs](#)

Web Console URL
<https://console-openshift-console.apps.pvccop.cdp.rdu2.scalelab.redhat.com>

Not able to access the Web Console?

Username
 kubeadmin

Password

Download and save your kubeconfig file in a safe place. This file will be automatically deleted from Assisted Installer's service in 30 days.

Add new hosts by generating a new Discovery ISO under your cluster's "Add hosts" tab on scalelab.redhat.com/openshift.

Hostname	Role	Status	Discovered on	CPU Cores	Memory	Total stor...
d29-h11-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installed	05/11/2025, 10:15:19	112	512.00 GB	45.29 TB
d29-h13-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installed	05/11/2025, 10:15:56	112	512.00 GB	45.29 TB
d30-h05-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installed	05/11/2025, 10:09:18	112	512.00 GB	45.29 TB
d30-h07-000-r750.cdp.rdu2.scalelab.redhat.com	Worker	Installed	05/11/2025, 10:09:34	112	512.00 GB	45.29 TB
f36-h17-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node (bootstrap)	Installed	05/11/2025, 10:19:50	80	384.00 GB	479.56 GB
f36-h18-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node	Installed	05/11/2025, 10:08:26	80	384.00 GB	479.56 GB
f36-h19-000-r640.cdp.rdu2.scalelab.redhat.com	Control plane node	Installed	05/11/2025, 10:26:14	80	384.00 GB	479.56 GB

Cluster summary

[Back to all clusters](#)

<https://console.redhat.com/openshift/assisted-installer/clusters/415eb0de-ca80-47d8-a96c-151f60f58d20>

Create OpenShift cluster with the Agent Based Installer (ABI)

The Agent Based Installer is used to create an OpenShift cluster for CDP Data Services. This method of OpenShift installation is used in place of the Assisted Installer.

- [Chapter 1. Preparing to install with the Agent-based Installer](#)
- [4.2. Installing OpenShift Container Platform with the Agent-based Installer](#)
- [4.2.1. Downloading the Agent-based Installer](#)
- [4.2.3. Creating the preferred configuration inputs](#)

STEP 1: Navigate to the [Red Hat Hybrid Cloud Console](#) using your login credentials.

STEP 2: Click **Download Installer** to download and extract the install program.

For this installation, we download the OpenShift installer for the latest stable release of **4.17**. Using this version of the installer will install the same version of OpenShift.

Download and extract the OpenShift installer using the following commands:


```
[root@ipaserver ~]# wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable-4.17/
openshift-install-linux.tar.gz
[root@ipaserver ~]# tar xvf openshift-install-linux.tar.gz
[root@ipaserver ~]# chmod +x openshift-install
[root@ipaserver ~]# mv openshift-install /usr/local/bin/openshift-install
```

STEP 3: Download or copy the pull secret by clicking on *Download pull secret* or *Copy pull secret*. This will be used when the install-config.yaml file is created.

STEP 4: Click Download command-line tools and place the openshift-install binary in a directory that is on your PATH.

```
[root@ipaserver ~]# wget
https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable/openshift-client-li
nux.tar.gz
[root@ipaserver ~]# tar xvf openshift-client-linux.tar.gz
[root@ipaserver ~]# chmod +x oc
[root@ipaserver ~]# mv oc /usr/local/sbin/oc
```

STEP 5: Install the nmstate dependency by running the following command:

```
[root@ipaserver ~]# dnf install /usr/bin/nmstatectl -y
```

install-config.yaml

STEP 6: Create the install-config.yaml file

```
[root@ipaserver ~]# cat > install-config.yaml << EOF
```

```
apiVersion: v1
baseDomain: rdu2.scalelab.redhat.com
compute:
- architecture: amd64
  name: worker
  replicas: 13
controlPlane:
  architecture: amd64
  name: master
  replicas: 3
metadata:
  name: cdp
```



```
networking:
  clusterNetwork:
    - cidr: 10.128.0.0/14
      hostPrefix: 23
  machineNetwork:
    - cidr: 10.1.48.0/23
  networkType: OVNKubernetes
  serviceNetwork:
    - 172.30.0.0/16
platform:
  baremetal:
    hosts:
      - name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:00
      - name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:01
      - name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
        role: master
        bootMACAddress: 02:00:e4:00:03:02
      - name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:03
      - name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:04
      - name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:05
      - name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:06
      - name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:07
      - name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:08
      - name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:09
      - name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
        role: worker
        bootMACAddress: 02:00:e4:00:03:0a
      - name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
```

```
role: worker
bootMACAddress: 02:00:e4:00:03:0b
- name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
  role: worker
  bootMACAddress: 02:00:e4:00:03:0c
- name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
  role: worker
  bootMACAddress: 02:00:e4:00:03:0d
- name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
  role: worker
  bootMACAddress: 02:00:e4:00:03:0e
- name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
  role: worker
  bootMACAddress: 02:00:e4:00:03:0f
apiVIP: "10.1.49.3"
ingressVIP: "10.1.49.4"
```

pullSecret:

```
'{"auths":{"cloud.openshift.com":{"auth":"b3B1bnNoawZ0LXJlbGVhc2UtZGV2K29jbV9hY2Nlc3NmZU4NDUyN2M1MWNiNDYzMd1jNDJhYjI0NTcxZDhi0TE6V0YzRFZOSV1VS1BET0RKT0JISVNWSkJTIV0t0WVNYV1JBOTVRSEo3NU5DV1JMS0s4RldJRE5GQ09TVldET1VYUQ==","email":"pfarley@redhat.com"},"quay.io":{"auth":"b3B1bnNoawZ0LXJlbGVhc2UtZGV2K29jbV9hY2Nlc3NmZU4NDUyN2M1MWNiNDYzMd1jNDJhYjI0NTcxZDhi0TE6V0YzRFZOSV1VS1BET0RKT0JISVNWSkJTIV0t0WVNYV1JBOTVRSEo3NU5DV1JMS0s4RldJRE5GQ09TVldET1VYUQ==","email":"pfarley@redhat.com"},"registry.connect.redhat.com":{"auth":"fHVoYy1wb29sLTcxZWm2OWMwLTBiMDAtNDg5Yi1hOTYyLTE4NzExNTkyOTA2NjpleUpoYkdjaU9pSlNve1V4TW1KOS5leUp6ZFdJaU9pSm1PVEU0WxpjNU1ESX1Na1kwTTJ0a09HTTROemMzWXPsa1lqTmhaRGhoTn1KOS5DZjZCdU1tdVN2aldjUHVdZmlxVmwxawRTSUJhSlZ3V0JIdnRfU05STEFhRW9MeXFtUloxSW1ROG1laE9KaXhZSnoybn10dkVza0Mwak1UTTY5YzA1QVNBR0puQTdsTX1HZUhpVhUdzZTUJDRFdBtGZWYjhTOFNyOXQtZFP5Z11ndm84WHFBnk5ubEl1RWl3QkxuZWZRaZW12bGVud2VjdXBuYTNvM2U0THJPM31fV2J0MzVuR1NhZ0Y5Sk1mRGUtR1ZsbGNHa3p1WlFiexJQZldyZmZjVkf1V3dqZUpXLUpvdkdKZ25LaJB1V2ZqMnFTY19sMW1WU31HeGVYVV96S2tmaUF6dmFuY0NsZ09wT01TRS1BVmx60FprRDVsNFBGZzNzdmU2S1FhbXVaLTJaeDM3bHJUajF0WEJVOHB2dEZYS2V3QzZcZn2k2LWJkeEZTWBvT3JlS21ydFdpT3lpTm1DekR2YU11X1BjSkx4cXNFUEJRZUdPeWd4aDVORHNjRVB5QmtZYTdlOHYtdVFxV3VldzNYSXo5d28yZjhsSlc1cGVPamJzZi1jcWxmX191NUZ4UWI5M24zOWw1X291Ykh1NHFPcW9IM2U2Z2FCWERhZmx1UE1kYzRENkU2MDNHZkxXY1NQcFFMNVE1QTFaa3Nhc183NU11ejVkyU5UM1padTVoOER3YVY2dTQ2TGszNU90blkybzRoUWhXYkxPUmptalo5LVplVnN2STkxaXIweVks5YkdyZ3MyQ1ZUN252MFowU3pNdEhpR0o0aWpzYXpjY3dMbVUtM2poSudBaXREaEN2eEjiY0plUGVWaENwZ3dGTkstMfdZT0V1cElHN1RmSjE5WHRlV1FvBDNzaTlQVlg5Q2RHdU93RHViV1NMQXJ5RXAtTVlZdw==","email":"pfarley@redhat.com"},"registry.redhat.io":{"auth":"fHVoYy1wb29sLTcxZWm2OWMwLTBiMDAtNDg5Yi1hOTYyLTE4NzExNTkyOTA2NjpleUpoYkdjaU9pSlNve1V4TW1KOS5leUp6ZFdJaU9pSm1PVEU0WxpjNU1ESX1Na1kwTTJ0a09HTTROemMzWXPsa1lqTmhaRGhoTn1KOS5DZjZCdU1tdVN2aldjUHVdZmlxVmwxawRTSUJhSlZ3V0JIdnRfU05STEFhRW9MeXFtUloxSW1ROG1laE9KaXhZSnoybn10dkVza0Mwak1UTTY5YzA1QVNBR0puQTdsTX1HZUhpVhUdzZTUJDRFdBtGZWYjhTOFNyOXQtZFP5Z11ndm84WHFBnk5ubEl1RWl3QkxuZWZRaZW12bGVud2VjdXBuYTNvM2U0THJPM31fV2J0MzVuR1NhZ0Y5Sk1mRGUtR1ZsbGNHa3p1WlFiexJQZldyZmZjVkf1V3dqZUpXLUpvdkdKZ25LaJB1V2ZqMnFTY19sMW1WU31HeGVYVV96S2tmaUF6dmFuY0NsZ09wT01TRS1BVmx60FprRDVsNFBGZzNzdmU2S1FhbXVaLTJaeDM
```

```
3bHJUajF0WEJVOHB2dEZYSGV3QzczN2k2LWJkeEZTWTBvT3JlS21ydFdpT3lpTm1DekR2YU11X1Bj
Skx4cXNFUEJRZUdPeWd4aDVORHNjRVB5QmtZYTd1OHYtdVFXV3VldzNYSXo5d28yZjhsSlc1cGVPa
mJzZi1jcWxmX19INUZ4UWI5M24zOWw1X291Ykh1NHFPcW9IM2U2Z2FCWERhZmx1UE1kYzRENKU2MD
NHZkxXY1NQcFFMVE1QTFaa3Nhc183NU11ejVkJYU5UM1padTVo0ER3YVY2dTQ2TGszNU90b1kybzR
oUWhXYkxPUmptaLo5LVp1VnN2STkxaXIweVk5YkdyZ3MyQlZUN252MFowU3pNdEhpR0o0aWpzYXpj
Y3dMbVUtM2poSudBaXREaEN2eEJiY0plUGVWaENwZ3dGTkstMFdZT0V1cElHN1RmSjE5WHR1V1FVb
DNzaTlQVlg5Q2RHdU93RHViV1NMQXJ5RXAtTVlZdw==", "email": "pfarley@redhat.com"}}}'
sshKey: |
  ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQGC5f46fvSfwLSXx0zCC52qFa21AS0kvJoolggoGL+CUWGdq6
Kf4ExWCJ2d1iDkAcPt6y7jGshujhfUR6DE1kuFpozTOgqBjwZl7ICtA5lmkxh7sxHuYYNFAZNli6D
38IbdMSfth23I2ypBpcQA0WweCzfPnjsjaniLG5EfTWw3K75QQOMxLvE4W0uKbA/EKg9F8Eq2ImKL
HZekB8nCFoi2mVgwEYGTSM5D5zAz56tuSbuppbkXfgf+4FFb/LFp3TJyCPQvu396ZRbEF64EWvbMC
01aAYNM3NpRqSqsGWWJoTnRzXSRKb096tOxWmYTyRiF0s/nuRA1Sb3kU0EN4qC5YQs9wobpnLQ5T
3Um8QP4pQxWhWIHEAW3SWM01IfeBjT8TPS80CG3EFLRVxJFdNZc/IfD3CGnevW8ovlSek//HfspwM
MMauXQfY570fXhn9BxVtnC6ymGLDBv9hoa82qtvigfkmU1u0xD9Ep7r114vprcQ5PF18yp09DPsKj
4Etc= root@ipaserver.cdp.rdu2.scalelab.redhat.com
```

EOF

agent-config.yaml

STEP 7: Create the agent-config.yaml file

Hostname	CPU	RAM	Disk	MAC Address	IP Address
pvcocp-master01.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:00	10.1.49.200/23
pvcocp-master02.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:01	10.1.49.201/23
pvcocp-master03.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:02	10.1.49.202/23
pvcocp-worker01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:03	10.1.49.203/23
pvcocp-worker02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:04	10.1.49.204/23
pvcocp-worker03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:05	10.1.49.205/23
pvcocp-worker04.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:06	10.1.49.206/23
pvcocp-worker05.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:07	10.1.49.207/23
pvcocp-worker06.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:08	10.1.49.208/23
pvcocp-worker07.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:09	10.1.49.209/23
pvcocp-worker08.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0a	10.1.49.210/23
pvcocp-worker09.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0b	10.1.49.211/23

pvcocp-worker10.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0c	10.1.49.212/23
pvcocp-infra01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0d	10.1.49.213/23
pvcocp-infra02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0e	10.1.49.214/23
pvcocp-infra03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0f	10.1.49.215/23

<domain_name> for the RH scale lab will be: cdp.rdu2.scalelab.redhat.com

```
[root@ipaserver ~]# cat > agent-config.yaml << EOF
```

```
apiVersion: v1alpha1
kind: AgentConfig
metadata:
  name: cdp
rendezvousIP: 10.1.49.200
hosts:
- hostname: master01.cdp.rdu2.scalelab.redhat.com
  interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:00 # Change for each host
  networkConfig:
    interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:00 # Change for each host
      ipv4:
        enabled: true
        address:
        - ip: 10.1.49.200 # Change for each host
          prefix-length: 23
        dhcp: false
    dns-resolver:
      config:
        server:
        - 10.1.49.1
    routes:
      config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
```

```

        table-id: 254
- hostname: master02.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:01 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:01 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.201 # Change for each host
            prefix-length: 23
        dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
- hostname: master03.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:02 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:02 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.202 # Change for each host
            prefix-length: 23
        dhcp: false
      dns-resolver:

```

```

    config:
      server:
        - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker01.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:03 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:03 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.203 # Change for each host
            prefix-length: 23
        dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
        table-id: 254
- hostname: worker02.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:04 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up

```

```

    mac-address: 02:00:e4:00:03:04 # Change for each host
    ipv4:
      enabled: true
      address:
        - ip: 10.1.49.204 # Change for each host
          prefix-length: 23
      dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
  - hostname: worker03.cdp.rdu2.scalelab.redhat.com
    interfaces:
      - name: ens192
        macAddress: 02:00:e4:00:03:05 # Change for each host
    networkConfig:
      interfaces:
        - name: ens192
          type: ethernet
          mtu: 9000
          state: up
          mac-address: 02:00:e4:00:03:05 # Change for each host
          ipv4:
            enabled: true
            address:
              - ip: 10.1.49.205 # Change for each host
                prefix-length: 23
            dhcp: false
          dns-resolver:
            config:
              server:
                - 10.1.49.1
          routes:
            config:
              - destination: 0.0.0.0/0
                next-hop-address: 10.1.49.254
                next-hop-interface: ens192
                table-id: 254
  - hostname: worker04.cdp.rdu2.scalelab.redhat.com
    interfaces:

```

```

- name: ens192
  macAddress: 02:00:e4:00:03:06 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:06 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.206 # Change for each host
            prefix-length: 23
        dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0
            next-hop-address: 10.1.49.254
            next-hop-interface: ens192
            table-id: 254
- hostname: worker05.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:07 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:07 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.207 # Change for each host
            prefix-length: 23
        dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1

```



```

    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker06.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:08 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:08 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.208 # Change for each host
            prefix-length: 23
        dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
        table-id: 254
- hostname: worker07.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:09 # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:09 # Change for each host
      ipv4:
        enabled: true

```

```

    address:
      - ip: 10.1.49.209 # Change for each host
        prefix-length: 23
    dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
        table-id: 254
- hostname: worker08.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0a # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0a # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.210 # Change for each host
              prefix-length: 23
          dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker09.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0b # Change for each host
  networkConfig:

```

```

interfaces:
  - name: ens192
    type: ethernet
    mtu: 9000
    state: up
    mac-address: 02:00:e4:00:03:0b # Change for each host
    ipv4:
      enabled: true
      address:
        - ip: 10.1.49.211 # Change for each host
          prefix-length: 23
      dhcp: false
    dns-resolver:
      config:
        server:
          - 10.1.49.1
    routes:
      config:
        - destination: 0.0.0.0/0
          next-hop-address: 10.1.49.254
          next-hop-interface: ens192
          table-id: 254
- hostname: worker10.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:0c # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:0c # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.212 # Change for each host
            prefix-length: 23
        dhcp: false
      dns-resolver:
        config:
          server:
            - 10.1.49.1
      routes:
        config:
          - destination: 0.0.0.0/0

```

```

        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
        table-id: 254
- hostname: infra01.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0d # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0d # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.213 # Change for each host
              prefix-length: 23
          dhcp: false
        dns-resolver:
          config:
            server:
              - 10.1.49.1
        routes:
          config:
            - destination: 0.0.0.0/0
              next-hop-address: 10.1.49.254
              next-hop-interface: ens192
              table-id: 254
- hostname: infra02.cdp.rdu2.scalelab.redhat.com
  interfaces:
    - name: ens192
      macAddress: 02:00:e4:00:03:0e # Change for each host
  networkConfig:
    interfaces:
      - name: ens192
        type: ethernet
        mtu: 9000
        state: up
        mac-address: 02:00:e4:00:03:0e # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.49.214 # Change for each host
              prefix-length: 23

```

```

        dhcp: false
    dns-resolver:
        config:
            server:
                - 10.1.49.1
    routes:
        config:
            - destination: 0.0.0.0/0
              next-hop-address: 10.1.49.254
              next-hop-interface: ens192
              table-id: 254
- hostname: infra03.cdp.rdu2.scalelab.redhat.com
interfaces:
  - name: ens192
    macAddress: 02:00:e4:00:03:0f # Change for each host
networkConfig:
  interfaces:
    - name: ens192
      type: ethernet
      mtu: 9000
      state: up
      mac-address: 02:00:e4:00:03:0f # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.49.215 # Change for each host
            prefix-length: 23
        dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: ens192
        table-id: 254

```

EOF

STEP 8: Create Install Dir

```
[root@ipaserver ~]# mkdir ocp-install-dir
```

STEP 9: Create backup copies of the yaml manifests

```
[root@ipaserver ~]# cp agent-config.yaml ocp-install-dir/agent-config.yaml
[root@ipaserver ~]# cp install-config.yaml
ocp-install-dir/install-config.yaml
```

STEP 10: Create ISO agent image

```
[root@ipaserver ~]# openshift-install --dir ocp-install-dir agent create
image
```

INFO Configuration has 3 master replicas and 13 worker replicas

WARNING hosts from install-config.yaml are ignored

INFO The rendezvous host IP (node0 IP) is 10.1.49.200

INFO Extracting base ISO from release payload

INFO Verifying cached file

INFO Using cached Base ISO /root/.cache/agent/image_cache/coreos-x86_64.iso

INFO Consuming Install Config from target directory

INFO Consuming Agent Config from target directory

INFO Generated ISO at ocp-install-dir/agent.x86_64.iso.

STEP 11: Upload the Discovery ISO image to a PVC using the virtctl utility:

```
[root@ipaserver ~]# oc project cdppvcds
```

```
[root@ipaserver ~]# virtctl image-upload dv discovery-image-cdp --size=2Gi
--image-path=ocp-install-dir/agent.x86_64.iso --insecure
--storage-class=ocs-storagecluster-ceph-rbd
```

PVC cdp-ds/discovery-image-cdp not found

DataVolume cdp-ds/discovery-image-cdp created

Waiting for PVC discovery-image-cdp upload pod to be ready...

Pod now ready

Uploading data to https://cdi-uploadproxy-openshift-cnv.apps.vlan601.rdu2.scalelab.redhat.com

1.24 GiB / 1.24 GiB

[-----]

-----] 100.00% 161.43 MiB p/s

8.1s

Uploading data completed successfully, waiting for processing to complete, you can hit ctrl-c without interrupting the progress

Processing completed successfully

Uploading ocp-install-dir/agent.x86_64.iso completed successfully

Create VMs for OpenShift Cluster (OpenShift Virtualization)

The following table is a reference of the VirtualMachines that will be created for the OpenShift cluster for CDP Data Services:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
pvcocp-master01.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:00	10.1.49.200/23
pvcocp-master02.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:01	10.1.49.201/23
pvcocp-master03.<domain_name>	8	16 GiB	120 GiB	02:00:e4:00:03:02	10.1.49.202/23
pvcocp-worker01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:03	10.1.49.203/23
pvcocp-worker02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:04	10.1.49.204/23
pvcocp-worker03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:05	10.1.49.205/23
pvcocp-worker04.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:06	10.1.49.206/23
pvcocp-worker05.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:07	10.1.49.207/23
pvcocp-worker06.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:08	10.1.49.208/23
pvcocp-worker07.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:09	10.1.49.209/23
pvcocp-worker08.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0a	10.1.49.210/23
pvcocp-worker09.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0b	10.1.49.211/23
pvcocp-worker10.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0c	10.1.49.212/23
pvcocp-infra01.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0d	10.1.49.213/23
pvcocp-infra02.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0e	10.1.49.214/23
pvcocp-infra03.<domain_name>	32	64 GiB	120 GiB	02:00:e4:00:03:0f	10.1.49.215/23

<domain_name> for the RH scale lab will be: `cdp.rdu2.scalelab.redhat.com`

STEP 13: Create the VM's for the OpenShift cluster using the following VM YAML files:

```
[root@bastion ~]# oc new-project cdp-ds
[root@bastion ~]# oc create -f <nodename>-vm.yaml
```

```
for i in {100..199}; do oc create -f <nodename>-vm.yaml; done
```

pvcocp-master01.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
```



```
vm.kubevirt.io/flavor: small
vm.kubevirt.io/os: rhel9
vm.kubevirt.io/workload: server
creationTimestamp: null
labels:
  kubevirt.io/domain: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
  kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 8
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
        - bootOrder: 2
          cdrom:
            bus: sata
            name: installation-cdrom
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:00'
          name: eth0
      rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 16Gi
    resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
```

```
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
  - dataVolume:
      name: pvcocp-master01.cdp.rdu2.scalelab.redhat.com-cdrom
      name: installation-cdrom
```

pvcocp-master02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
```

```
    storage:
      resources:
        requests:
          storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
              name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:01'
            name: eth0
          rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
```

```

    type: pc-q35-rhel9.4.0
    memory:
      guest: 16Gi
    resources: {}
    networks:
      - multus:
          networkName: default/localnet-network
          name: eth0
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
      - dataVolume:
          name: pvcocp-master02.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom

```

pvcocp-master03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:

```

```
        storage: 120Gi
- metadata:
  name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com-cdrom
  spec:
    source:
      pvc:
        name: discovery-image-cdp
        namespace: cdppvcds
    storage:
      resources:
        requests:
          storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 8
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
              name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:02'
            name: eth0
          rng: {}
```

```

features:
  acpi: {}
  smm:
    enabled: true
firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.4.0
memory:
  guest: 16Gi
resources: {}
networks:
- multus:
    networkName: default/localnet-network
    name: eth0
terminationGracePeriodSeconds: 180
volumes:
- dataVolume:
    name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com
    name: rootdisk
- dataVolume:
    name: pvcocp-master03.cdp.rdu2.scalelab.redhat.com-cdrom
    name: installation-cdrom

```

pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
  - apiVersion: cdi.kubevirt.io/v1beta1
    kind: DataVolume
    metadata:
      annotations:
        cdi.kubevirt.io/storage.bind.immediate.requested: 'true'

```

```
creationTimestamp: null
name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
spec:
  source:
    blank: {}
  storage:
    resources:
      requests:
        storage: 120Gi
- metadata:
  name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-cdrom
spec:
  source:
    pvc:
      name: discovery-image-cdp
      namespace: cdppvcds
  storage:
    resources:
      requests:
        storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
  labels:
    kubevirt.io/domain: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
        - bootOrder: 2
          cdrom:
```

```

        bus: sata
        name: installation-cdrom
    interfaces:
      - bridge: {}
        model: virtio
        macAddress: '02:00:e4:00:03:03'
        name: eth0
    rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 64Gi
    resources: {}
    networks:
      - multus:
          networkName: default/localnet-network
          name: eth0
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
      - dataVolume:
          name: pvcocp-worker01.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom

```

pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com

```



```
namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
          sockets: 32
          threads: 1
```

```

devices:
  disks:
    - bootOrder: 1
      disk:
        bus: virtio
        name: rootdisk
    - bootOrder: 2
      cdrom:
        bus: sata
        name: installation-cdrom
  interfaces:
    - bridge: {}
      model: virtio
      macAddress: '02:00:e4:00:03:04'
      name: eth0
  rng: {}
features:
  acpi: {}
  smm:
    enabled: true
firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.4.0
memory:
  guest: 64Gi
resources: {}
networks:
  - multus:
      networkName: default/localnet-network
      name: eth0
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
  - dataVolume:
      name: pvcocp-worker02.cdp.rdu2.scalelab.redhat.com-cdrom
      name: installation-cdrom

```

pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:

```

```
labels:
  app: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
  kubevirt.io/dynamic-credentials-support: 'true'
  vm.kubevirt.io/template: rhel9-server-small
  vm.kubevirt.io/template.namespace: openshift
  vm.kubevirt.io/template.revision: '1'
  vm.kubevirt.io/template.version: v0.32.2
name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
```

```
kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
        - bootOrder: 2
          cdrom:
            bus: sata
            name: installation-cdrom
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:05'
          name: eth0
      rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 64Gi
    resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
    - dataVolume:
```

```
name: pvcocp-worker03.cdp.rdu2.scalelab.redhat.com-cdrom
name: installation-cdrom
```

pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
```

```
metadata:
  annotations:
    vm.kubevirt.io/flavor: small
    vm.kubevirt.io/os: rhel9
    vm.kubevirt.io/workload: server
  creationTimestamp: null
  labels:
    kubevirt.io/domain: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
        - bootOrder: 2
          cdrom:
            bus: sata
            name: installation-cdrom
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:06'
          name: eth0
      rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 64Gi
    resources: {}
  networks:
    - multus:
```

```

    networkName: default/localnet-network
    name: eth0
    terminationGracePeriodSeconds: 180
    volumes:
      - dataVolume:
          name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com
          name: rootdisk
      - dataVolume:
          name: pvcocp-worker04.cdp.rdu2.scalelab.redhat.com-cdrom
          name: installation-cdrom

```

pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:

```

```
    name: discovery-image-cdp
    namespace: cdppvcds
  storage:
    resources:
      requests:
        storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
              name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:07'
            name: eth0
          rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
```



```

    efi: {}
  machine:
    type: pc-q35-rhel9.4.0
  memory:
    guest: 64Gi
  resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
    - dataVolume:
        name: pvcocp-worker05.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom

```

pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:

```

```
resources:
  requests:
    storage: 120Gi
- metadata:
  name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-cdrom
spec:
  source:
    pvc:
      name: discovery-image-cdp
      namespace: cdppvcds
  storage:
    resources:
      requests:
        storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
  labels:
    kubevirt.io/domain: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
              name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:08'
```

```

      name: eth0
    rng: {}
  features:
    acpi: {}
    smm:
      enabled: true
  firmware:
    bootloader:
      efi: {}
  machine:
    type: pc-q35-rhel9.4.0
  memory:
    guest: 64Gi
  resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
    - dataVolume:
        name: pvcocp-worker06.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom

```

pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:

```

```

    annotations:
      cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
    creationTimestamp: null
    name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
  spec:
    source:
      blank: {}
    storage:
      resources:
        requests:
          storage: 120Gi
- metadata:
  name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-cdrom
  spec:
    source:
      pvc:
        name: discovery-image-cdp
        namespace: cdppvcds
    storage:
      resources:
        requests:
          storage: 2Gi
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk

```

```

    - bootOrder: 2
      cdrom:
        bus: sata
        name: installation-cdrom
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:09'
          name: eth0
      rng: {}
      features:
        acpi: {}
        smm:
          enabled: true
      firmware:
        bootloader:
          efi: {}
      machine:
        type: pc-q35-rhel9.4.0
      memory:
        guest: 64Gi
      resources: {}
      networks:
        - multus:
            networkName: default/localnet-network
            name: eth0
      terminationGracePeriodSeconds: 180
      volumes:
        - dataVolume:
            name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com
            name: rootdisk
        - dataVolume:
            name: pvcocp-worker07.cdp.rdu2.scalelab.redhat.com-cdrom
            name: installation-cdrom

```

pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'

```

```
vm.kubevirt.io/template.version: v0.32.2
name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
namespace: cdppvcds
spec:
  dataVolumeTemplates:
  - apiVersion: cdi.kubevirt.io/v1beta1
    kind: DataVolume
    metadata:
      annotations:
        cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
      creationTimestamp: null
      name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
    spec:
      source:
        blank: {}
      storage:
        resources:
          requests:
            storage: 120Gi
  - metadata:
      name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-cdrom
    spec:
      source:
        pvc:
          name: discovery-image-cdp
          namespace: cdppvcds
      storage:
        resources:
          requests:
            storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
      labels:
        kubevirt.io/domain: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
        kubevirt.io/size: small
    spec:
      architecture: amd64
      domain:
        cpu:
          cores: 1
```

```

sockets: 32
threads: 1
devices:
  disks:
    - bootOrder: 1
      disk:
        bus: virtio
        name: rootdisk
    - bootOrder: 2
      cdrom:
        bus: sata
        name: installation-cdrom
  interfaces:
    - bridge: {}
      model: virtio
      macAddress: '02:00:e4:00:03:0a'
      name: eth0
  rng: {}
features:
  acpi: {}
  smm:
    enabled: true
firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.4.0
memory:
  guest: 64Gi
resources: {}
networks:
  - multus:
      networkName: default/localnet-network
      name: eth0
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
  - dataVolume:
      name: pvcocp-worker08.cdp.rdu2.scalelab.redhat.com-cdrom
      name: installation-cdrom

```

pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
```

```
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
  runStrategy: RerunOnFailure
  template:
    metadata:
      annotations:
        vm.kubevirt.io/flavor: small
        vm.kubevirt.io/os: rhel9
        vm.kubevirt.io/workload: server
      creationTimestamp: null
```



```
labels:
  kubevirt.io/domain: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
  kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
        - bootOrder: 2
          cdrom:
            bus: sata
            name: installation-cdrom
      interfaces:
        - bridge: {}
          model: virtio
          macAddress: '02:00:e4:00:03:0b'
          name: eth0
      rng: {}
    features:
      acpi: {}
      smm:
        enabled: true
    firmware:
      bootloader:
        efi: {}
    machine:
      type: pc-q35-rhel9.4.0
    memory:
      guest: 64Gi
    resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com
```

```
    name: rootdisk
  - dataVolume:
      name: pvcocp-worker09.cdp.rdu2.scalelab.redhat.com-cdrom
    name: installation-cdrom
```

pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-vms.yaml

```
apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
```

```
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
    labels:
      kubevirt.io/domain: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
      kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
      cpu:
        cores: 1
        sockets: 32
        threads: 1
      devices:
        disks:
          - bootOrder: 1
            disk:
              bus: virtio
              name: rootdisk
          - bootOrder: 2
            cdrom:
              bus: sata
              name: installation-cdrom
        interfaces:
          - bridge: {}
            model: virtio
            macAddress: '02:00:e4:00:03:0c'
            name: eth0
          rng: {}
        features:
          acpi: {}
          smm:
            enabled: true
        firmware:
          bootloader:
            efi: {}
        machine:
          type: pc-q35-rhel9.4.0
        memory:
          guest: 64Gi
        resources: {}
```

```

networks:
  - multus:
      networkName: default/localnet-network
      name: eth0
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
  - dataVolume:
      name: pvcocp-worker10.cdp.rdu2.scalelab.redhat.com-cdrom
      name: installation-cdrom

```

pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-cdrom

```

```
spec:
  source:
    pvc:
      name: discovery-image-cdp
      namespace: cdppvcds
    storage:
      resources:
        requests:
          storage: 2Gi
- metadata:
  name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-disk1
spec:
  source:
    blank: {}
  storage:
    accessModes:
      - ReadWriteMany
    resources:
      requests:
        storage: 2000Gi
    storageClassName: ocs-storagecluster-ceph-rbd
    volumeMode: Block
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
  labels:
    kubevirt.io/domain: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/size: small
spec:
  architecture: amd64
  domain:
    cpu:
      cores: 1
      sockets: 32
      threads: 1
    devices:
      disks:
        - bootOrder: 1
          disk:
            bus: virtio
            name: rootdisk
```

```

    - bootOrder: 2
      cdrom:
        bus: sata
        name: installation-cdrom
    - bootOrder: 3
      disk:
        bus: virtio
        name: disk-1
  interfaces:
    - bridge: {}
      model: virtio
      macAddress: '02:00:e4:00:03:0d'
      name: eth0
  rng: {}
  features:
    acpi: {}
    smm:
      enabled: true
  firmware:
    bootloader:
      efi: {}
  machine:
    type: pc-q35-rhel9.4.0
  memory:
    guest: 64Gi
  resources: {}
  networks:
    - multus:
        networkName: default/localnet-network
        name: eth0
  terminationGracePeriodSeconds: 180
  volumes:
    - dataVolume:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com
        name: rootdisk
    - dataVolume:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-cdrom
        name: installation-cdrom
    - dataVolume:
        name: pvcocp-infra01.cdp.rdu2.scalelab.redhat.com-disk1
        name: disk-1

```

pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine

```

```
metadata:
  labels:
    app: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
    - apiVersion: cdi.kubevirt.io/v1beta1
      kind: DataVolume
      metadata:
        annotations:
          cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
        creationTimestamp: null
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
      spec:
        source:
          blank: {}
        storage:
          resources:
            requests:
              storage: 120Gi
    - metadata:
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-cdrom
      spec:
        source:
          pvc:
            name: discovery-image-cdp
            namespace: cdppvcds
        storage:
          resources:
            requests:
              storage: 2Gi
    - metadata:
        name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-disk1
      spec:
        source:
          blank: {}
        storage:
          accessModes:
            - ReadWriteMany
          resources:
```

```
      requests:
        storage: 2000Gi
        storageClassName: ocs-storagecluster-ceph-rbd
        volumeMode: Block
    runStrategy: RerunOnFailure
    template:
      metadata:
        annotations:
          vm.kubevirt.io/flavor: small
          vm.kubevirt.io/os: rhel9
          vm.kubevirt.io/workload: server
        creationTimestamp: null
        labels:
          kubevirt.io/domain: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
          kubevirt.io/size: small
      spec:
        architecture: amd64
        domain:
          cpu:
            cores: 1
            sockets: 32
            threads: 1
          devices:
            disks:
              - bootOrder: 1
                disk:
                  bus: virtio
                  name: rootdisk
              - bootOrder: 2
                cdrom:
                  bus: sata
                  name: installation-cdrom
              - bootOrder: 3
                disk:
                  bus: virtio
                  name: disk-1
            interfaces:
              - bridge: {}
                model: virtio
                macAddress: '02:00:e4:00:03:0e'
                name: eth0
            rng: {}
          features:
            acpi: {}
            smm:
              enabled: true
```



```

firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.4.0
memory:
  guest: 64Gi
resources: {}
networks:
- multus:
    networkName: default/localnet-network
    name: eth0
terminationGracePeriodSeconds: 180
volumes:
- dataVolume:
    name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com
    name: rootdisk
- dataVolume:
    name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-cdrom
    name: installation-cdrom
- dataVolume:
    name: pvcocp-infra02.cdp.rdu2.scalelab.redhat.com-disk1
    name: disk-1

```

pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-vms.yaml

```

apiVersion: kubevirt.io/v1
kind: VirtualMachine
metadata:
  labels:
    app: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/dynamic-credentials-support: 'true'
    vm.kubevirt.io/template: rhel9-server-small
    vm.kubevirt.io/template.namespace: openshift
    vm.kubevirt.io/template.revision: '1'
    vm.kubevirt.io/template.version: v0.32.2
  name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
  namespace: cdppvcds
spec:
  dataVolumeTemplates:
  - apiVersion: cdi.kubevirt.io/v1beta1
    kind: DataVolume
    metadata:
      annotations:
        cdi.kubevirt.io/storage.bind.immediate.requested: 'true'
      creationTimestamp: null

```

```
name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
spec:
  source:
    blank: {}
  storage:
    resources:
      requests:
        storage: 120Gi
- metadata:
  name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-cdrom
  spec:
    source:
      pvc:
        name: discovery-image-cdp
        namespace: cdppvcfs
    storage:
      resources:
        requests:
          storage: 2Gi
- metadata:
  name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-disk1
  spec:
    source:
      blank: {}
    storage:
      accessModes:
        - ReadWriteMany
      resources:
        requests:
          storage: 2000Gi
      storageClassName: ocs-storagecluster-ceph-rbd
      volumeMode: Block
runStrategy: RerunOnFailure
template:
  metadata:
    annotations:
      vm.kubevirt.io/flavor: small
      vm.kubevirt.io/os: rhel9
      vm.kubevirt.io/workload: server
    creationTimestamp: null
  labels:
    kubevirt.io/domain: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
    kubevirt.io/size: small
  spec:
    architecture: amd64
    domain:
```

```
cpu:
  cores: 1
  sockets: 32
  threads: 1
devices:
  disks:
    - bootOrder: 1
      disk:
        bus: virtio
        name: rootdisk
    - bootOrder: 2
      cdrom:
        bus: sata
        name: installation-cdrom
    - bootOrder: 3
      disk:
        bus: virtio
        name: disk-1
  interfaces:
    - bridge: {}
      model: virtio
      macAddress: '02:00:e4:00:03:0f'
      name: eth0
  rng: {}
features:
  acpi: {}
  smm:
    enabled: true
firmware:
  bootloader:
    efi: {}
machine:
  type: pc-q35-rhel9.4.0
memory:
  guest: 64Gi
resources: {}
networks:
  - multus:
      networkName: default/localnet-network
      name: eth0
terminationGracePeriodSeconds: 180
volumes:
  - dataVolume:
      name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com
      name: rootdisk
  - dataVolume:
```

```
name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-cdrom
name: installation-cdrom
- dataVolume:
  name: pvcocp-infra03.cdp.rdu2.scalelab.redhat.com-disk1
  name: disk-1
---
```

STEP 14: Track the installation progress.

```
[root@bastion ~]# openshift-install --dir ocp-install-dir agent wait-for
install-complete --log-level=debug
```

```
[root@ipaserver ocp-agent-install]# openshift-install --dir ocp-install-dir agent
wait-for install-complete --log-level=debug
```

```
DEBUG OpenShift Installer 4.17.42
DEBUG Built from commit f4a8c8c1eed4b425c434d2300c1a9a387789ed9
DEBUG asset directory: ocp-install-dir
DEBUG Loading Agent Config...
DEBUG Using Agent Config Loaded from state file
DEBUG Loading Agent Manifests...
DEBUG Loading Agent PullSecret...
DEBUG Loading Agent Workflow...
DEBUG Using Agent Workflow Loaded from state file
DEBUG Loading Agent Installer ClusterInfo...
DEBUG Loading Agent Workflow...
DEBUG Loading AddNodes Config...
DEBUG Using AddNodes Config Loaded from state file
DEBUG Using Agent Installer ClusterInfo Loaded from state file
DEBUG Loading Install Config...
DEBUG Using Install Config Loaded from state file
DEBUG Using Agent PullSecret Loaded from state file
DEBUG Loading InfraEnv Config...
DEBUG Loading Agent Workflow...
DEBUG Loading Agent Installer ClusterInfo...
DEBUG Loading Install Config...
DEBUG Loading Agent Config...
DEBUG Using InfraEnv Config Loaded from state file
DEBUG Loading NMState Config...
DEBUG Loading Agent Workflow...
DEBUG Loading Agent Installer ClusterInfo...
DEBUG Loading Agent Hosts...
DEBUG Loading Agent Workflow...
DEBUG Loading AddNodes Config...
DEBUG Loading Install Config...
DEBUG Loading Agent Config...
DEBUG Using Agent Hosts Loaded from state file
DEBUG Loading Install Config...
DEBUG Using NMState Config Loaded from state file
DEBUG Loading AgentClusterInstall Config...
DEBUG Loading Agent Workflow...
DEBUG Loading Install Config...
DEBUG Loading Agent Hosts...
DEBUG Using AgentClusterInstall Config Loaded from state file
DEBUG Loading ClusterDeployment Config...
```

```

DEBUG Loading Agent Workflow...
DEBUG Loading Install Config...
DEBUG Using ClusterDeployment Config Loaded from state file
DEBUG Loading ClusterImageSet Config...
DEBUG Loading Agent Workflow...
DEBUG Loading Agent Installer ClusterInfo...
DEBUG Loading Release Image Pull Spec...
DEBUG Using Release Image Pull Spec Loaded from state file
DEBUG Loading Install Config...
DEBUG Using ClusterImageSet Config Loaded from state file
DEBUG Using Agent Manifests Loaded from state file
DEBUG Loading Install Config...
DEBUG Loading SSH Key...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Cluster Name...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Pull Secret...
DEBUG Loading Platform...
DEBUG Loading Agent Hosts...
DEBUG RendezvousIP from the AgentConfig 10.1.49.200
DEBUG Loading Agent Installer API Auth Config...
DEBUG Loading Agent Installer InfraEnv ID...
DEBUG Using Agent Installer InfraEnv ID Loaded from state file
DEBUG Loading Agent Workflow...
DEBUG Using Agent Workflow Loaded from state file
DEBUG Loading AddNodes Config...
DEBUG Using AddNodes Config Loaded from state file
DEBUG Using Agent Installer API Auth Config Loaded from state file
DEBUG Agent Rest API Initialized
INFO Cluster is not ready for install. Check validations
WARNING Cluster validation: The cluster has hosts that are not ready to install.
DEBUG Cluster validation: The cluster has the exact amount of dedicated control plane nodes.
DEBUG Cluster validation: API virtual IPs are defined.
DEBUG Cluster validation: api vips 10.1.49.3 belongs to the Machine CIDR and is not in use.
DEBUG Cluster validation: The Cluster Network CIDR is defined.
DEBUG Cluster validation: The base domain is defined.
DEBUG Cluster validation: Ingress virtual IPs are defined.
DEBUG Cluster validation: ingress vips 10.1.49.4 belongs to the Machine CIDR and is not in use.
DEBUG Cluster validation: The Machine Network CIDR is defined.
DEBUG Cluster validation: The Cluster Machine CIDR is equivalent to the calculated CIDR.
DEBUG Cluster validation: The Cluster Network prefix is valid.
DEBUG Cluster validation: The cluster has a valid network type
DEBUG Cluster validation: Same address families for all networks.
DEBUG Cluster validation: No CIDRS are overlapping.
DEBUG Cluster validation: No ntp problems found
DEBUG Cluster validation: The Service Network CIDR is defined.
DEBUG Cluster validation: cnv is disabled
DEBUG Cluster validation: lso is disabled
DEBUG Cluster validation: lvm is disabled
DEBUG Cluster validation: mce is disabled
DEBUG Cluster validation: odf is disabled
DEBUG Cluster validation: Platform requirements satisfied
DEBUG Cluster validation: The pull secret is set.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Hostname worker08.cdp.rdu2.scalelab.redhat.com is unique in cluster
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Hostname worker08.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the host inventory
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Media device is connected

```

```

DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
93bed28b-df8f-558c-81f9-6422039241dd
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker08.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Hostname master02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Hostname master02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
64c50e5e-5bfd-5a20-b59b-a03a9a1458a1
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host

```

```

DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Hostname worker02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Hostname worker02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
00089653-134d-5a4c-bc25-e3b77da8fbec
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Hostname master01.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Hostname master01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.

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DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
462a303e-0cb3-5fe4-a266-a580b4fcfa76
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Hostname worker06.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Hostname worker06.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
eb194bfd-9bbb-5e78-852b-b9649fe65211
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker06.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Hostname worker05.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Hostname worker05.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk

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DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
f7b7f1b9-817a-5e9e-85ee-4d05d0d8f3be
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker05.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Hostname worker04.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Hostname worker04.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
2ec0e61a-c7ec-5f37-917c-88a8a486ace9
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: lso is disabled

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DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: Lvm is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker04.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Hostname worker10.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Hostname worker10.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
dfd6e101-e113-5938-9c9f-b5598c386584
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: Lvm is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker10.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role master
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role master
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Hostname master03.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Hostname master03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host master03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service

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DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
3833b526-a7bf-5672-a028-f59d7d680ec8
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host master03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Hostname worker01.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Hostname worker01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
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successfully or no attempt was made to pull them
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
b1cfd435-1098-50fe-a95f-ab3765e345e9
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Hostname infra01.cdp.rdu2.scalelab.redhat.com is unique in
cluster

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DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Hostname infra01.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra01.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled successfully
or no attempt was made to pull them
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
1ddf58df-3d52-50e0-bd87-9391b5b9f1dc
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra01.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Hostname worker09.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Hostname worker09.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
422bcd9f-8ff0-57a3-9c2e-e575ff094535
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled

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DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Lso is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: Lvm is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker09.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Hostname worker03.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Hostname worker03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the
host inventory
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
def36ddf-7a48-5a98-860a-1ca00fe0ffec
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Lso is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: Lvm is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the
cluster
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled
successfully or no attempt was made to pull them
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Network latency requirement has been satisfied.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: DNS wildcard check was successful
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
4ae3ae26-a3ef-5845-8a63-f0481cb9bffa

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```

DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role worker
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role worker
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Hostname worker07.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Hostname worker07.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host worker07.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: calculated role is worker
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra02.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: All required container images were either pulled successfully
or no attempt was made to pull them
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Packet Loss requirement has been satisfied.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the
*.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host
fd61237d-f691-5b68-8def-e0e42f360e98
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role auto-assign
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role auto-assign
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Hostname infra02.cdp.rdu2.scalelab.redhat.com is unique in
cluster
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Hostname infra02.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra02.cdp.rdu2.scalelab.redhat.com validation: All disks that have skipped formatting are present in the
host inventory
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: calculated role is worker
INFO Host infra01.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Host 590a24c7-af35-5db9-bddc-03da5310c1f0: Successfully registered
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Valid inventory exists for the host
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient minimum RAM
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient disk capacity
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient CPU cores for role auto-assign

```

```

DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Sufficient RAM for role auto-assign
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Hostname infra03.cdp.rdu2.scalelab.redhat.com is unique in cluster
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Hostname infra03.cdp.rdu2.scalelab.redhat.com is allowed
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Speed of installation disk has not yet been measured
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host is compatible with cluster platform baremetal
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host agent is compatible with the service
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: No request to skip formatting of the installation disk
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: ALL disks that have skipped formatting are present in the host inventory
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host is connected
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Media device is connected
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Machine Network CIDR is defined
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host belongs to all machine network CIDRs
WARNING Host infra03.cdp.rdu2.scalelab.redhat.com validation: No connectivity to the majority of hosts in the cluster
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Platform OpenShift Virtualization is allowed
WARNING Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host couldn't synchronize with any NTP server
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host clock is synchronized with service
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: ALL required container images were either pulled successfully or no attempt was made to pull them
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Network Latency requirement has been satisfied.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Packet loss requirement has been satisfied.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host has been configured with at least one default route.
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the api.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the api-int.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Domain name resolution for the *.apps.cdp.rdu2.scalelab.redhat.com domain was successful or not required
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host subnets are not overlapping
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: No IP collisions were detected by host 590a24c7-af35-5db9-bddc-03da5310c1f0
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: cnv is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: lso is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: lvm is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: mce is disabled
DEBUG Host infra03.cdp.rdu2.scalelab.redhat.com validation: odf is disabled
INFO Host infra03.cdp.rdu2.scalelab.redhat.com: calculated role is worker
INFO Host infra02.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Host infra03.cdp.rdu2.scalelab.redhat.com validation: Host has connectivity to the majority of hosts in the cluster
INFO Host infra03.cdp.rdu2.scalelab.redhat.com: updated status from insufficient to known (Host is ready to be installed)
INFO Cluster is ready for install
INFO Cluster validation: ALL hosts in the cluster are ready to install.
INFO Preparing cluster for installation
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from known to preparing-for-installation (Host finished successfully to prepare for installation)
INFO Host worker05.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.38 seconds; size: 447.71 Megabytes; download rate: 338.98 MBps
INFO Host worker10.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.05 seconds; size: 447.71 Megabytes; download rate: 449.24 MBps
INFO Host worker08.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.73 seconds; size: 447.71 Megabytes; download rate: 271.09 MBps
INFO Host worker03.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.25 seconds; size: 447.71 Megabytes; download rate: 375.09 MBps
INFO Host worker01.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.16 seconds; size: 447.71 Megabytes; download rate: 404.46 MBps
INFO Host worker04.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful (Host finished successfully to prepare for installation)
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.13 seconds; size: 447.71 Megabytes; download rate: 413.80 MBps
INFO Host worker08.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful (Host finished successfully to prepare for installation)
INFO Host worker06.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful (Host finished successfully to prepare for installation)

```

```

INFO Host worker03.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.06 seconds; size: 447.71 Megabytes; download rate: 442.97 MBps
INFO Host master02.cdp.rdu2.scalelab.redhat.com: New image status
quay.io/openshift-release-dev/ocp-v4.0-art-dev@sha256:eba47e7d1ab87091b9a4657e28d203a5bb96f03e118c863ce9ed846d44088cc5.
result: success. time: 1.28 seconds; size: 447.71 Megabytes; download rate: 366.99 MBps
INFO Host worker07.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra01.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Cluster installation in progress
INFO Host master02.cdp.rdu2.scalelab.redhat.com: updated status from preparing-for-installation to preparing-successful
(Host finished successfully to prepare for installation)
INFO Host infra02.cdp.rdu2.scalelab.redhat.com: updated status from preparing-successful to installing (Installation is
in progress)
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 6%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 14%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 21%
INFO Host: worker09.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk
INFO Host: worker09.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 7%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 45%
INFO Host: worker05.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 16%
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 67%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 10%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 14%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 20%
INFO Host: worker06.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 22%
INFO Host: worker03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 30%
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 6%
INFO Host: worker10.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 81%
INFO Host: worker10.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 93%
INFO Host: infra03.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 41%
INFO Host: worker01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 74%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 32%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 43%
INFO Host: infra02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 71%
INFO Host: infra02.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 79%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 60%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 61%
INFO Host: worker07.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 58%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 69%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 77%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 86%
INFO Host: infra01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 97%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 93%
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 100%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Writing image to disk: 100%
INFO Host: worker08.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for control plane
INFO Bootstrap Kube API Initialized
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Rebooting
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Rebooting
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for control plane: Waiting for
masters to join bootstrap control plane
INFO Host: master02.cdp.rdu2.scalelab.redhat.com, reached installation stage Configuring
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for bootkube
INFO Host: master03.cdp.rdu2.scalelab.redhat.com, reached installation stage Done
INFO Node master03.cdp.rdu2.scalelab.redhat.com has been rebooted 1 times before completing installation
INFO Node master02.cdp.rdu2.scalelab.redhat.com has been rebooted 1 times before completing installation
INFO Host: master01.cdp.rdu2.scalelab.redhat.com, reached installation stage Waiting for bootkube: waiting for ETCD
bootstrap to be complete
INFO Bootstrap configMap status is complete
INFO Bootstrap is complete
INFO cluster bootstrap is complete
DEBUG Still waiting for the cluster to initialize: Working towards 4.17.42: 791 of 901 done (87% complete)
DEBUG Still waiting for the cluster to initialize: Multiple errors are preventing progress:
DEBUG * Cluster operators authentication, image-registry, ingress, insights, kube-apiserver, machine-api, monitoring,
openshift-apiserver, openshift-samples, operator-lifecycle-manager-packageserver are not available
DEBUG * Could not update imagestream "openshift/driver-toolkit" (611 of 901): resource may have been deleted
DEBUG * Could not update oauthclient "console" (545 of 901): the server does not recognize this resource, check extension
API servers

```


[illegible]

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```
overwrite: true
contents:
  inline: |
    pool foreman.rdu2.scalelab.redhat.com iburst
    driftfile /var/lib/chrony/drift
    makestep 1.0 3
    rtcsync
    logdir /var/log/chrony
```

```
./butane 99-worker-chrony.bu -o 99-worker-chrony.yaml
```

```
cat 99-worker-chrony.yaml
```

```
# Generated by Butane; do not edit
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: worker
  name: 99-worker-chrony
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
        - contents:
            compression: ""
            source:
data:,pool%20foreman.rdu2.scalelab.redhat.com%20iburst%20%0Adriftfile%20%2Fva
r%2Flib%2Fchrony%2Fdrift%0Amakestep%201.0%203%0Artcsync%0Alogdir%20%2Fvar%2Fl
og%2Fchrony%0A
            mode: 420
            overwrite: true
            path: /etc/chrony.conf
```

Apply the MachineConfig using the following command:

```
oc create -f 99-worker-chrony.yaml
```

This will initiate a rolling reboot to update each worker node in the cluster. You can use the following commands to view the status of the nodes during the update:

```
oc get mcp
oc get nodes
```

```
oc describe machineconfig 99-worker-chrony
```

```
vi 99-master-chrony.bu
```

```
variant: openshift
version: 4.16.0
metadata:
  name: 99-master-chrony
  labels:
    machineconfiguration.openshift.io/role: master
storage:
  files:
  - path: /etc/chrony.conf
    mode: 0644
    overwrite: true
    contents:
      inline: |
        pool foreman.rdu2.scalelab.redhat.com iburst
        driftfile /var/lib/chrony/drift
        makestep 1.0 3
        rtcsync
        logdir /var/log/chrony
```

```
./butane 99-worker-chrony.bu -o 99-worker-chrony.yaml
```

```
cat 99-worker-chrony.yaml
```

```
# Generated by Butane; do not edit
apiVersion: machineconfiguration.openshift.io/v1
kind: MachineConfig
metadata:
  labels:
    machineconfiguration.openshift.io/role: master
  name: 99-master-chrony
spec:
  config:
    ignition:
      version: 3.4.0
    storage:
      files:
      - contents:
          compression: ""
          source:
            data: ,pool%20foreman.rdu2.scalelab.redhat.com%20iburst%20%0Adriftfile%20%2Fvar%2Flib%2Fchrony%2Fdrift%0Amakestep%201.0%203%0Artcsync%0Alogdir%20%2Fvar%2Flog%2Fchrony%0A
            mode: 420
            overwrite: true
            path: /etc/chrony.conf
```

```
oc create -f 99-master-chrony.yaml
```

This will initiate a rolling reboot to update each master node in the cluster. You can use the following commands to view the status of the nodes during the update:

```
oc get mcp  
oc get nodes
```

```
oc describe machineconfig 99-master-chrony
```

ALTERNATIVE CONFIGURATION STEPS

OPTIONAL Method to create VMs

STEP 1. Select **Virtualization > Catalog > Template Catalog > Red Hat Enterprise Linux 9 VM**

Project: pvcbase ▾

Create new VirtualMachine

Select an option to create a VirtualMachine from.

InstanceTypes **Template catalog**

Template project: All projects ▾

All templates

Default templates **Filter by keyword...**

12 items

Default templates

User templates

☐ Boot source available

Operating system

☐ CentOS

☐ Fedora

☐ Other

☐ RHEL

☐ Windows

Workload

☐ Desktop

☐ High performance

☐ Server

 CentOS Stream 8 VM centos-stream8-server-small Project openshift Boot source PVC Workload Server CPU 1 Memory 2 GiB	 CentOS Stream 9 VM centos-stream9-server-small Project openshift Boot source PVC (auto import) Workload Server CPU 1 Memory 2 GiB	 CentOS 7 VM centos7-server-small Project openshift Boot source PVC Workload Server CPU 1 Memory 2 GiB	 Fedora VM fedora-server-small Project openshift Boot source PVC (auto import) Workload Server CPU 1 Memory 2 GiB
 Red Hat Enterprise Linux 7 VM rhel7-server-small Project openshift Boot source PVC Workload Server CPU 1 Memory 2 GiB	 Red Hat Enterprise Linux 8 VM rhel8-server-small Project openshift Boot source PVC (auto import) Workload Server CPU 1 Memory 2 GiB	 Red Hat Enterprise Linux 9 VM rhel9-server-small Project openshift Boot source PVC (auto import) Workload Server CPU 1 Memory 2 GiB	 Microsoft Windows 10 VM windows10-desktop-medium Project openshift Boot source PVC Workload Desktop CPU 1 Memory 4 GiB

STEP 2. Edit the following parameters: VirtualMachine name, CPU | Memory, Disk size, Optional Parameters-CLOUD_USER_PASSWORD:



Red Hat Enterprise Linux 9 VM

rhel9-server-small

Server (quantity)

1

Description

Template for Red Hat Enterprise Linux 9 VM or newer. A PVC with the RHEL disk image must be available.

Documentation

[Refer to documentation](#)

CPU | Memory

32 CPU | 64 GiB Memory

Network interfaces (1)

Name	Network	Type
		r3dh4t!

Disk size *

- 1508 + GiB

Drivers

☐ Mount Windows drivers disk

Optional parameters

CLOUD_USER_PASSWORD

r3dh4t!

Quick create VirtualMachine

VirtualMachine name *

cldr-mngr

Project pvcbase

Public SSH key cdp-ssh

☒ Start this VirtualMachine after creation

Quick create VirtualMachine Customize VirtualMachine Cancel

VM configuration parameters:

Parameter	Setting/Value
VirtualMachine name	cldr-mngr.redhat.local
Project name	pvc-base
Operating System	Red Hat Enterprise Linux 9 VM
CPU	32 CPU
Memory	64 GiB
Storage:	
Disk source	Blank
Disk Size	1500 GiB

CLOUD_USER_PASSWORD

r3dh4t!

STEP 3. Select Customize Virtualmachine.

IMPORTANT: From the screen that follows, make sure that you edit several settings *before* you select **Create Virtual Machine**.

Project: pvcbase

Catalog

Customize and create VirtualMachine

YAML

Template: Red Hat Enterprise Linux 9 VM

Overview

YAML

Scheduling

Environment

Network interfaces

Disks

Scripts

Metadata

Name

ipaserver

Namespace

pvcbase

Description

Not available

Operating system

Red Hat Enterprise Linux 9 VM

CPU | Memory

16 CPU | 32 GiB Memory

Machine type

pc-q35-rhel9.4.0

Boot mode

UEFI (secure)

Network interfaces (1)

Name	Network	Type
default	Pod networking	Masquerade

Disks (2)

Name	Drive	Size
rootdisk	Disk	250 GiB
cloudinitdisk	Disk	-

Hardware devices

GPU devices

Not available

Host devices

Not available

Headless mode

Hostname

☒ Start this VirtualMachine after creation

Create VirtualMachine

Cancel

STEP 4. From the “*Network Interfaces*” screen, click on the three vertical dots to the far right of the “*default*” network interface, and click on “*Delete*”.

Project: pvc-ds ▾

Catalog

Customize and create VirtualMachine ☒ YAML

Template: Red Hat Enterprise Linux 9 VM

Overview [YAML](#) [Scheduling](#) [Environment](#) [Network interfaces](#) [Disks](#) [Scripts](#) [Metadata](#)

[Add network interface](#)

▼ Filter ▾ Name ▾ Search by name... /

Name ↑	Model ▾	Network ▾	Type ▾	MAC address ▾	
default	virtio	Pod networking	Masquerade	027a:32:00:00:04	⋮

Edit
Delete

☒ Start this VirtualMachine after creation (RerunOnFailure)

[Create VirtualMachine](#) [Cancel](#)

STEP 5. Then, confirm by clicking on “Delete”.

Delete NIC? ✕

Are you sure you want to delete **default** in namespace **pvc-ds**?

[Delete](#) [Cancel](#)

STEP 6. From the “*Network Interfaces*” screen, click on “*Add network interface*”.

Project: pvc-ds ▾

Catalog

Customize and create VirtualMachine  YAML

Template: Red Hat Enterprise Linux 9 VM

Overview YAML Scheduling Environment Network interfaces Disks Scripts Metadata

Add network interface

Not found

☒ Start this VirtualMachine after creation (RerunOnFailure)

Create VirtualMachine Cancel

STEP 7. From the “Add network interface” screen, enter the following parameters and then click on “Save”.

Network interface configuration parameters:

Parameter	Setting/Value
Network interface name	eth0
Model	virtio
Network	default/localnet-network

Add network interface ✕

Name *

eth0

Model

virtio ▼

Network * ?

Q default/localnet-network ✕ ▼

> Advanced

Save

Cancel

STEP 8. Next, select the **Scripts** tab; and click on **Edit** in the **Cloud-init** section:

Project: pvcbase ▾

Catalog

Customize and create VirtualMachine YAML

Template: Red Hat Enterprise Linux 9 VM

OverviewYAMLSchedulingEnvironmentNetwork interfacesDisksScriptsMetadata

⚠ Cloud-init and SSH key configurations will be applied to the VirtualMachine only at the first boot.
You must ensure that the configuration is correct before starting the VirtualMachine.

Cloud-init Edit

You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started.

The cloud-init service is enabled by default in Fedora and RHEL images. [Learn more](#)

User	Password	Network data
cloud-user	*****	Default

Public SSH key Linux only Edit

Store the key in a project secret.

The key will be stored after the machine is created

cdp-ssh

☒ Start this VirtualMachine after creation

Create VirtualMachine

Cancel

STEP 9. From the **Cloud-init** menu, select the checkbox next to **Add network data**, then enter the parameters as shown in the following screen for the **Ethernet name** and **Gateway address**.

Refer to the table provided at the beginning of this activity for the **IP address** to use for each VM. Select **Apply** when done.

Cloud-init



You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started.

The cloud-init service is enabled by default in Fedora and RHEL images.

[Learn more](#)

Configure via: ☒ Form view ☐ Script

User *

cloud-user

Password

r3dh4t!



Password for this username - [generate password](#)

☒ Add network data

check this option to add network data section to the cloud-init script.

Ethernet name

eth0

IP addresses

10.6.131.20/26

Use commas to separate between IP addresses

Gateway address

10.6.131.62

Apply

Cancel

Use the following table as a reference to create the VirtualMachines for CDP Base:

Hostname	CPU	RAM	Disk	MAC Address	IP Address
cldr-mngr.redhat.local	32	64G	1.5T	02:7c:71:05:00:01	192.168.2.190/24
pvcbase-master.redhat.local	32	64G	1T	02:7c:71:05:00:02	192.168.2.191/24
pvcbase-worker1.redhat.local	32	64G	1T	02:7c:71:05:00:03	192.168.2.192/24
pvcbase-worker2.redhat.local	32	64G	1T	02:7c:71:05:00:04	192.168.2.193/24
pvcbase-worker3.redhat.local	32	64G	1T	02:7c:71:05:00:05	192.168.2.194/24
pvcbase-worker4.redhat.local	32	64G	1T	02:7c:71:05:00:06	192.168.2.195/24
pvcbase-worker5.redhat.local	32	64G	1T	02:7c:71:05:00:07	192.168.2.196/24

Project: pvcbase

Catalog

Customize and create VirtualMachine
YAML

Template: Red Hat Enterprise Linux 9 VM

Overview
YAML
Scheduling
Environment
Network interfaces
Disks
Scripts
Metadata

⚠ Cloud-init and SSH key configurations will be applied to the VirtualMachine only at the first boot.
You must ensure that the configuration is correct before starting the VirtualMachine.

Cloud-init
Edit

You can use cloud-init to initialize the operating system with a specific configuration when the VirtualMachine is started.
The cloud-init service is enabled by default in Fedora and RHEL images. [Learn more](#)

User
Password
Network data

cloud-user

Custom

Public SSH key
Linux only
Edit

Store the key in a project secret.
The key will be stored after the machine is created

cdp-ssh

☒ Start this VirtualMachine after creation

Create VirtualMachine
Cancel

STEP 10. Select the **YAML** tab to edit the YAML definition for the VM. Add the highlighted **nameservers** section for DNS servers and search domains to match the provided section,

IMPORTANT: Make sure you click on **Save**, before you click on **Create VirtualMachine**. Otherwise, the changes made to the YAML will NOT be saved.

Project: pvc-base ▾

Catalog

Customize and create VirtualMachine

Template: Red Hat Enterprise Linux 9 VM

Overview **YAML** Scheduling Environment Network interfaces Disks Scripts Metadata

179
180
181
182
183
184
185
186
187
188
189
190
191
192
193
194
195
196
197
198
199
200
201
202

```
- cloudInitNoCloud:
  networkData: |
    ethernet0:
      addresses:
        - 10.6.131.21/26
      gateway4: 10.6.131.62
      nameservers:
        search: [cdppvcds.rdu3.labs.perfscale.redhat.com]
        addresses: [10.6.131.1]
      version: 2
    userData: |
      #cloud-config
      users: cloud-user
      password: r3dh4t!
      chpasswd:
        expire: false
      runcmd:
        - subscription-manager register --org=11009103 --activationkey=cdpbase
        - systemctl enable --now dnf-automatic-install.timer
      packages:
        - dnf-automatic
    name: cloudinitdisk
```

Save Reload Download

☒ Start this VirtualMachine after creation

Create VirtualMachine Cancel

Edit the YAML to include the following, under the **spec.template.spec.volumes.cloudInitNoCloud** section for the DNS servers and search domains:

```
spec:
  template:
    spec:
      volumes:
        - dataVolume:
            name: cldr-mngr
            name: rootdisk
        - cloudInitNoCloud:
            networkData: |
              ethernet0:
                addresses:
                  - 192.168.2.190/26
                gateway4: 192.168.2.1
                nameservers:
                  search: [redhat.local]
                  addresses: [192.168.1.210]
            version: 2
            userData: |
              #cloud-config
              user: cloud-user
```



```
password: r3dh4t!
chpasswd:
  expire: false
runcmd:
  - subscription-manager register --org=[your_org_#]
--activationkey=[your_activation_key]
  - systemctl enable --now dnf-automatic-install.timer
packages:
  - dnf-automatic
name: cloudinitdisk
```

NOTE: In case the VM does not register with Red Hat Insights, you can use the following command to manually register the VM: `rhc connect`

Example:

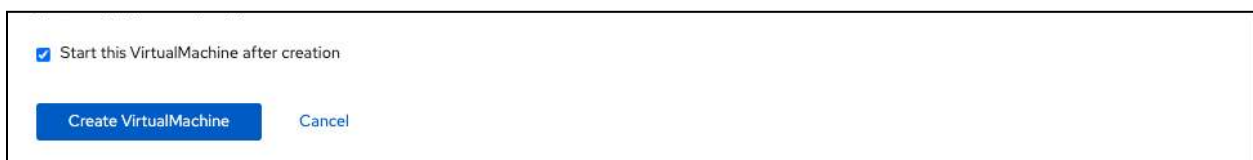
```
# rhc connect --activation-key [your_activation_key] --organization
[your_org_#]
```

STEP 11. When you have finished editing the VM, select the **Create VirtualMachine** button.

IMPORTANT: When editing the YAML, **make sure you click on the Save button** before you click on **Create VirtualMachine**. Otherwise, the changes made in the YAML editor will NOT be saved.

IMPORTANT: Some of the parameters, such as CloudInit only take effect after the **first** boot of the virtual machine.

If there are settings that still need to be made before this, then make sure to unselect the option to **'Start this VirtualMachine after creation'**.



☒ Start this VirtualMachine after creation

Create VirtualMachine Cancel

STEP 12. From the **Overview** tab, you can view the VM provisioning status.

Project: pvcbase

VirtualMachines > VirtualMachine details

VM cldr-mngr Running

Overview Metrics YAML Configuration Events Console Snapshots Diagnostics

Details

Name	cldr-mngr	VNC console
Status	Running	
Created	Sep 27, 2024, 5:31 PM (1 minute ago)	
Operating system	Red Hat Enterprise Linux 9.4 (Plow)	
CPU Memory	32 CPU 64 GiB Memory	
Time zone	EDT	
Template	rhel9-server-small	
Hostname	cldr-mngr	
Machine type	pc-q35-rhel9.4.0	Open web console

Alerts (0)

General

Namespace NS pvcbase

Node N bb37-h37-000-r750.cdppvc...

VirtualMachineInstance VM cldr-mngr

Pod P virt-launcher-cldr-mngr-c89...

Owner No owner

Snapshots (0) Take snapshot

No snapshots found

Network (1)

Utilization

CPU	Memory	Storage	Network transfer
			Breakdown by network

ADD NODES TO THE CLUSTER

6.4.3. Adding nodes to your cluster

vi nodes-config.yaml file

```
hosts:
- hostname: f36-h03-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:79:19:f1 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:79:19:f1 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.48.113 # Change for each host
            prefix-length: 23
        dhcp: false
```

```

dns-resolver:
  config:
    server:
      - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: eno2np1
        table-id: 254
- hostname: f36-h05-000-r640
rootDeviceHints:
  deviceName: /dev/sda
interfaces:
  - name: eno2np1
    macAddress: bc:97:e1:7c:f6:31 # Change for each host
networkConfig:
  interfaces:
    - name: eno2np1
      type: ethernet
      state: up
      mac-address: bc:97:e1:7c:f6:31 # Change for each host
      ipv4:
        enabled: true
        address:
          - ip: 10.1.48.114 # Change for each host
            prefix-length: 23
        dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: eno2np1
        table-id: 254
- hostname: f36-h07-000-r640
rootDeviceHints:
  deviceName: /dev/sda
interfaces:
  - name: eno2np1
    macAddress: bc:97:e1:79:2c:11 # Change for each host
networkConfig:
  interfaces:

```

```
- name: eno2np1
  type: ethernet
  state: up
  mac-address: bc:97:e1:79:2c:11 # Change for each host
  ipv4:
    enabled: true
    address:
      - ip: 10.1.48.115 # Change for each host
        prefix-length: 23
    dhcp: false
  dns-resolver:
    config:
      server:
        - 10.1.49.1
  routes:
    config:
      - destination: 0.0.0.0/0
        next-hop-address: 10.1.49.254
        next-hop-interface: eno2np1
        table-id: 254
- hostname: f36-h09-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:78:e8:61 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:78:e8:61 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.116 # Change for each host
              prefix-length: 23
          dhcp: false
        dns-resolver:
          config:
            server:
              - 10.1.49.1
        routes:
          config:
            - destination: 0.0.0.0/0
              next-hop-address: 10.1.49.254
```

```

        next-hop-interface: eno2np1
        table-id: 254
- hostname: f36-h10-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:78:e7:a1 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:78:e7:a1 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.117 # Change for each host
              prefix-length: 23
          dhcp: false
        dns-resolver:
          config:
            server:
              - 10.1.49.1
        routes:
          config:
            - destination: 0.0.0.0/0
              next-hop-address: 10.1.49.254
              next-hop-interface: eno2np1
              table-id: 254
- hostname: f36-h11-000-r640
  rootDeviceHints:
    deviceName: /dev/sda
  interfaces:
    - name: eno2np1
      macAddress: bc:97:e1:69:92:31 # Change for each host
  networkConfig:
    interfaces:
      - name: eno2np1
        type: ethernet
        state: up
        mac-address: bc:97:e1:69:92:31 # Change for each host
        ipv4:
          enabled: true
          address:
            - ip: 10.1.48.118 # Change for each host

```

```
        prefix-length: 23
        dhcp: false
    dns-resolver:
        config:
            server:
                - 10.1.49.1
    routes:
        config:
            - destination: 0.0.0.0/0
              next-hop-address: 10.1.49.254
              next-hop-interface: eno2np1
              table-id: 254
```

```
[root@ipaserver]# oc adm node-image create nodes-config.yaml
```

```
[root@ipaserver]# ll *.iso
```

```
-rw-r--r--. 1 root root 1326184448 Nov  4 18:01 node.x86_64.iso
```

```
cp node.x86_64.iso /var/www/html/.
```

Verify that the discovery image is available at:
http://10.1.49.1/node.x86_64.iso

BADFISH

Configure the servers to boot from the Discovery image:

```
export USER=quads
export PASS=rdu2@4618
```

```
dnf install -y podman
```

```
vi workerlist
```

```
10.1.64.196
```

```
10.1.65.226
```

```
10.1.65.96
```

```
10.1.65.139
```

```
10.1.64.17
```

```
10.1.65.3
```

```
10.1.64.139
```

```
10.1.65.220
```

10.1.65.210

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-off;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --unmount-virtual-media;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --mount-virtual-media  
http://10.1.49.1/node.x86_64.iso;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --boot-to-virtual-media;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-on;done
```

```
for HOST in `cat ~/workerlist`;do podman run -it --rm quay.io/quads/badfish  
-H $HOST -u $USER -p $PASS --power-state;done
```

- - -

NodeNetworkConfigurationPolicy

```
apiVersion: nmstate.io/v1  
kind: NodeNetworkConfigurationPolicy  
metadata:  
  name: f36-h03-000-r640  
  #namespace:  
spec:  
  nodeSelector:  
    node-role.kubernetes.io/worker: ""  
    kubernetes.io/hostname: f36-h03-000-r640  
  desiredState:  
    interfaces:  
      - name: eno1np0  
        description: iSCSI network  
        type: ethernet  
        ethernet:  
          state: down  
          ipv4:  
            enabled: false  
          ipv6:  
            enabled: false  
---
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h05-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h05-000-r640
  desiredState:
    interfaces:
      - name: eno1np0
        description: iSCSI network
        type: ethernet
        ethernet:
          state: down
          ipv4:
            enabled: false
          ipv6:
            enabled: false
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h07-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h07-000-r640
  desiredState:
    interfaces:
      - name: eno1np0
        description: iSCSI network
        type: ethernet
        ethernet:
          state: down
          ipv4:
            enabled: false
          ipv6:
            enabled: false
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h09-000-r640
```



```
#namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h09-000-r640
  desiredState:
    interfaces:
      - name: eno1np0
        description: iSCSI network
        type: ethernet
        ethernet:
          state: down
          ipv4:
            enabled: false
          ipv6:
            enabled: false
---
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h10-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h10-000-r640
  desiredState:
    interfaces:
      - name: eno1np0
        description: iSCSI network
        type: ethernet
        ethernet:
          state: down
          ipv4:
            enabled: false
          ipv6:
            enabled: false
---
```

```
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h11-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
```

```

    kubernetes.io/hostname: f36-h11-000-r640
  desiredState:
    interfaces:
      - name: eno1np0
        description: iSCSI network
        type: ethernet
        ethernet:
          state: down
          ipv4:
            enabled: false
          ipv6:
            enabled: false
    ---
  apiVersion: nmstate.io/v1
  kind: NodeNetworkConfigurationPolicy
  metadata:
    name: f36-h13-000-r640
    #namespace:
  spec:
    nodeSelector:
      node-role.kubernetes.io/worker: ""
      kubernetes.io/hostname: f36-h13-000-r640
    desiredState:
      interfaces:
        - name: eno1np0
          description: iSCSI network
          type: ethernet
          ethernet:
            state: down
            ipv4:
              enabled: false
            ipv6:
              enabled: false
      ---
    apiVersion: nmstate.io/v1
    kind: NodeNetworkConfigurationPolicy
    metadata:
      name: f36-h14-000-r640
      #namespace:
    spec:
      nodeSelector:
        node-role.kubernetes.io/worker: ""
        kubernetes.io/hostname: f36-h14-000-r640
      desiredState:
        interfaces:
          - name: eno1np0

```

```

    description: iSCSI network
    type: ethernet
    ethernet:
    state: down
    ipv4:
      enabled: false
    ipv6:
      enabled: false
---
apiVersion: nmstate.io/v1
kind: NodeNetworkConfigurationPolicy
metadata:
  name: f36-h15-000-r640
  #namespace:
spec:
  nodeSelector:
    node-role.kubernetes.io/worker: ""
    kubernetes.io/hostname: f36-h15-000-r640
  desiredState:
    interfaces:
    - name: eno1np0
      description: iSCSI network
      type: ethernet
      ethernet:
      state: down
      ipv4:
        enabled: false
      ipv6:
        enabled: false
---

```

OPTIONAL METHOD TO CREATE VM YAML

STEP 6. Switch to the OpenShift console, and navigate to Virtualization > Catalog > Create a VM > from Catalog > RHEL 9. Enter the VM configuration parameters from the table provided below, and then select “*Customize VirtualMachine*” to further customize the VM.

Project: pvc-ds

Create new VirtualMachine

Select an option to create a VirtualMachine from.

InstanceTypes

Template catalog

Template project

All projects

Filter by keyword...

Default templates

User templates

☒ Hide deprecated templates

☐ Boot source available

Operating system

☐ CentOS
 ☐ Fedora
 ☐ Other
 ☐ RHEL
 ☐ Windows

Workload

☐ Desktop
 ☐ High performance
 ☐ Server

CentOS Stream 9 VM

centos-stream9-server-small

Project openshift

Boot source PVC (auto import)

Workload Server

CPU 1

Memory 2 GiB

Red Hat Enterprise Linux 9 VM

rhel9-server-small

Project openshift

Boot source PVC (auto import)

Workload Server

CPU 1

Memory 2 GiB

Microsoft Windows Server 2019 VM

Red Hat Enterprise Linux 9 VM

rhel9-server-small

Template info

Operating system

Red Hat Enterprise Linux 9 VM

Workload type

Server (default)

Description

Template for Red Hat Enterprise Linux 9 VM or newer. A PVC with the RHEL disk image must be available.

Documentation

Refer to documentation

CPU | Memory

4 CPU | 16 GiB Memory

Network interfaces (1)

Name	Network	Type
default	Pod networking	Masquerade

Disks (3)

Name	Drive	Size
rootdisk	Disk	120 GiB
cloudinitdisk	Disk	-
installation-cdrom	CD-ROM	2 GiB

Storage

☒ Boot from CD

CD source

URL (creates PVC)

Image URL

http://192.168.1.210/discovery_image_ocp-ds.iso

Enter URL to download. For example: https://access.redhat.com/downloads/content/479/ver/rhel--9/9.0/x86_64/product-software

Disk size

- 2 + GiB

Disk source

Blank

Disk size

- 120 + GiB

Drivers

☐ Mount Windows drivers disk

Quick create VirtualMachine

VirtualMachine name

master0.ocp-ds.redhat.local

Project

pvc-ds

Public SSH key

cdp-ssh

☒ Start this VirtualMachine after creation (RerunOnFailure)

Quick create VirtualMachine

Customize VirtualMachine

Cancel

VM configuration parameters:

Parameter	Setting/Value
VirtualMachine name	master0.cdp.rdu2.scalelab.redhat.com
Project name	cdp-ds
Operating System	Red Hat Enterprise Linux 9 VM
CPU	4 CPU
Memory	16 GiB
Storage:	
Boot from CD	Selected (enabled)
CD source	URL (creates PVC)
Image URL	http://10.1.49.1/discovery_image_cdpds.iso

Disk Size	2 GiB
Disk source	Blank
Disk Size	120 GiB

STEP 7. From the Customize and create VirtualMachine screen, select the **“Network Interfaces”** tab.

Project: pvc-ds

Catalog

Customize and create VirtualMachine
YAML

Template: Red Hat Enterprise Linux 9 VM

Overview
YAML
Scheduling
Environment
Network interfaces
Disks
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Metadata

Name
master0.ocp-ds.redhat.local

Namespace
pvc-ds

Description
Not available

Operating system
Red Hat Enterprise Linux 9 VM

CPU | Memory
4 CPU | 16 GiB Memory

Machine type
pc-q35-rhel9.4.0

Boot mode
UEFI (secure)

Start in pause mode

Workload profile
Server

Network interfaces (1)

Name	Network	Type
default	Pod networking	Masquerade

Disks (3)

Name	Drive	Size
rootdisk	Disk	120 GiB
installation-cdrom	CD-ROM	2 GiB
cloudinitdisk	Disk	-

Hardware devices

GPU devices
Not available

Host devices
Not available

Headless mode

Hostname
master0.ocp-ds.redhat.local

Guest system log access

☒ Start this VirtualMachine after creation (RerunOnFailure)

Create VirtualMachine
Cancel

STEP 8. From the **“Network Interfaces”** screen, click on the three vertical dots to the far right of the **“default”** network interface, and click on **“Delete”**.

Project: pvc-ds ▾

Catalog

Customize and create VirtualMachine ☐ YAML

Template: Red Hat Enterprise Linux 9 VM

Overview YAML Scheduling Environment Network interfaces Disks Scripts Metadata

[Add network interface](#)

▼ Filter ▾ Name ▾ Search by name... /

Name ↑	Model ▾	Network ▾	Type ▾	MAC address ▾	
default	virtio	Pod networking	Masquerade	027a:32:00:00:04	⋮

Edit

Delete

☒ Start this VirtualMachine after creation (RerunOnFailure)

[Create VirtualMachine](#) [Cancel](#)

STEP 9. Then, confirm by clicking on “Delete”.

Delete NIC? ✕

Are you sure you want to delete **default** in namespace **pvc-ds**?

[Delete](#) [Cancel](#)

STEP 10. From the “*Network Interfaces*” screen, click on “*Add network interface*”.

Project: pvc-ds ▾

Catalog

Customize and create VirtualMachine  YAML

Template: Red Hat Enterprise Linux 9 VM

Overview YAML Scheduling Environment Network interfaces Disks Scripts Metadata

Add network interface

Not found

☒ Start this VirtualMachine after creation (RerunOnFailure)

Create VirtualMachine

Cancel

STEP 11. From the “Add network interface” screen, enter the following parameters and then click on “Save”.

Network interface configuration parameters:

Parameter	Setting/Value
Network interface name	eth0
Model	virtio
Network	default/localnet-network

Add network interface ✕

Name *

eth0

Model

virtio ▼

Network * ?

Q default/localnet-network ✕ ▼

> Advanced

Save

Cancel

STEP 12. Select the “Overview” tab, and confirm the configuration settings. Then click on “Create VirtualMachine”.

Project: pvc-ds

Catalog

Customize and create VirtualMachine ☐ YAML

Template: Red Hat Enterprise Linux 9 VM

OverviewYAMLSchedulingEnvironmentNetwork interfacesDisksScriptsMetadata

Name

master0.ocp-ds.redhat.local

Namespace

pvc-ds

Description

Not available

Operating system

Red Hat Enterprise Linux 9 VM

CPU | Memory

4 CPU | 16 GiB Memory

Machine type

pc-q35-rhel9.4.0

Boot mode

UEFI (secure)

Start in pause mode

☐

Workload profile

Server

Network interfaces (1)

Name	Network	Type
eth0	default/localnet-network	Bridge

Disks (3)

Name	Drive	Size
rootdisk	Disk	120 GiB
installation-cdrom	CD-ROM	2 GiB
cloudinitdisk	Disk	-

Hardware devices

GPU devices

Not available

Host devices

Not available

Headless mode

☐

Hostname

master0.ocp-ds.redhat.local

Guest system log access

☐

☒ Start this VirtualMachine after creation (RerunOnFailure)

Create VirtualMachineCancel